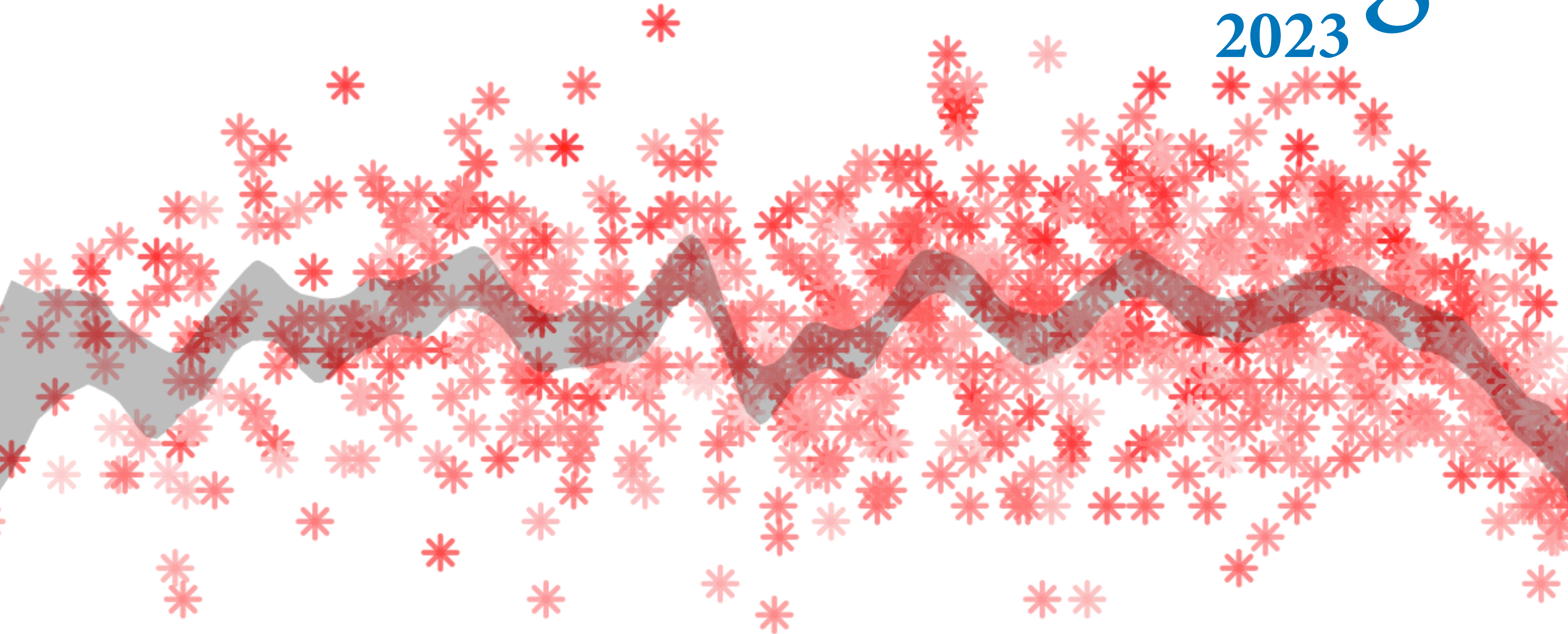


Statistical Rethinking

2023

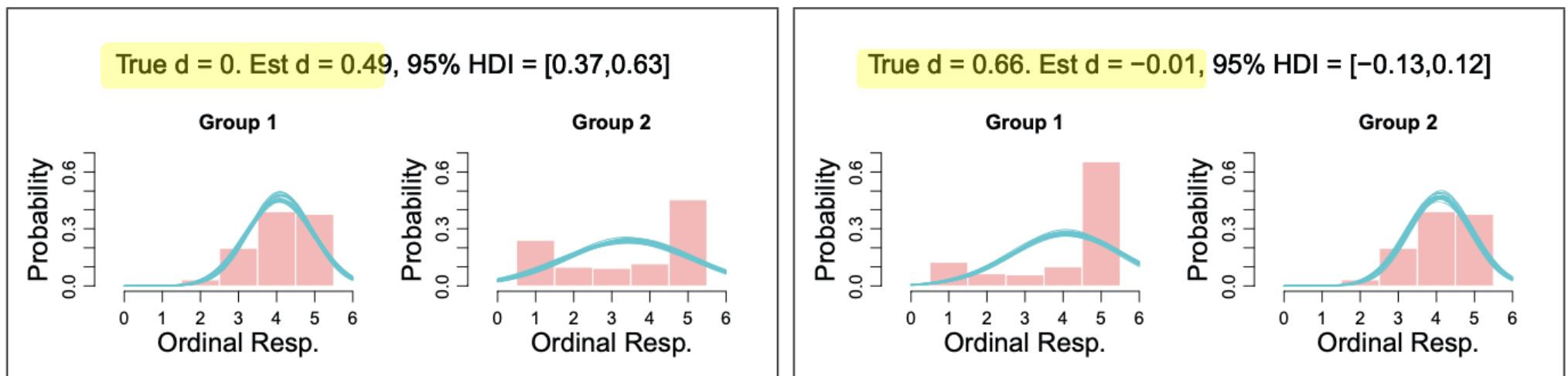


11. Ordered Categories

Ordinal data are often analyzed as if they were metric

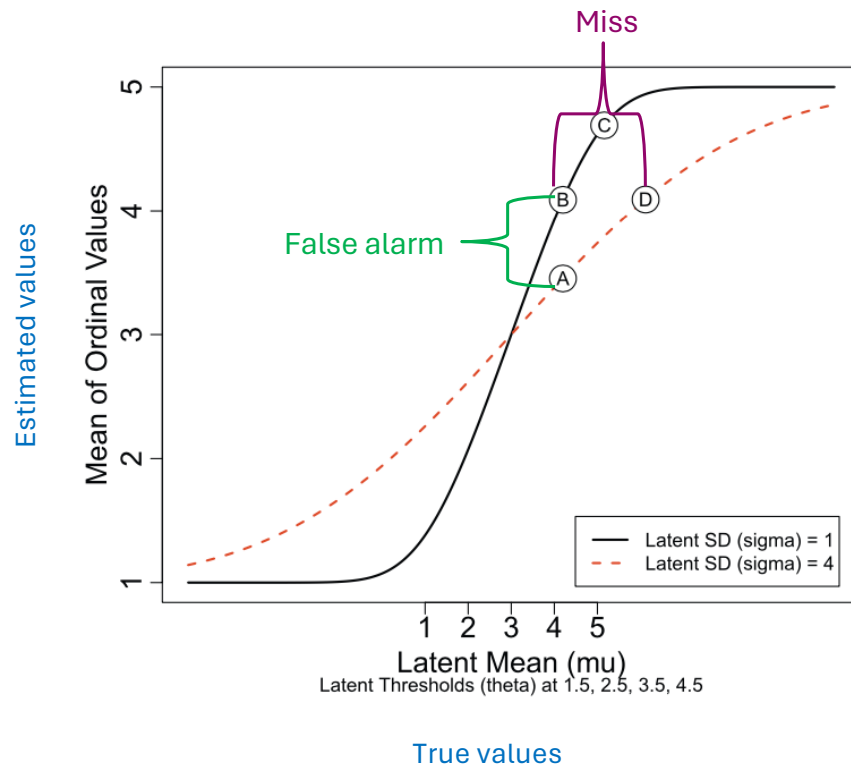
- Liddell and Kruschke 2018
 - <https://doi.org/10.1016/j.jesp.2018.08.009>
- All “top psych” journal papers using a Likert scale (N = 68) analyzed the data as if they were metric instead of ordinal. (in 2016)
 - *Journal of Personality and Social Psychology* (JPSP),
 - *Psychological Science* (PS),
 - *Journal of Experimental Psychology: General* (JEP:G)
- Does making a metric assumption about ordinal data matter?

Assuming ordinal data are metric can lead to **false positive and false negative** results

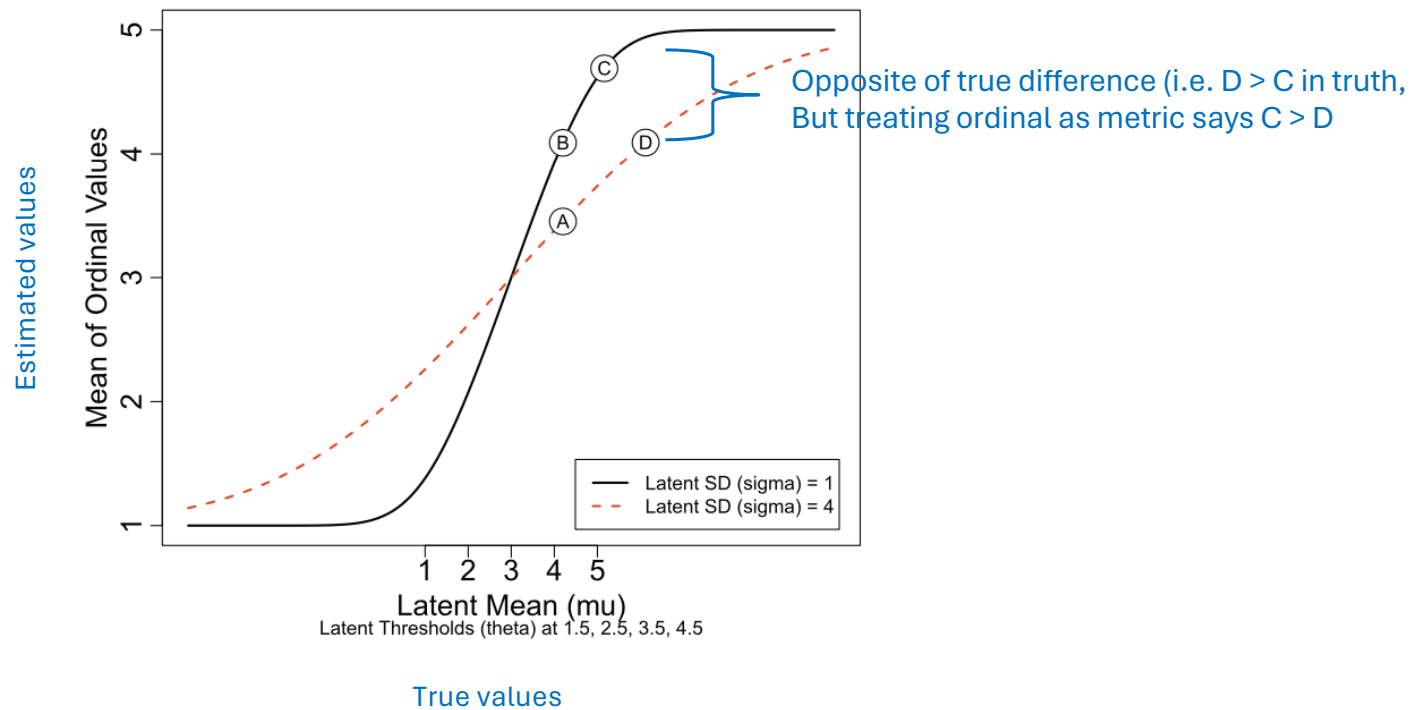


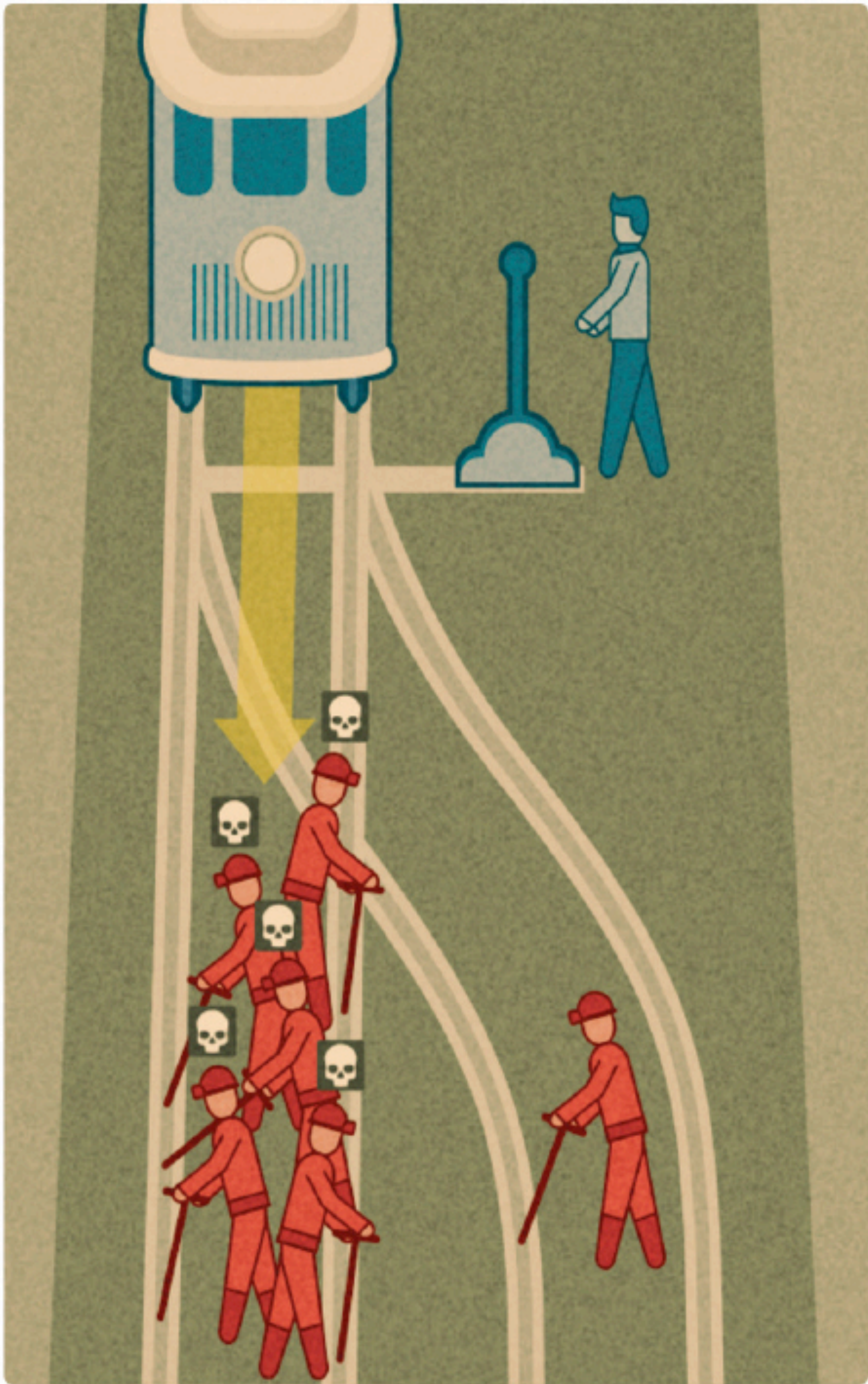
*This example is about testing means. The mean response may not be relevant or meaningful in some cases.

Reversed effects are also possible

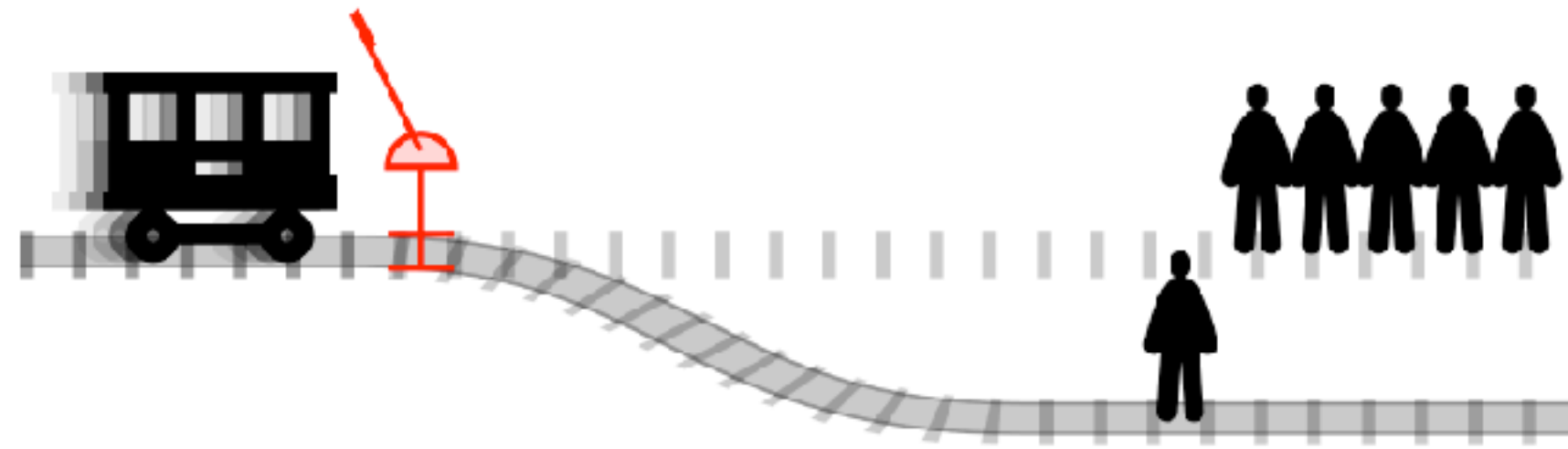


Reversed effects are also possible

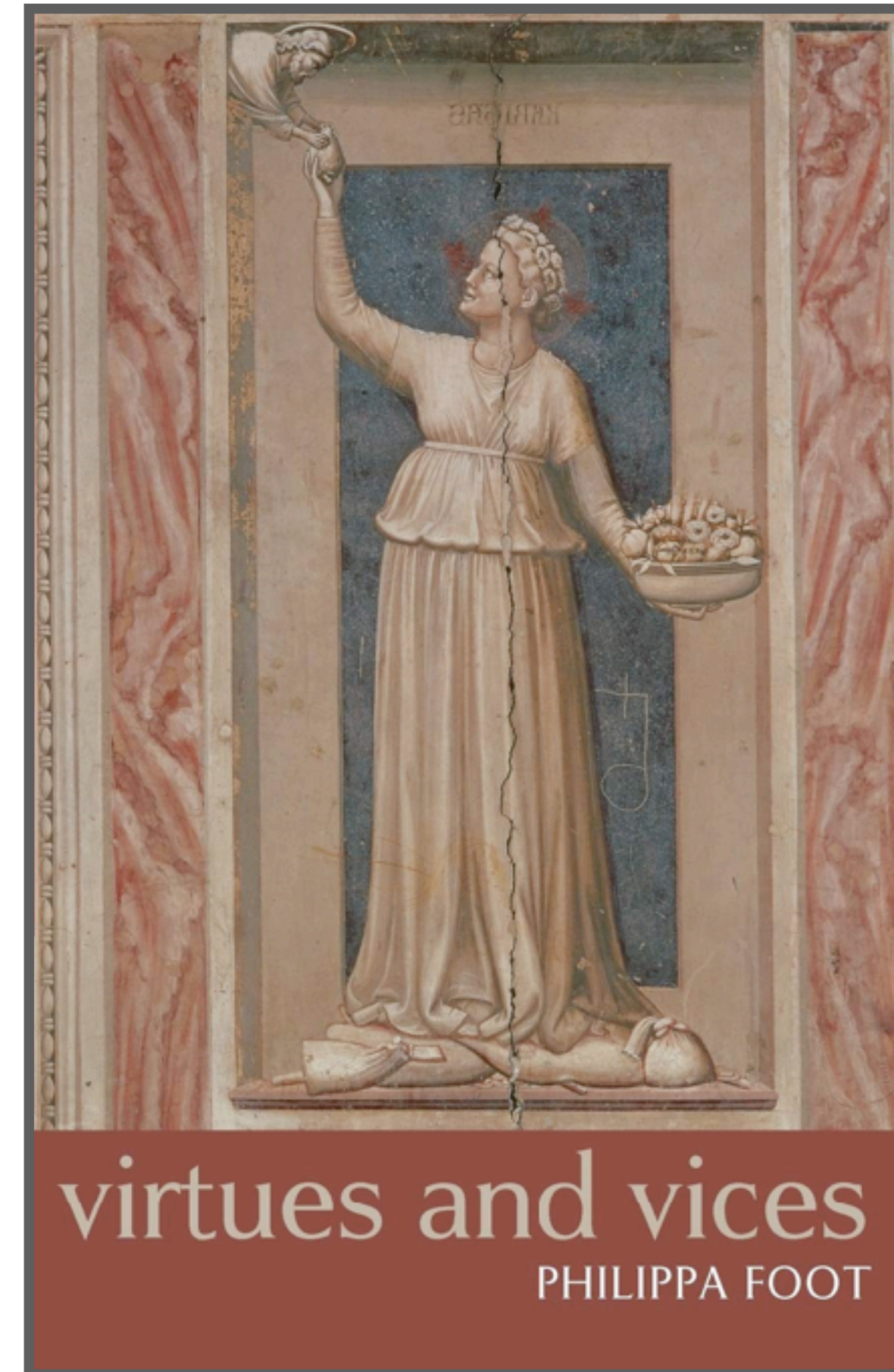




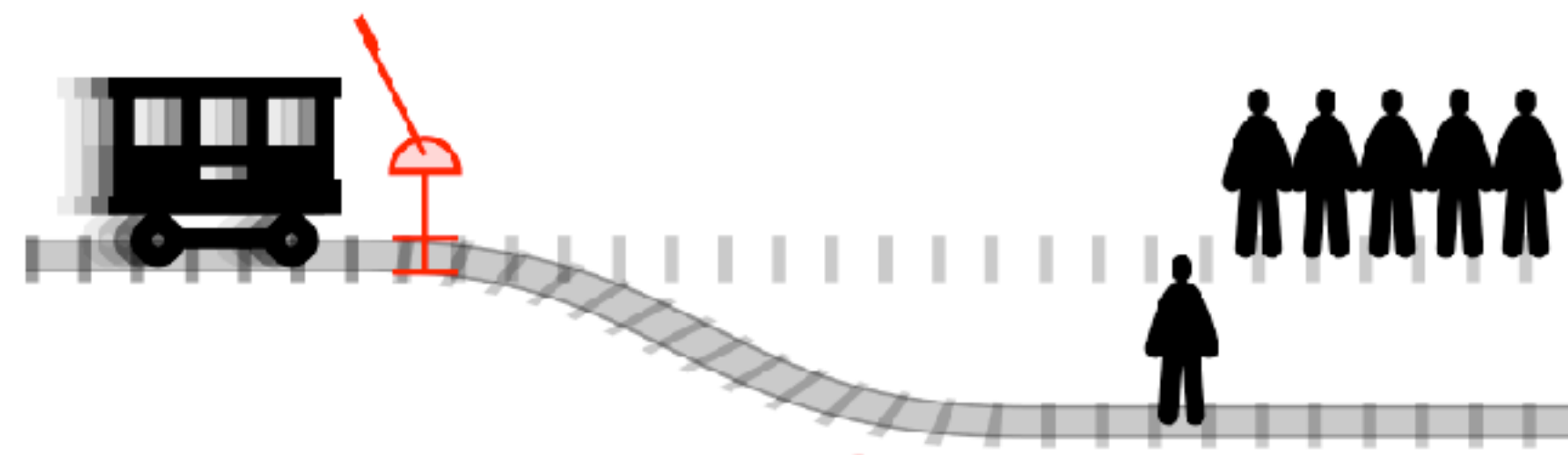
1. The switch
Foot, 1967



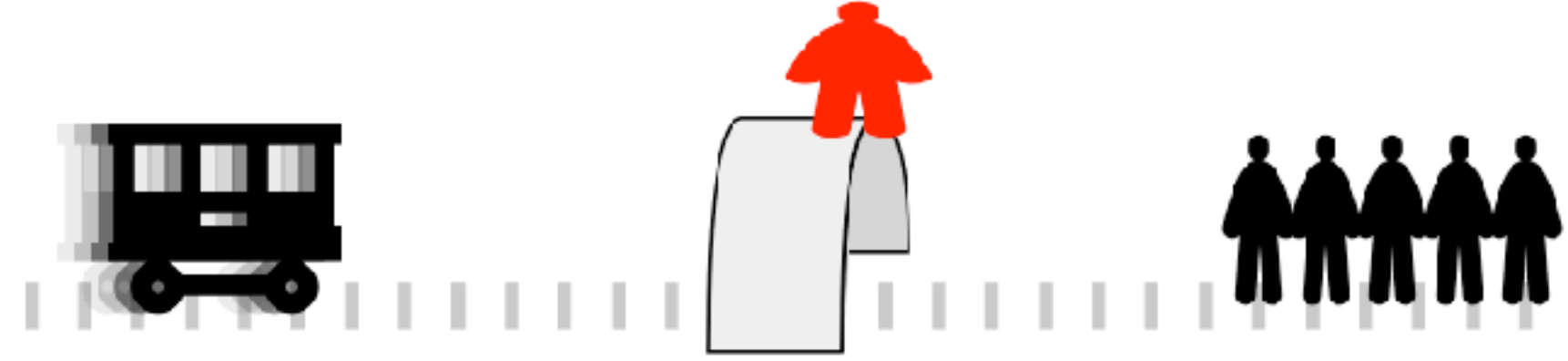
Philippa Foot (1920–2010)



1. The switch
Foot, 1967



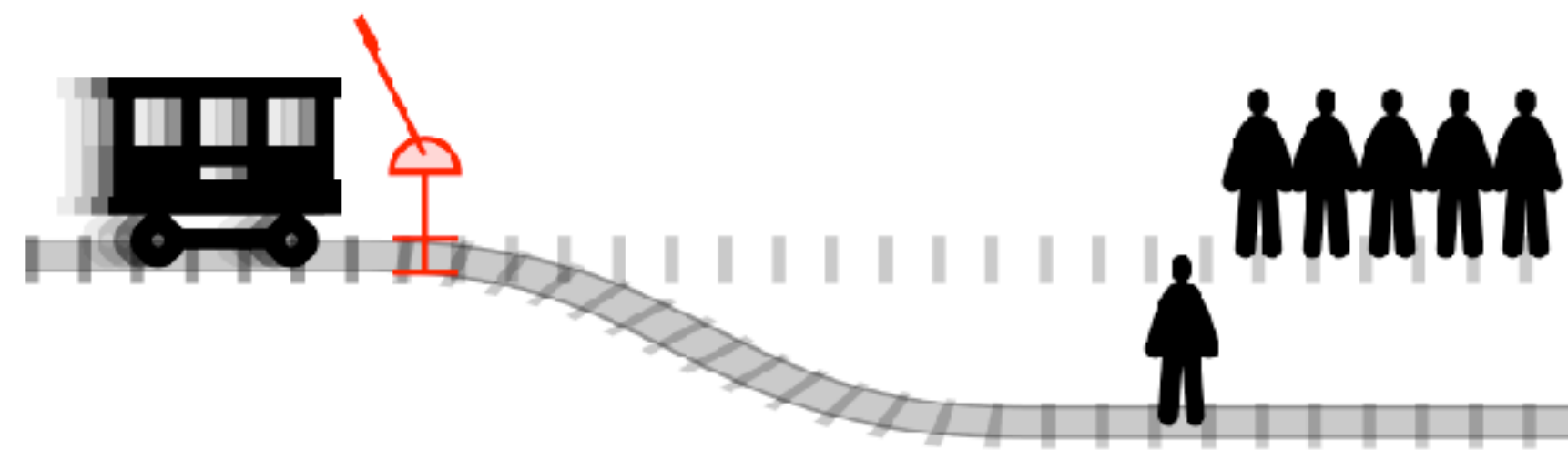
2. The fat man
Thomson, 1976



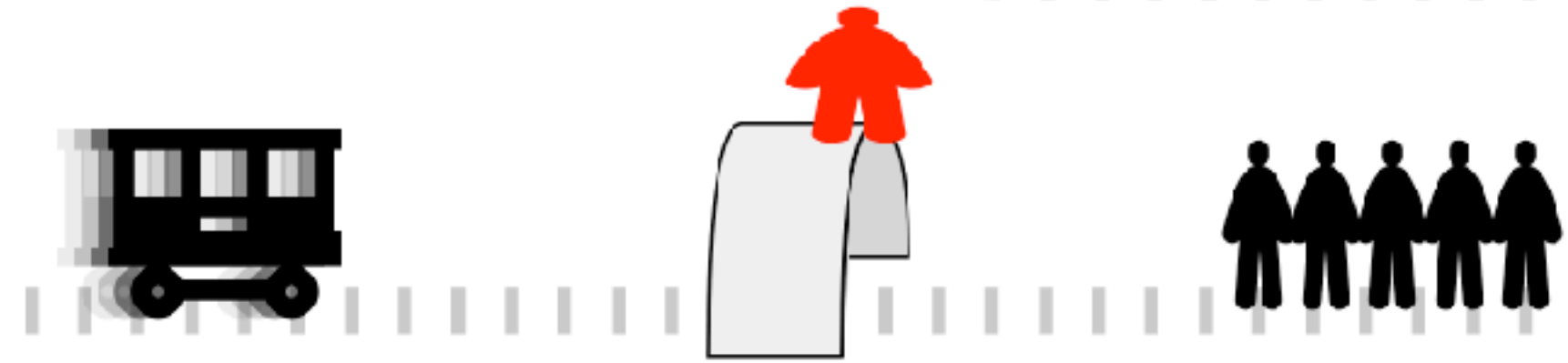
3. The fat villain



1. The switch
Foot, 1967



2. The fat man
Thomson, 1976



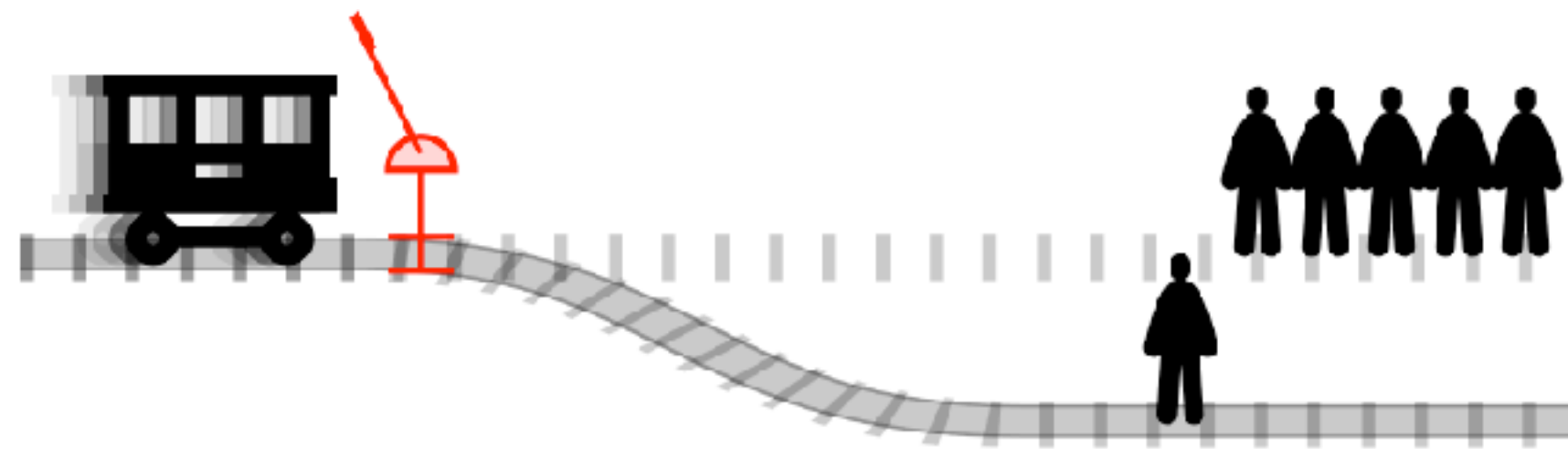
3. The fat villain



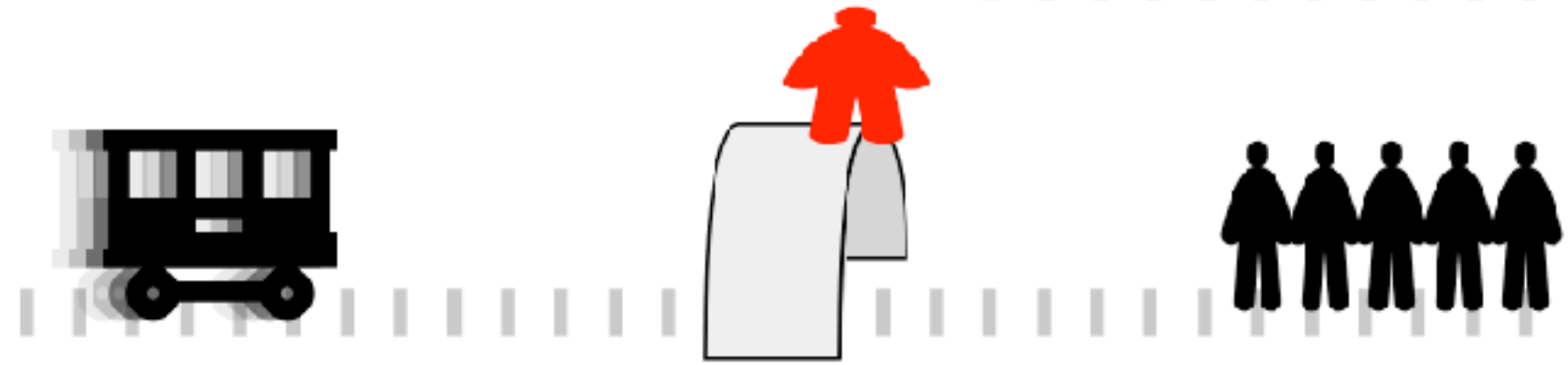
4. The loop
Costa, 1987



1. The switch
Foot, 1967



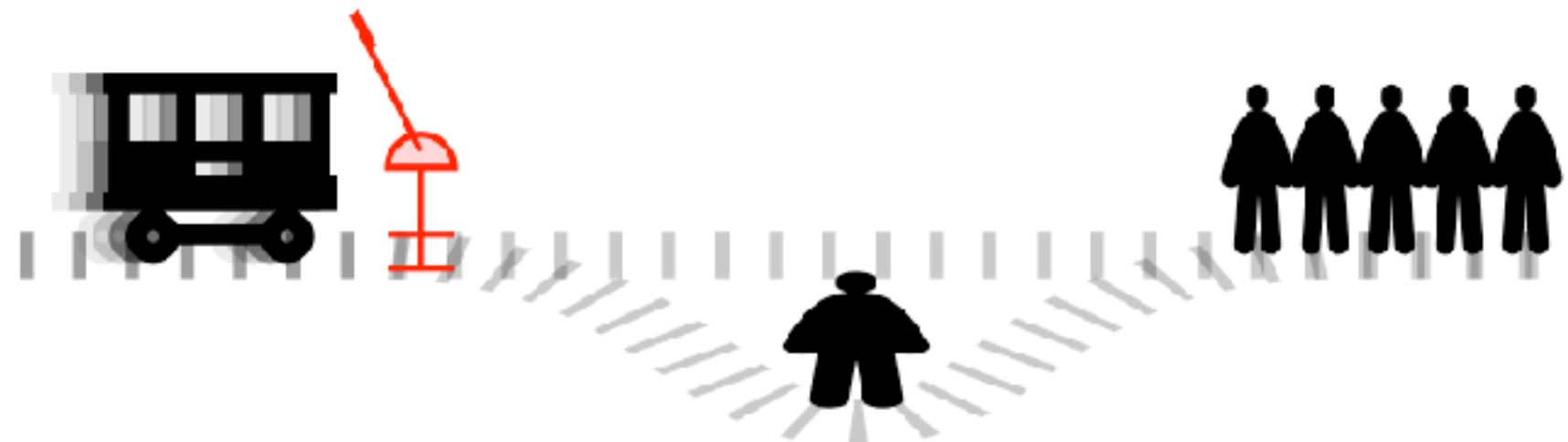
2. The fat man
Thomson, 1976



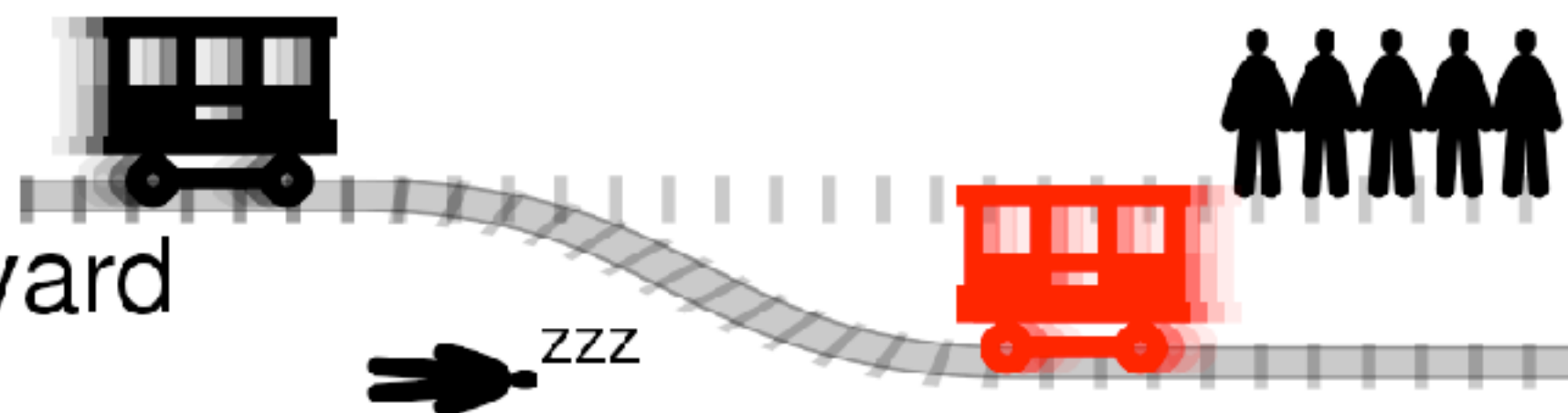
3. The fat villain



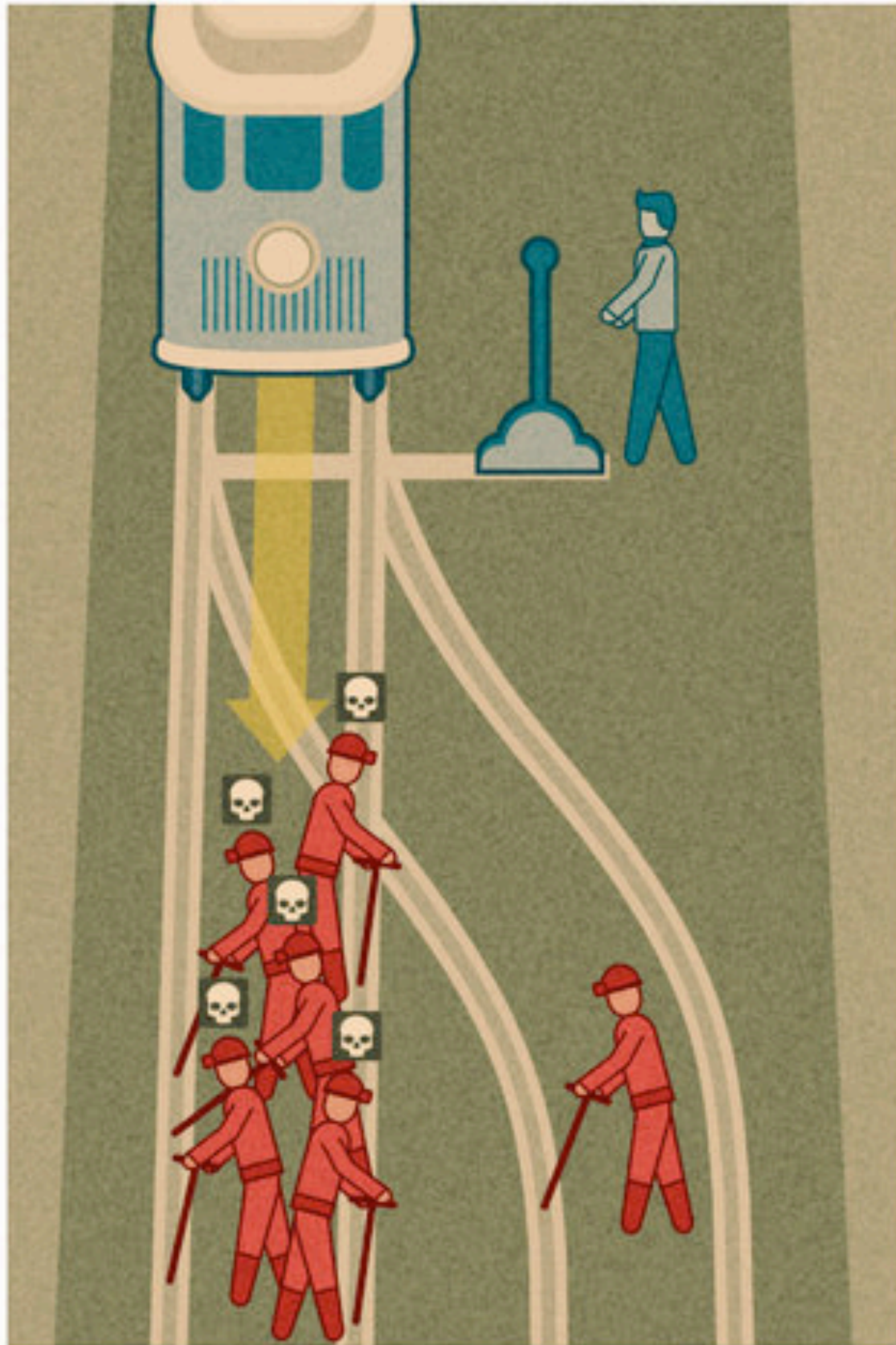
4. The loop
Costa, 1987



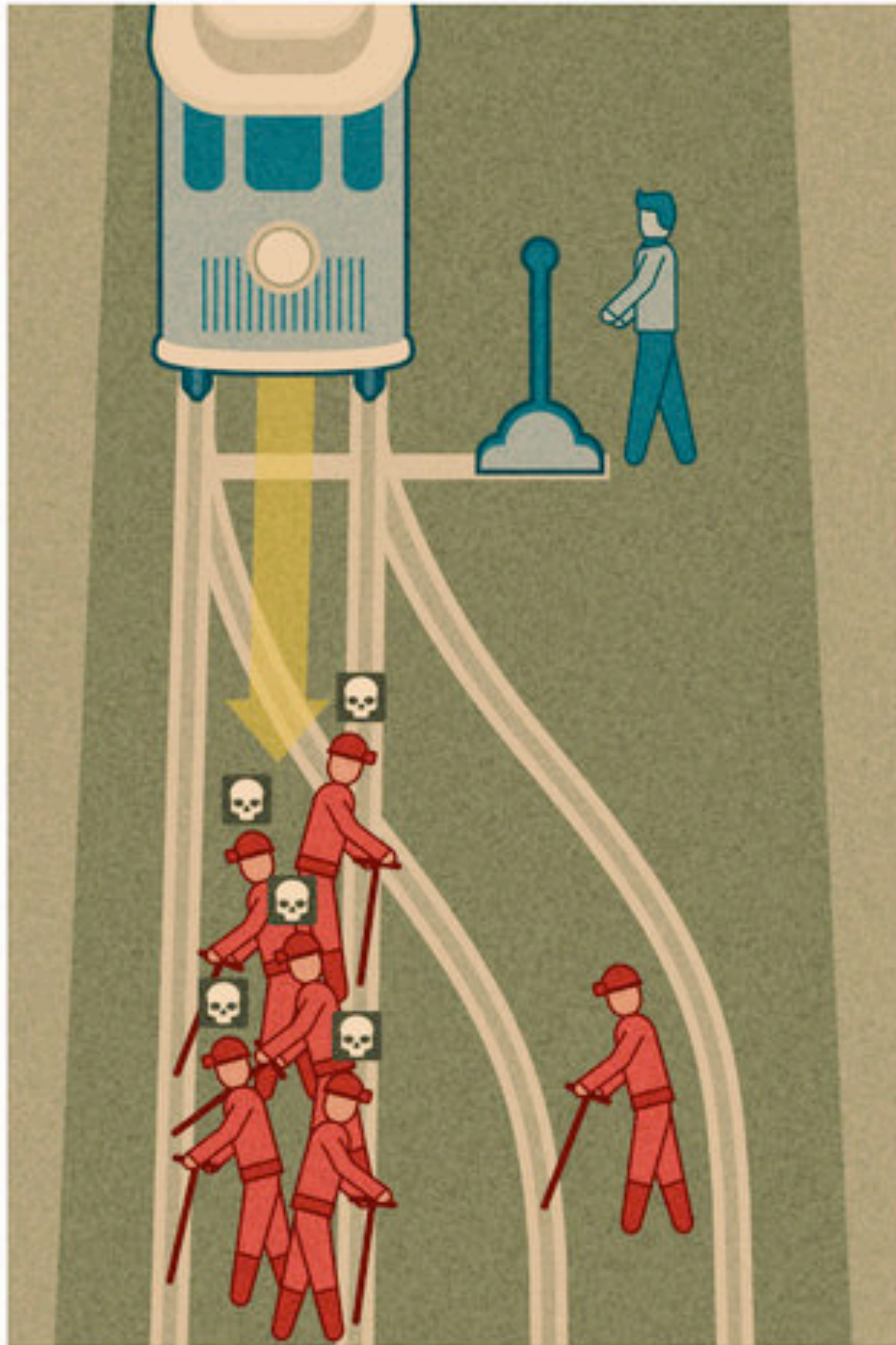
5. The man in the yard
Unger, 1992



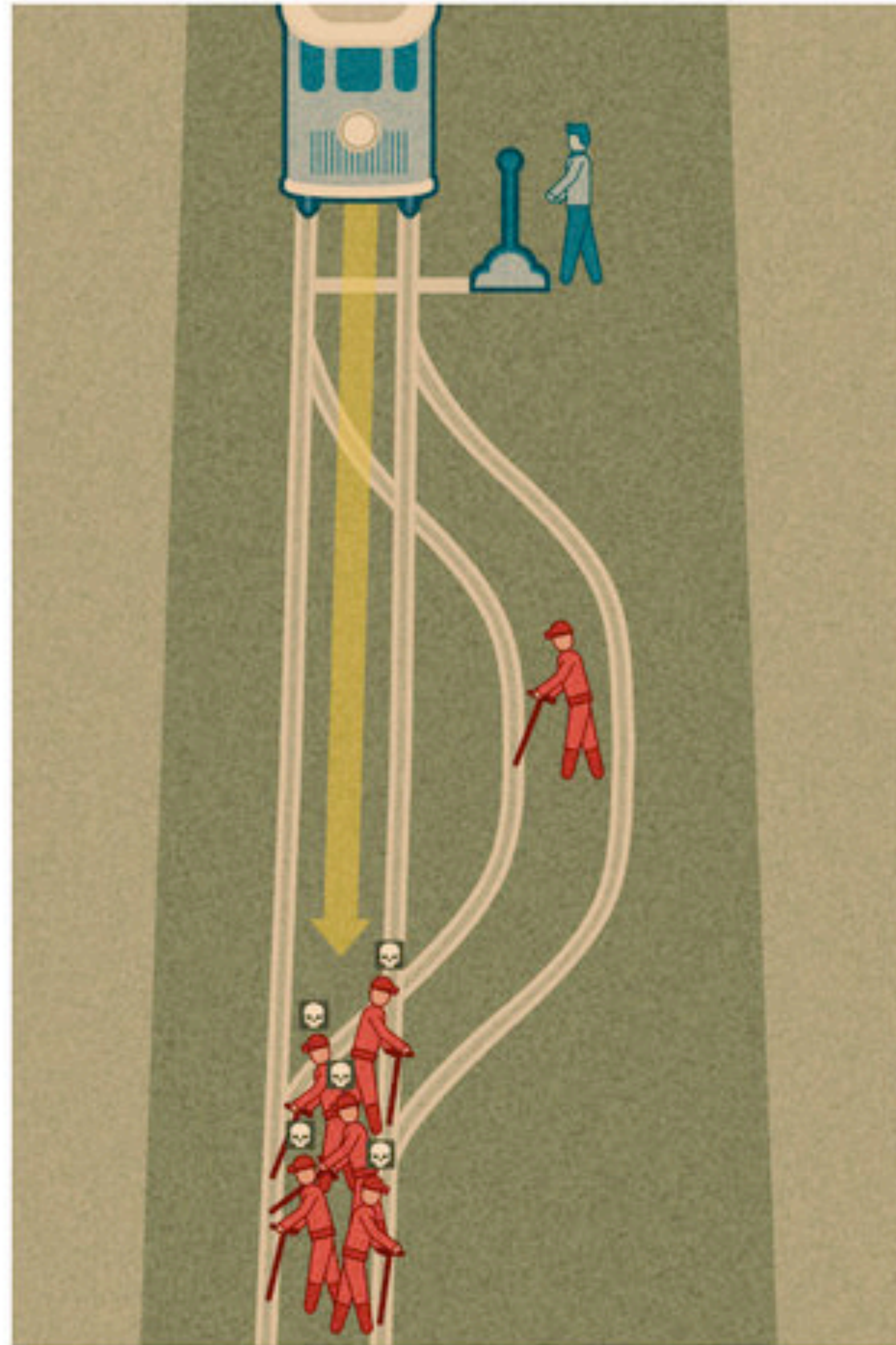
Action



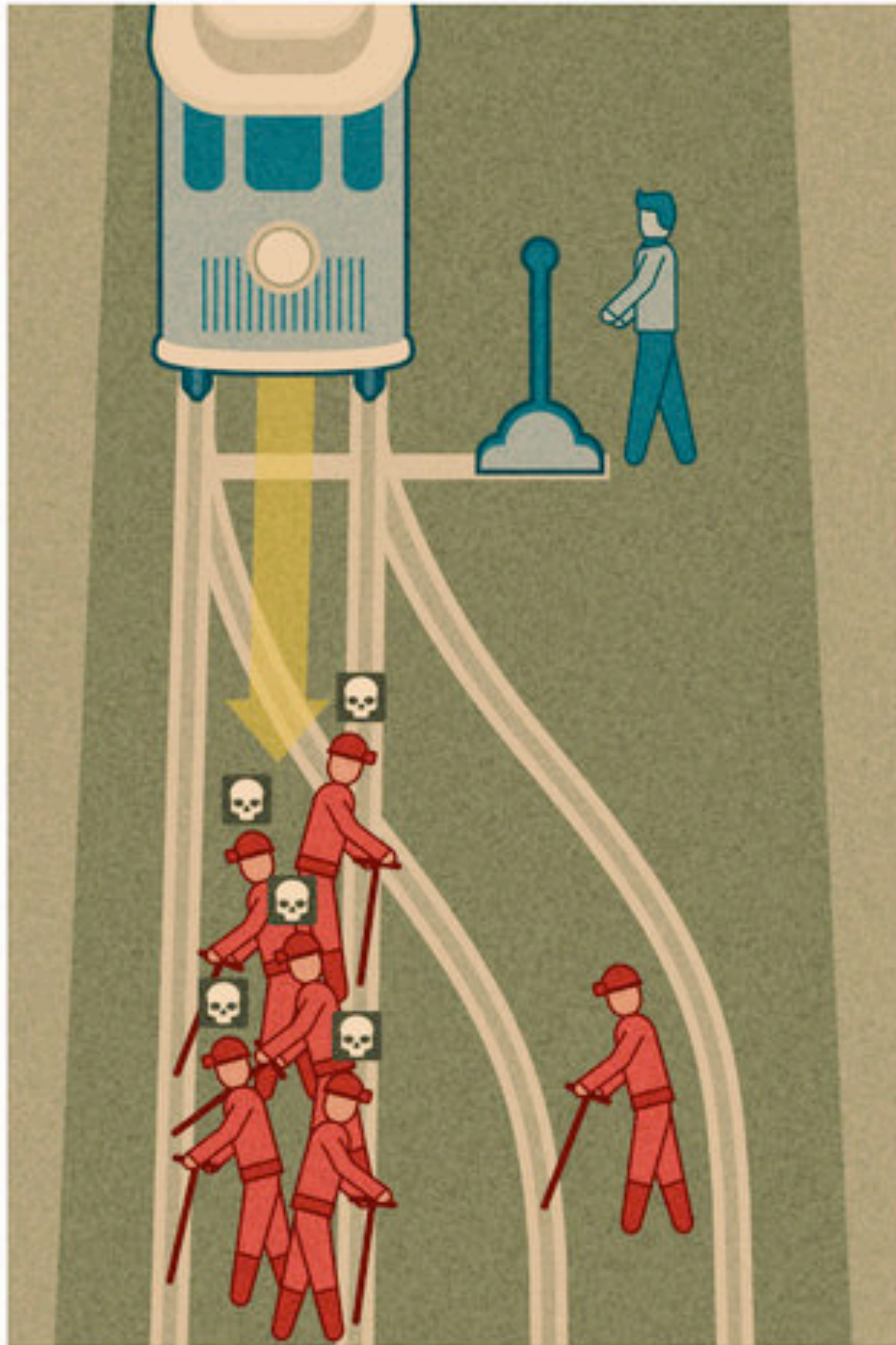
Action



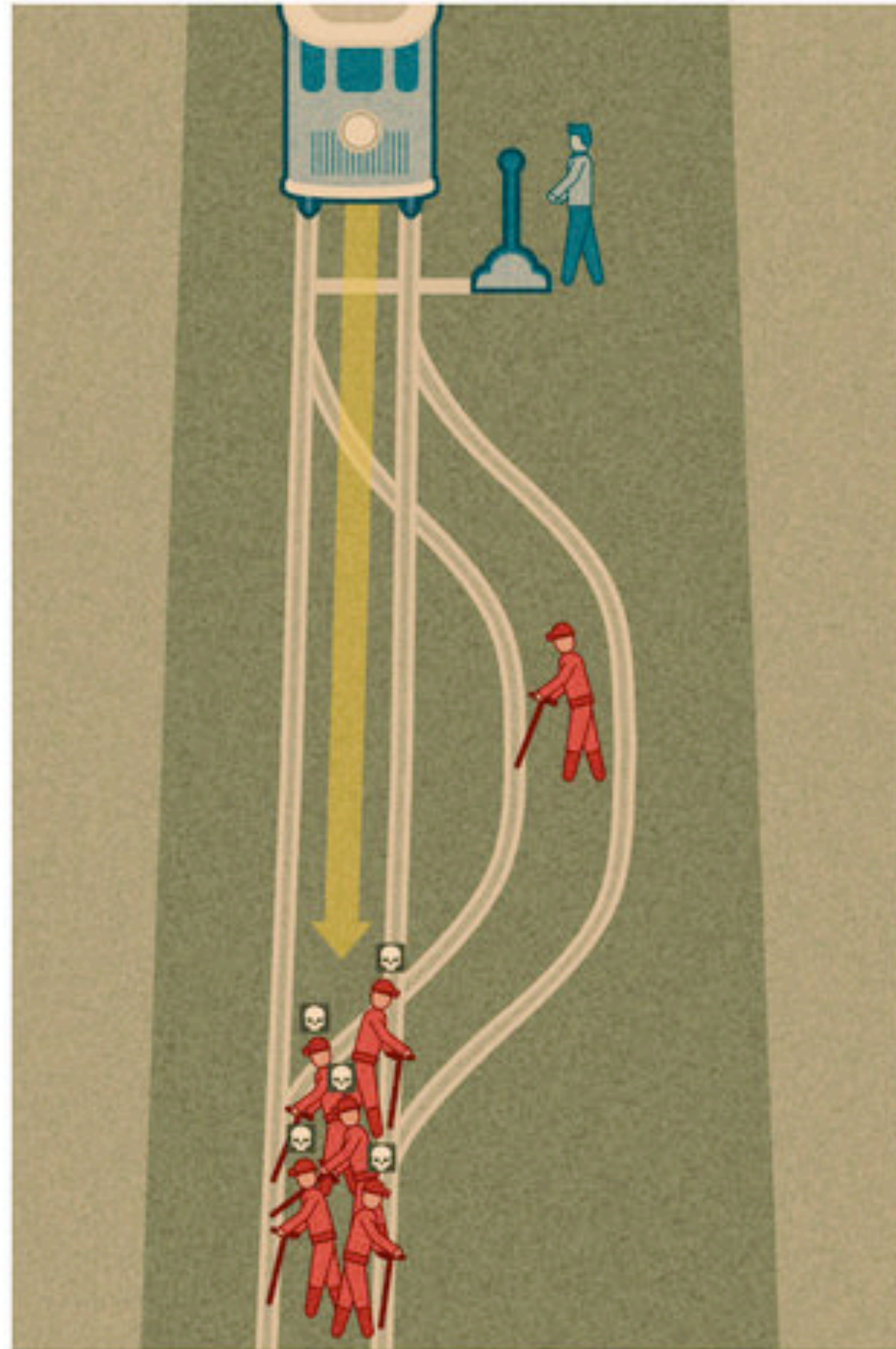
Intention



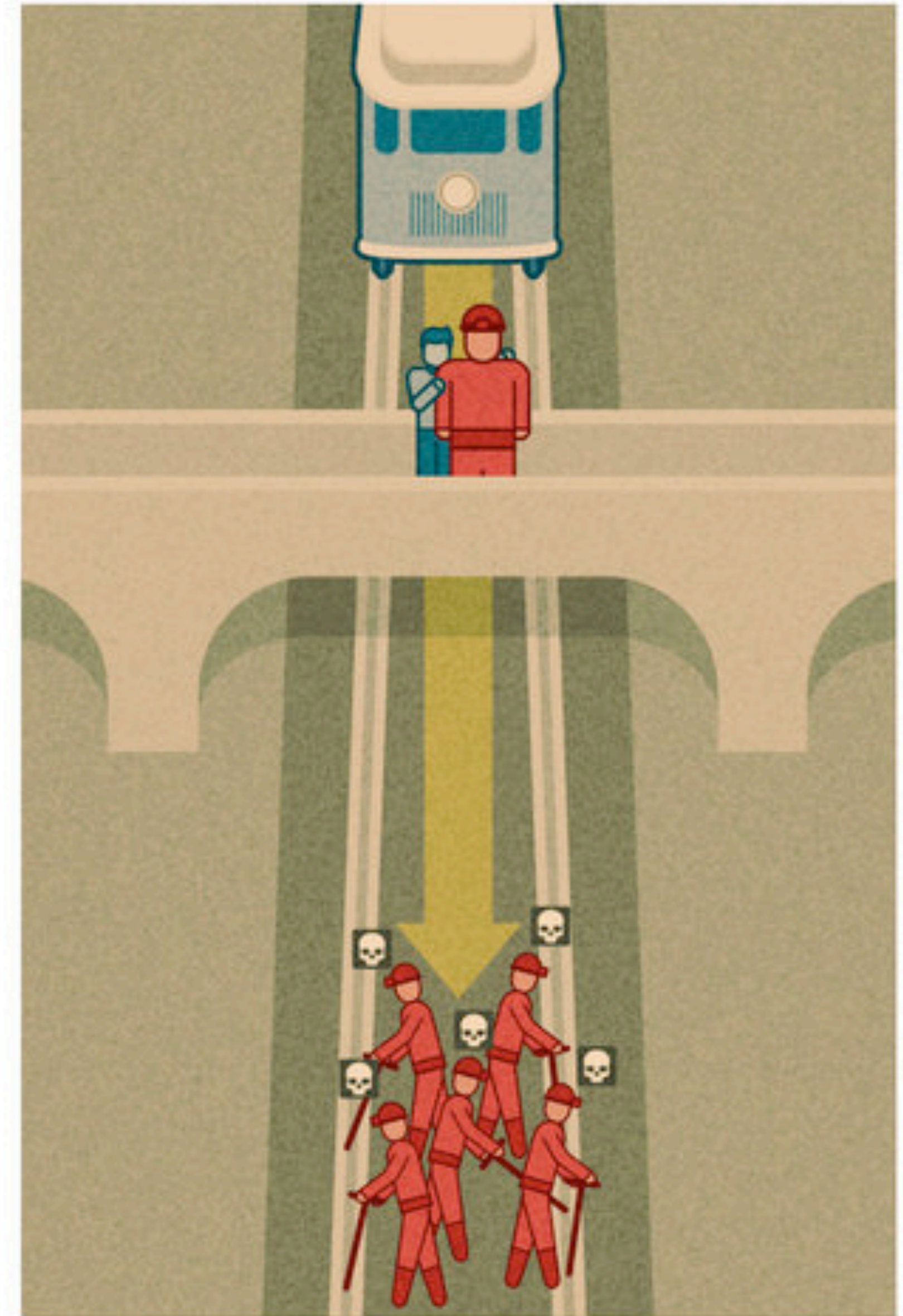
Action



Intention



Contact



Trolley Problems

data(Trolley)

331 individuals (age, gender, edu)

Voluntary participation (online)

30 different trolley problems

action / intention / contact

9930 responses:

How appropriate (from 1 to 7)?



Trolley Problems

data(Trolley)

331 individuals (age, gender, edu)

Voluntary participation (online)

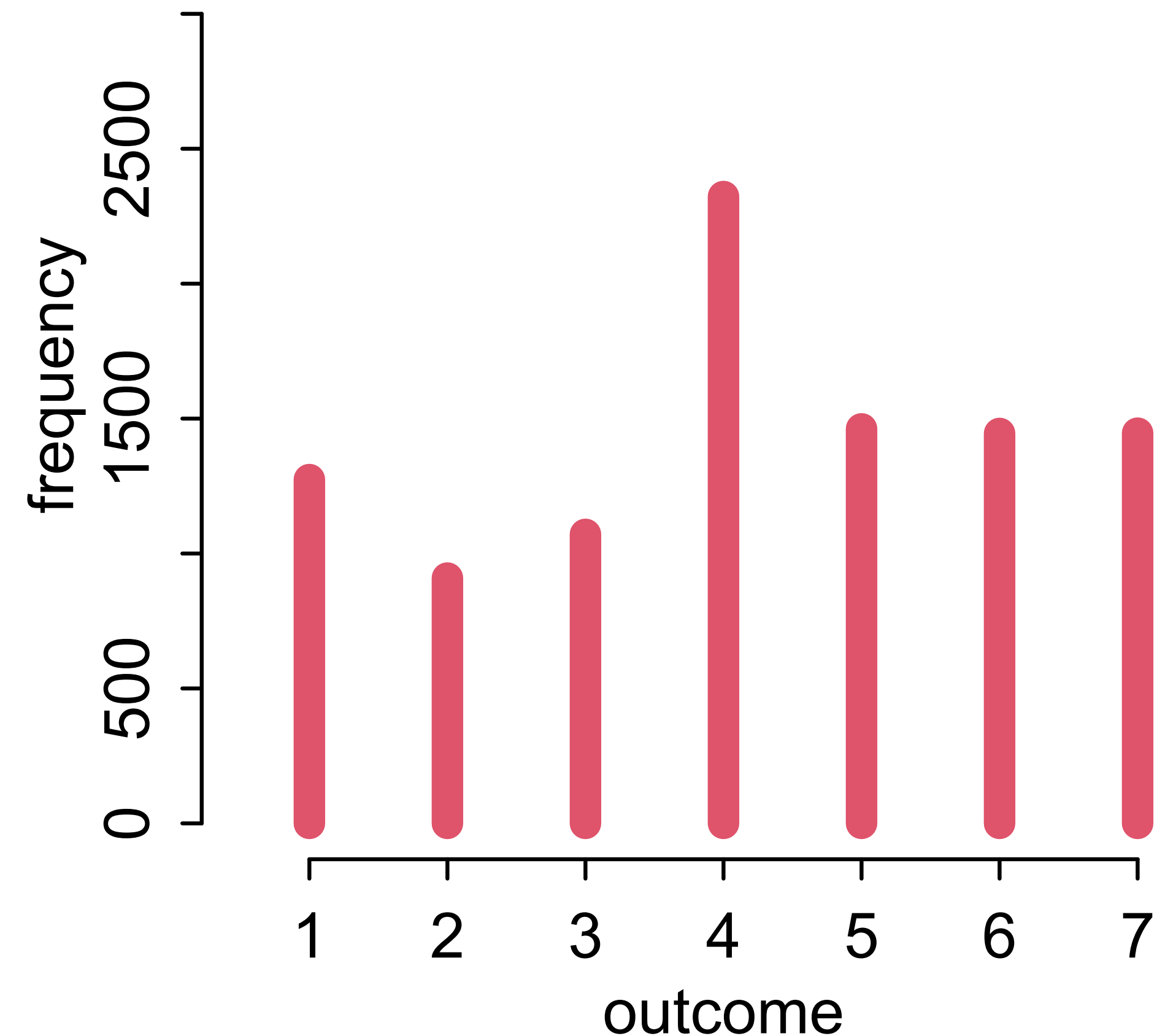
30 different trolley problems

action / intention / contact

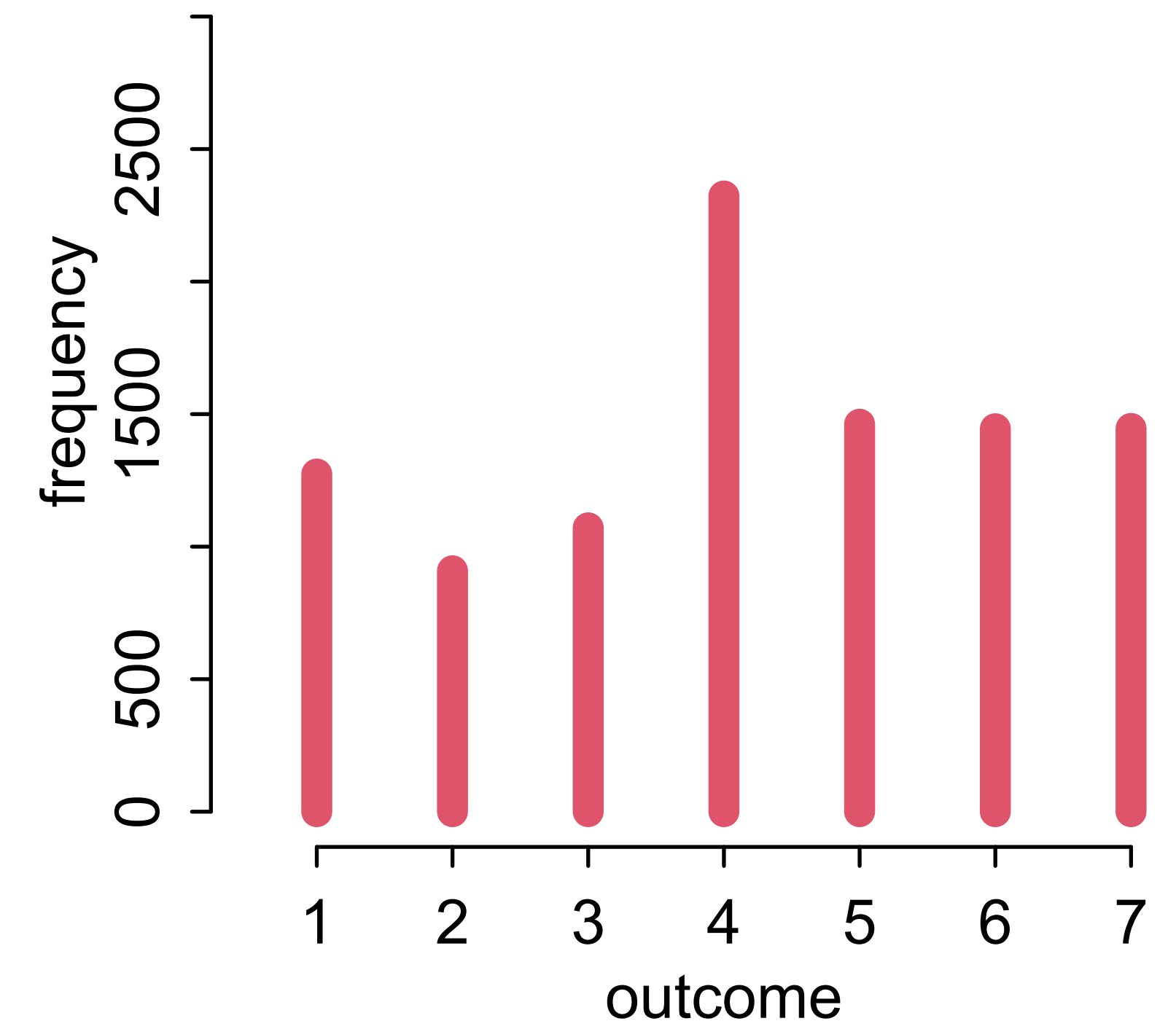
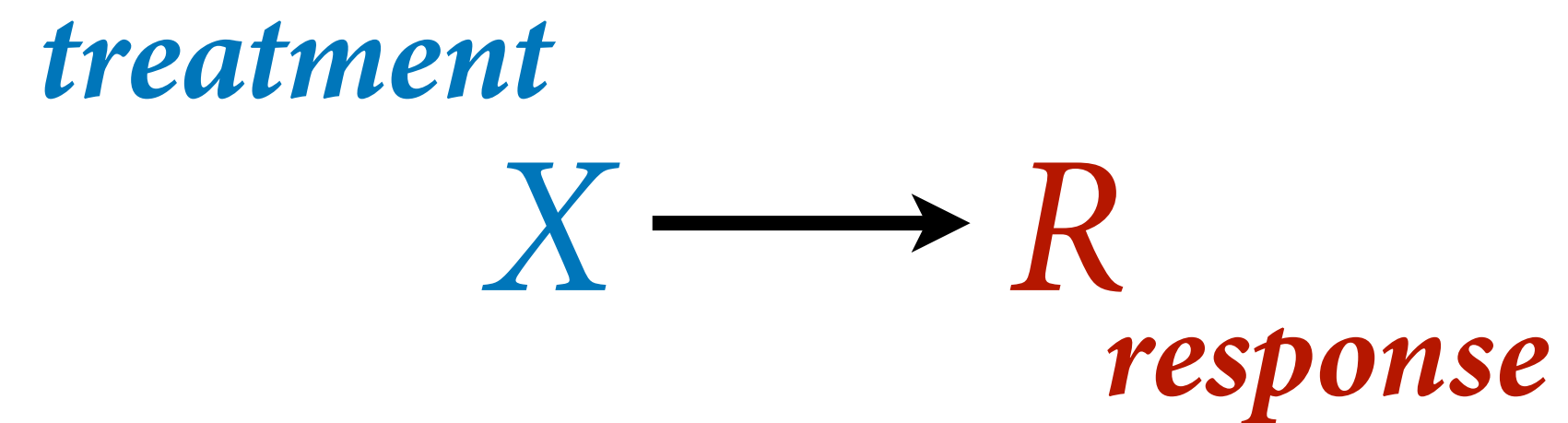
9930 responses:

How appropriate (from 1 to 7)?

Ordered categorical

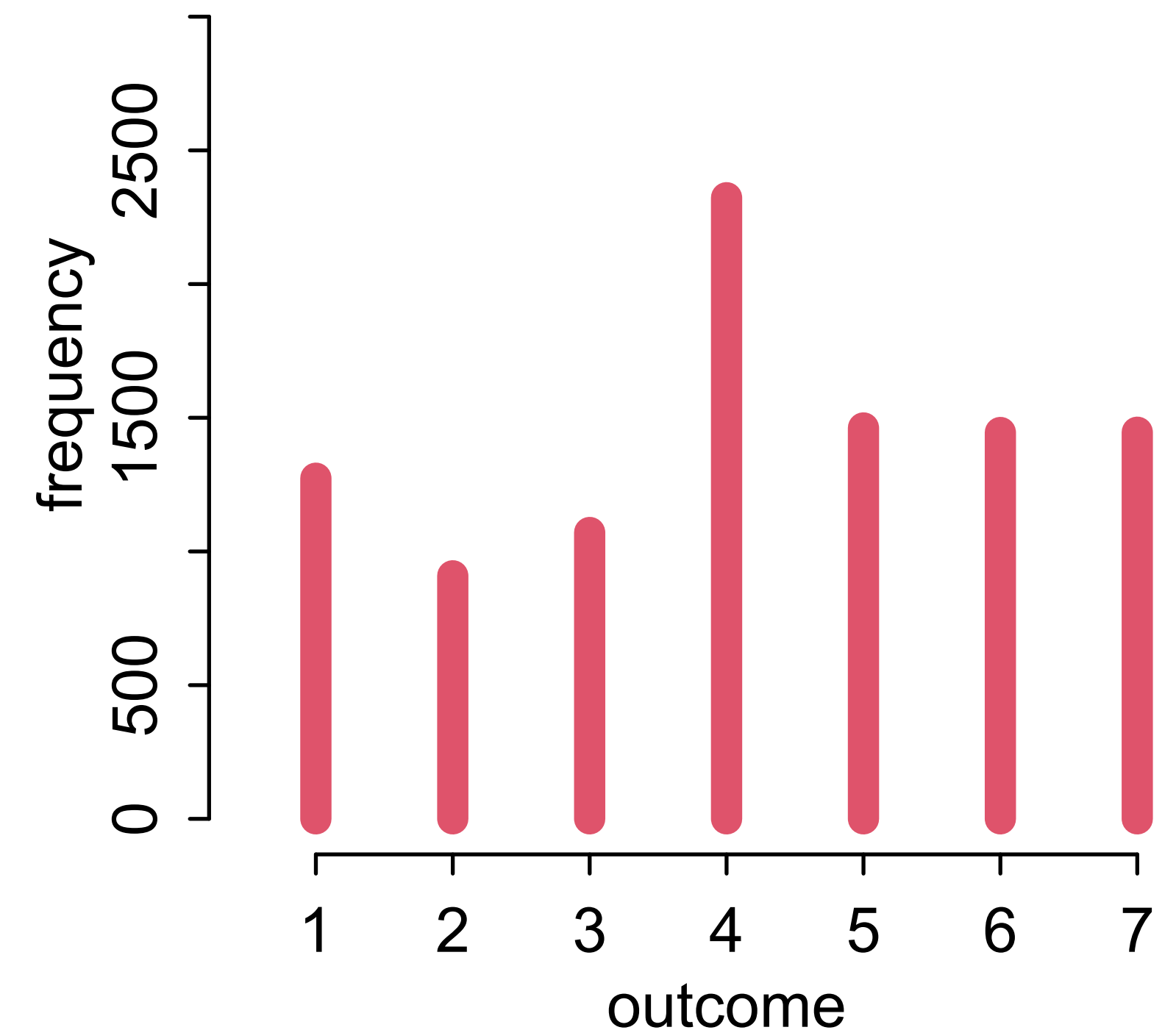
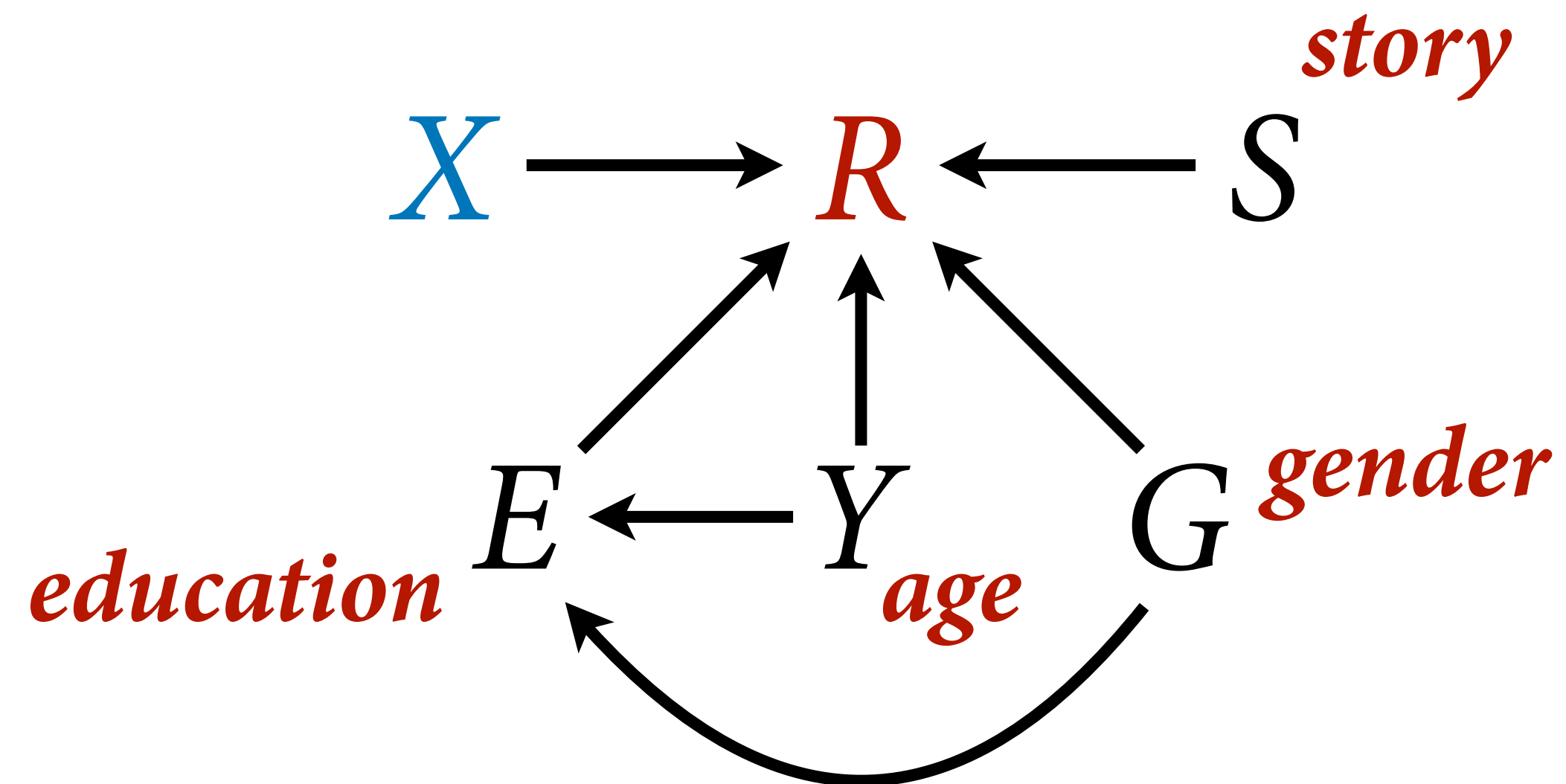


Estimand: How do **action**, **intention**, **contact** influence **response** to a trolley story?



Estimand: How do **action**, **intention**, **contact** influence **response** to a trolley story?

How are influences of A/I/C associated with other variables?



Ordered categories

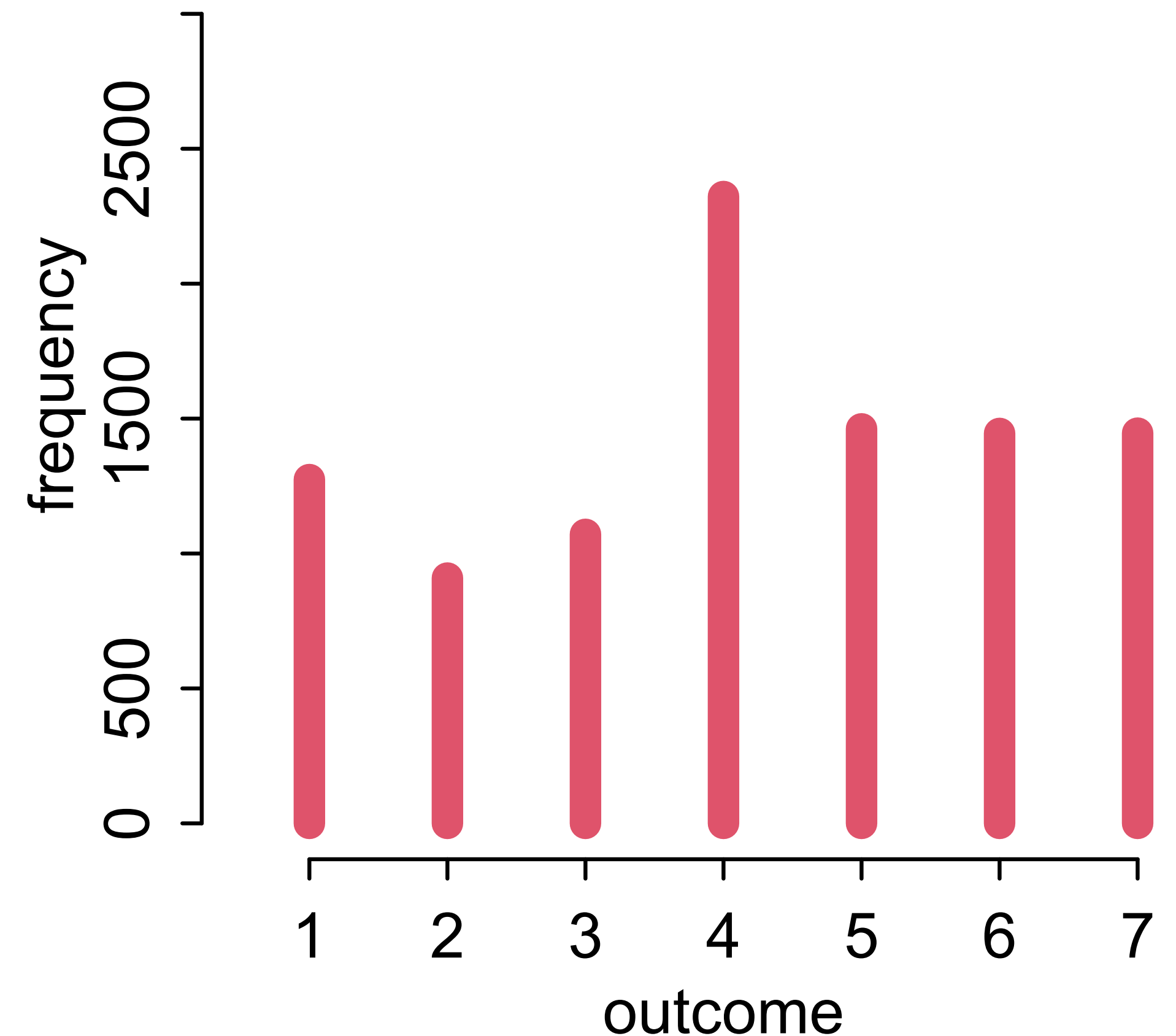
Categories: Discrete types

cat, dog, chicken

Ordered categories: Discrete types
with ordered relationships

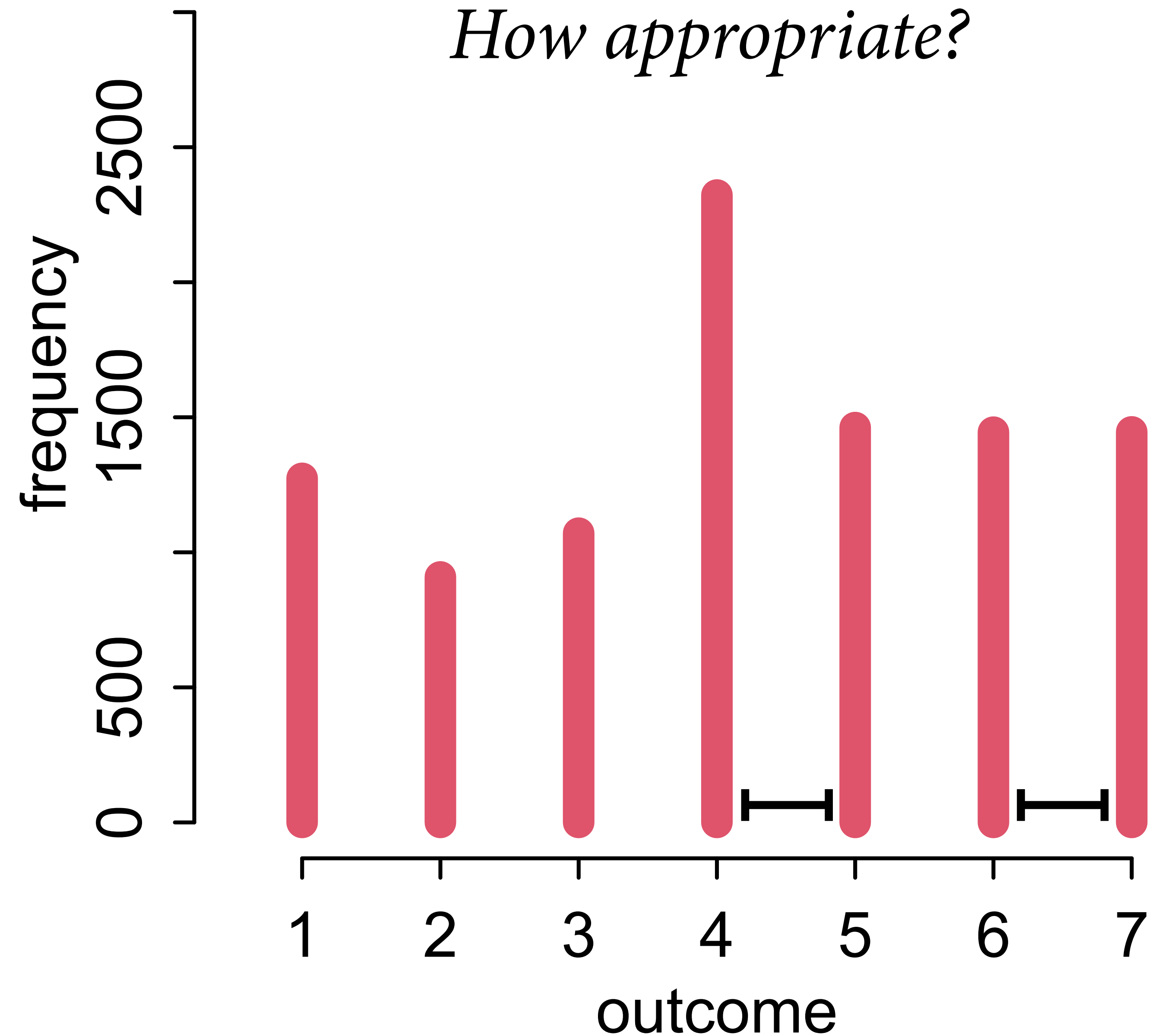
bad, good, excellent

Ordered categorical



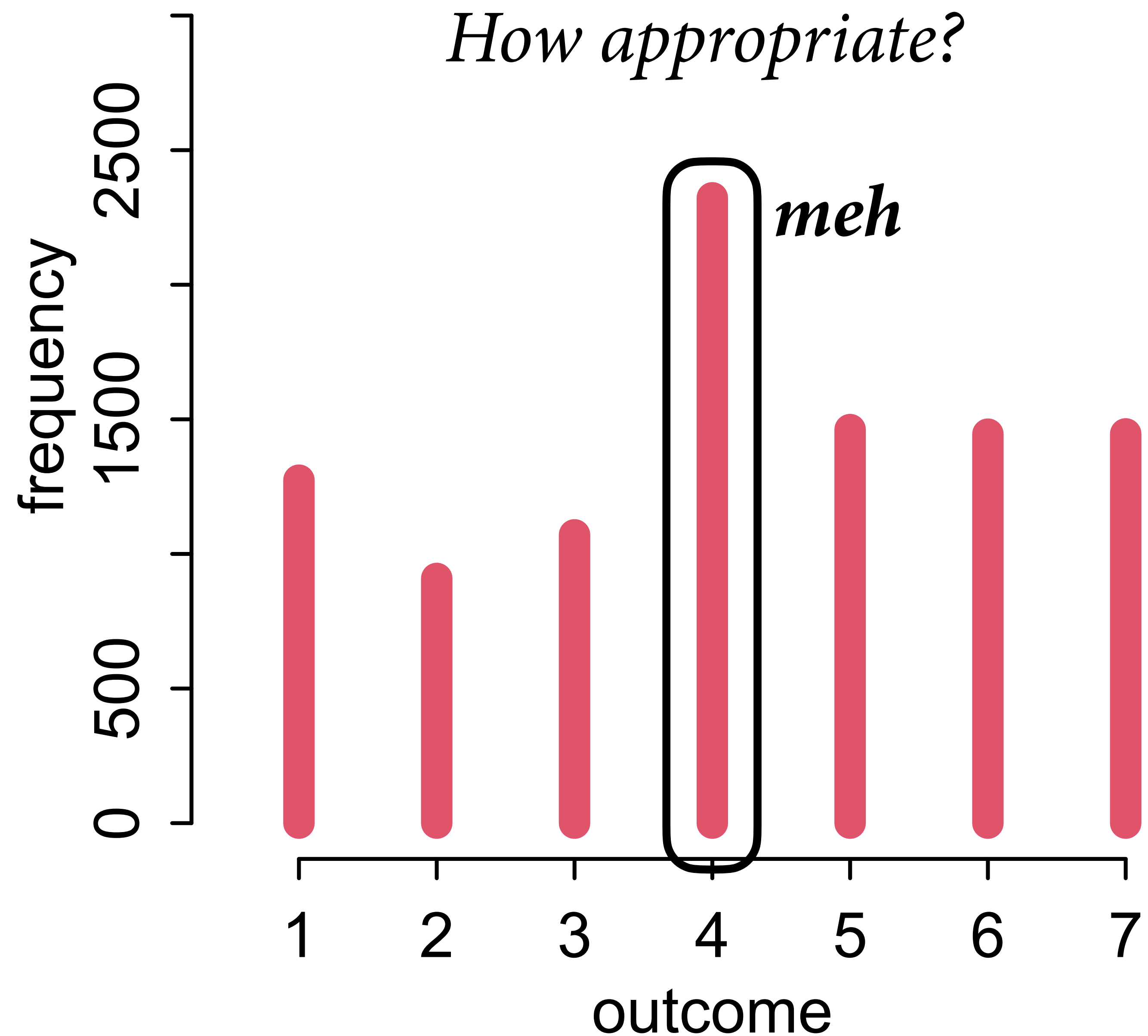
Distance between values
not constant

Probably much easier to
go from 4 to 5 than from
6 to 7

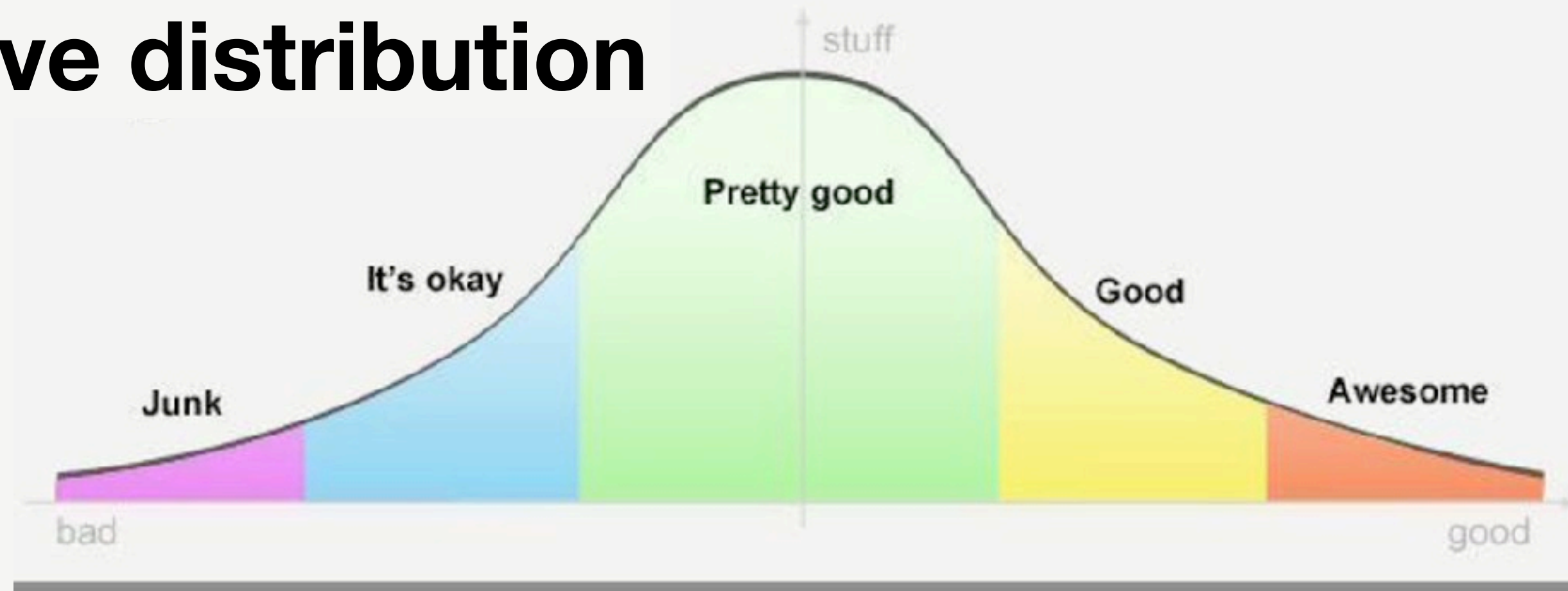


Anchor points common

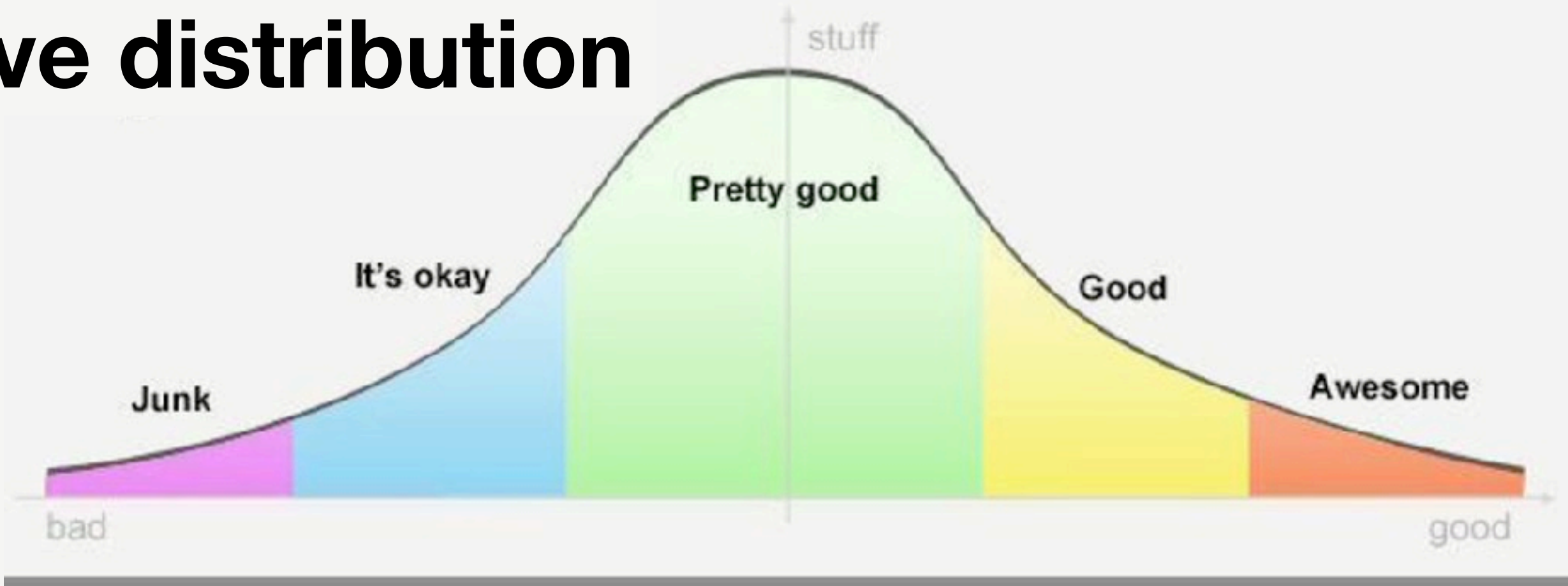
Not everyone shares the
same anchor points



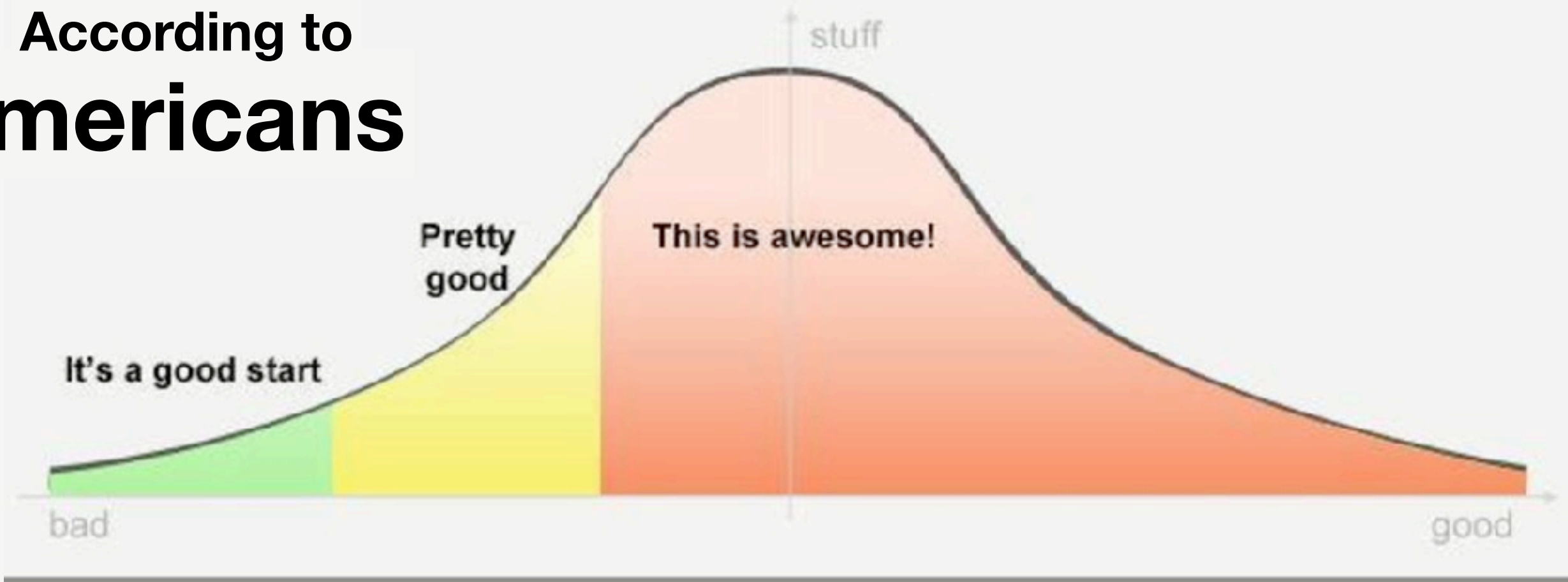
Objective distribution



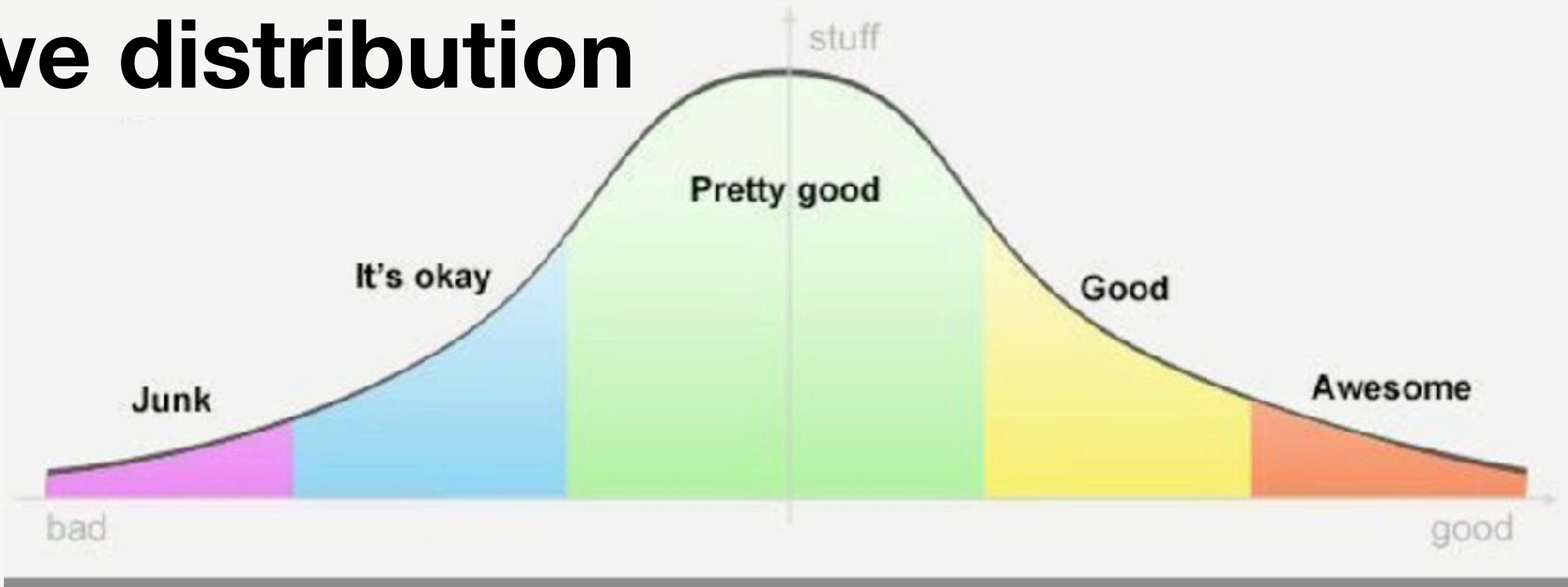
Objective distribution



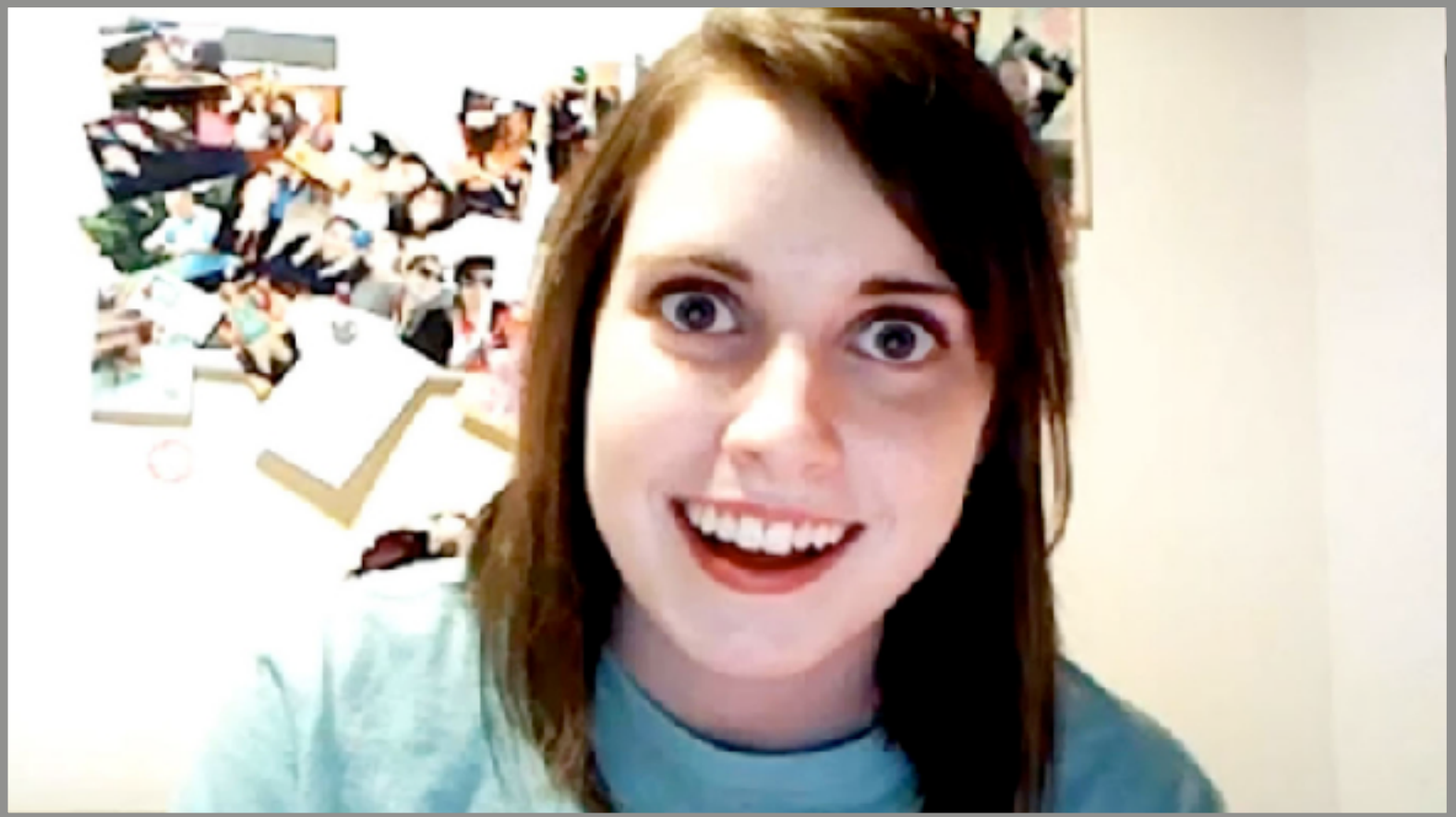
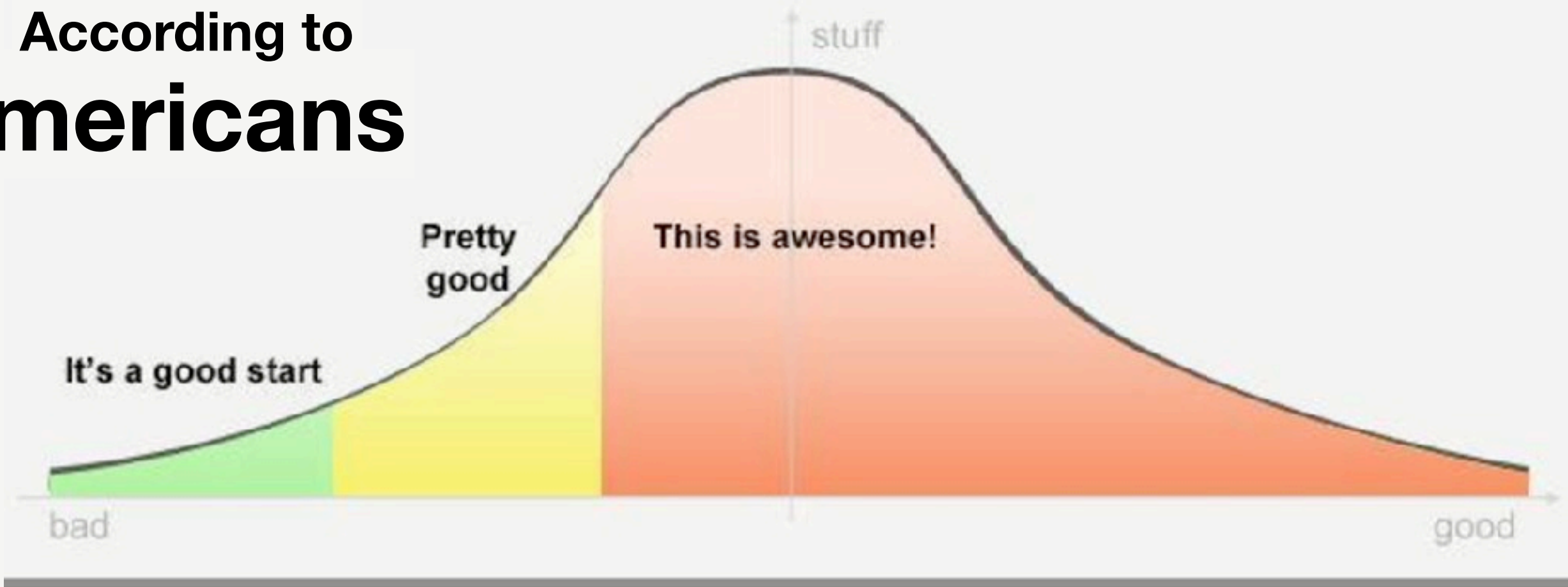
According to
Americans



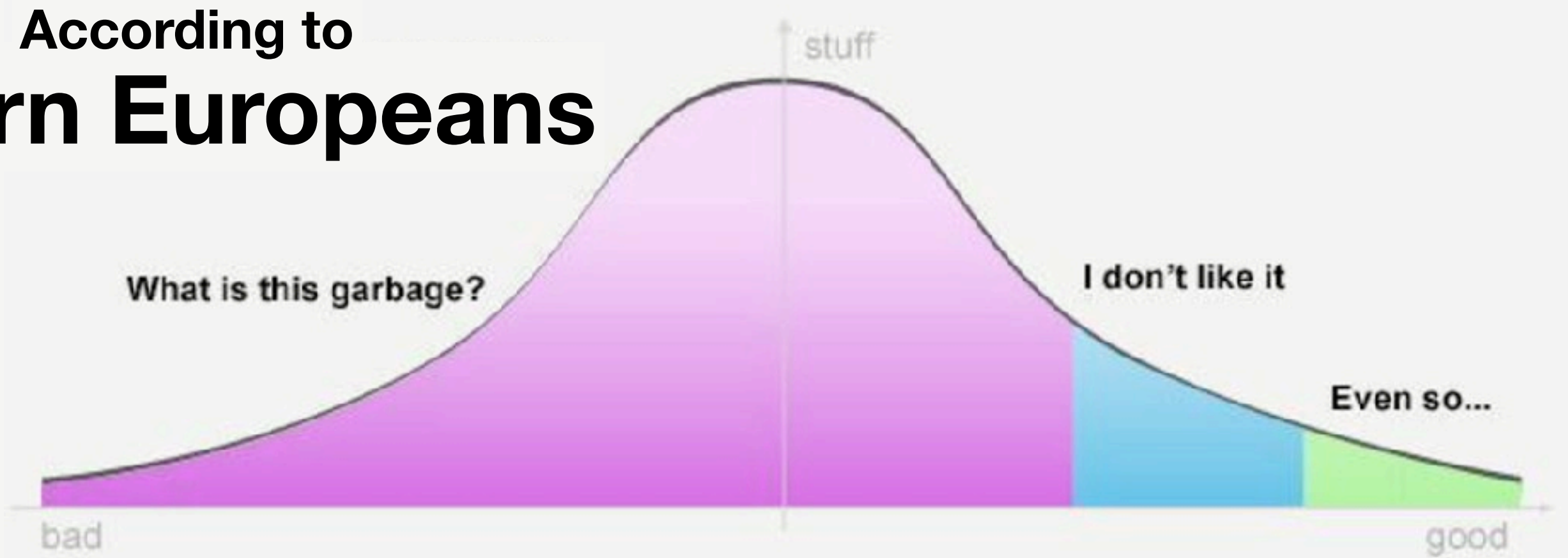
Objective distribution

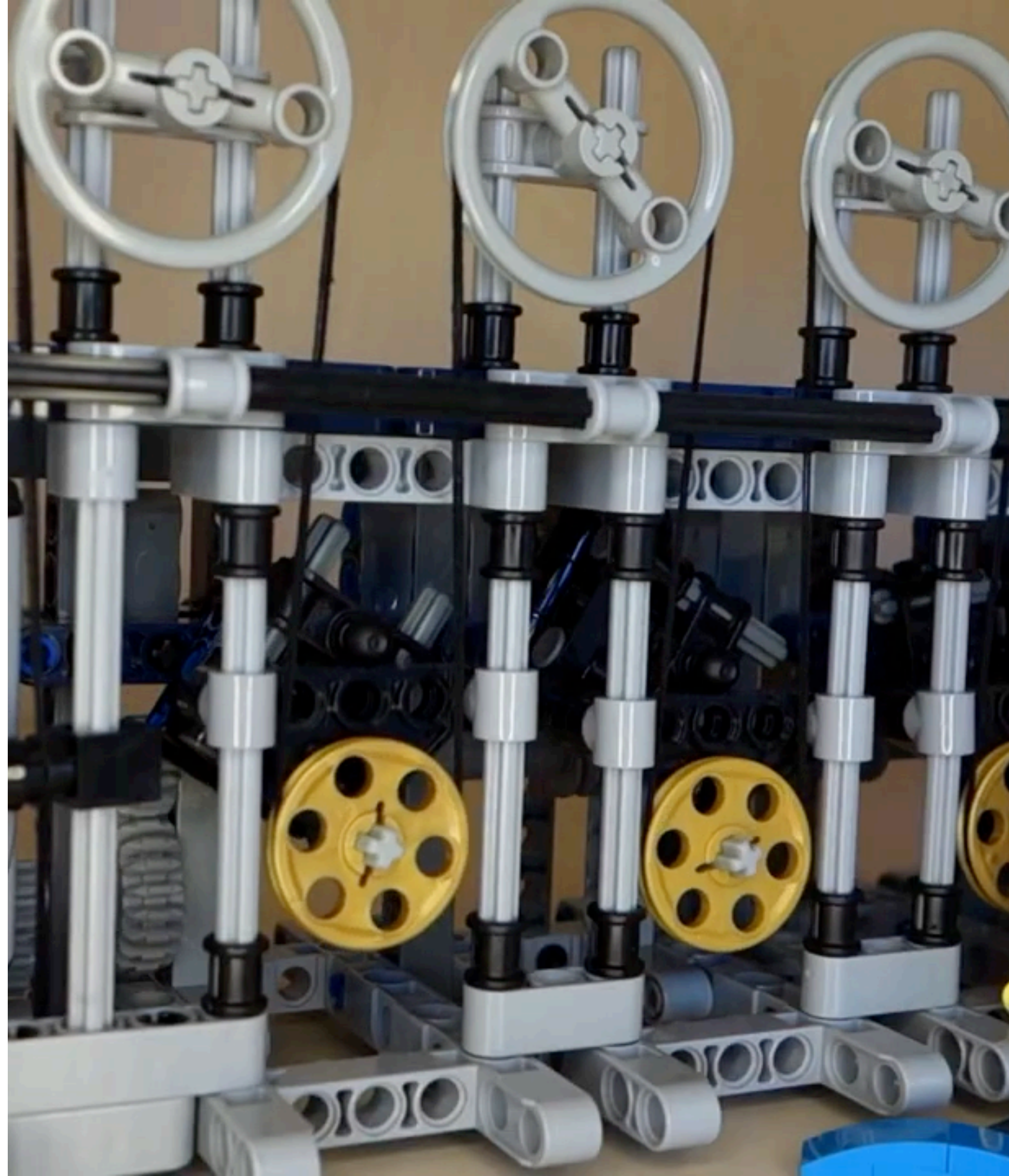
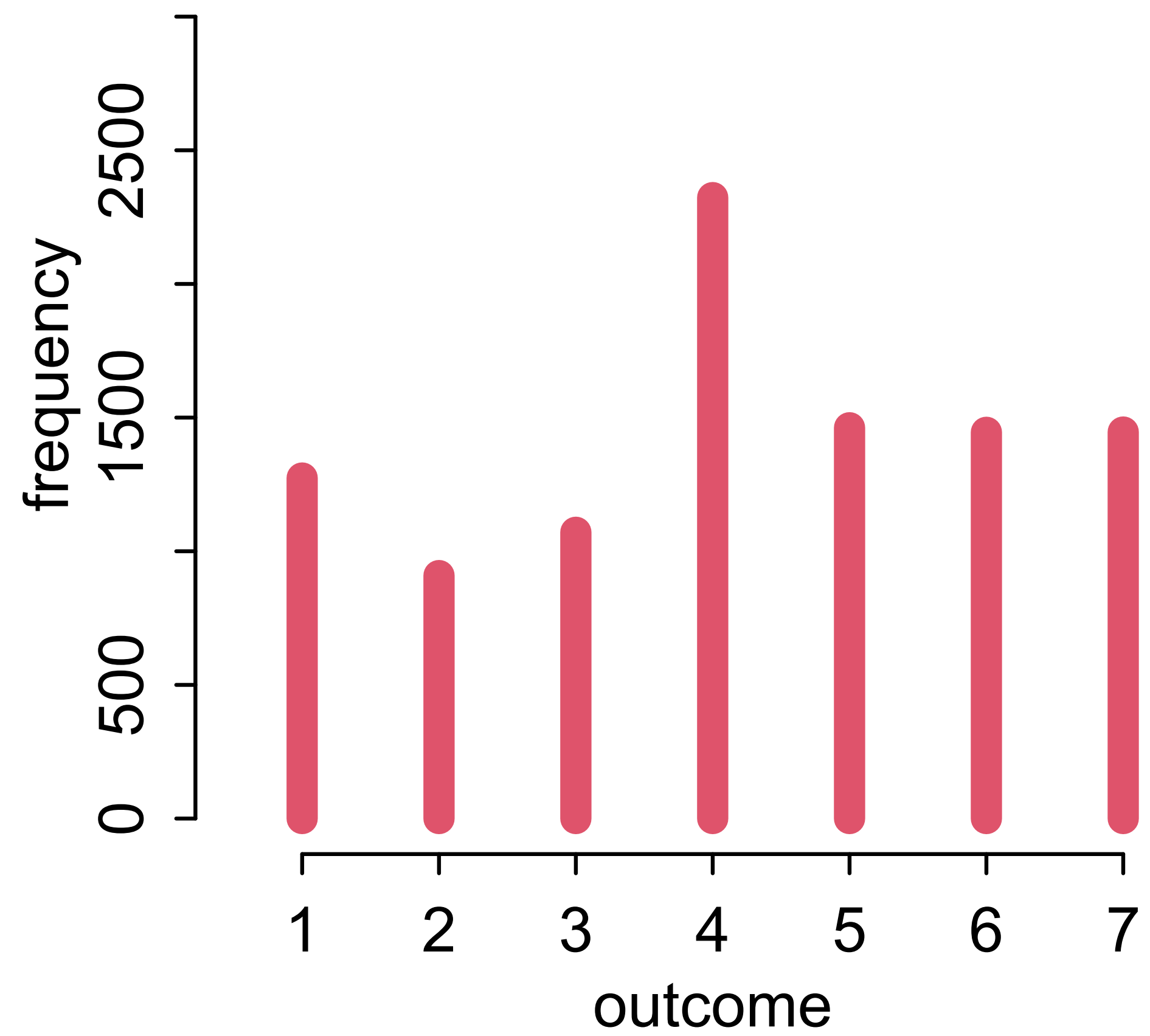


According to Americans



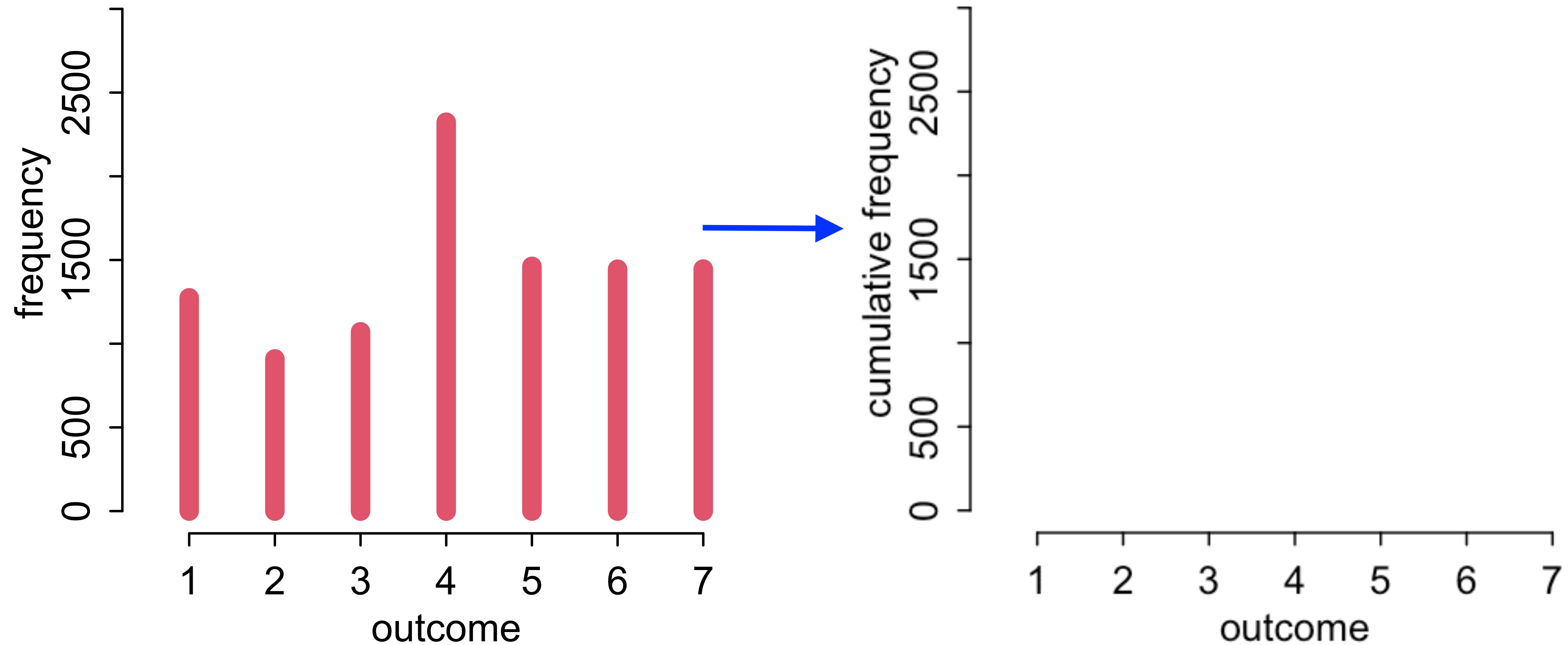
According to Eastern Europeans

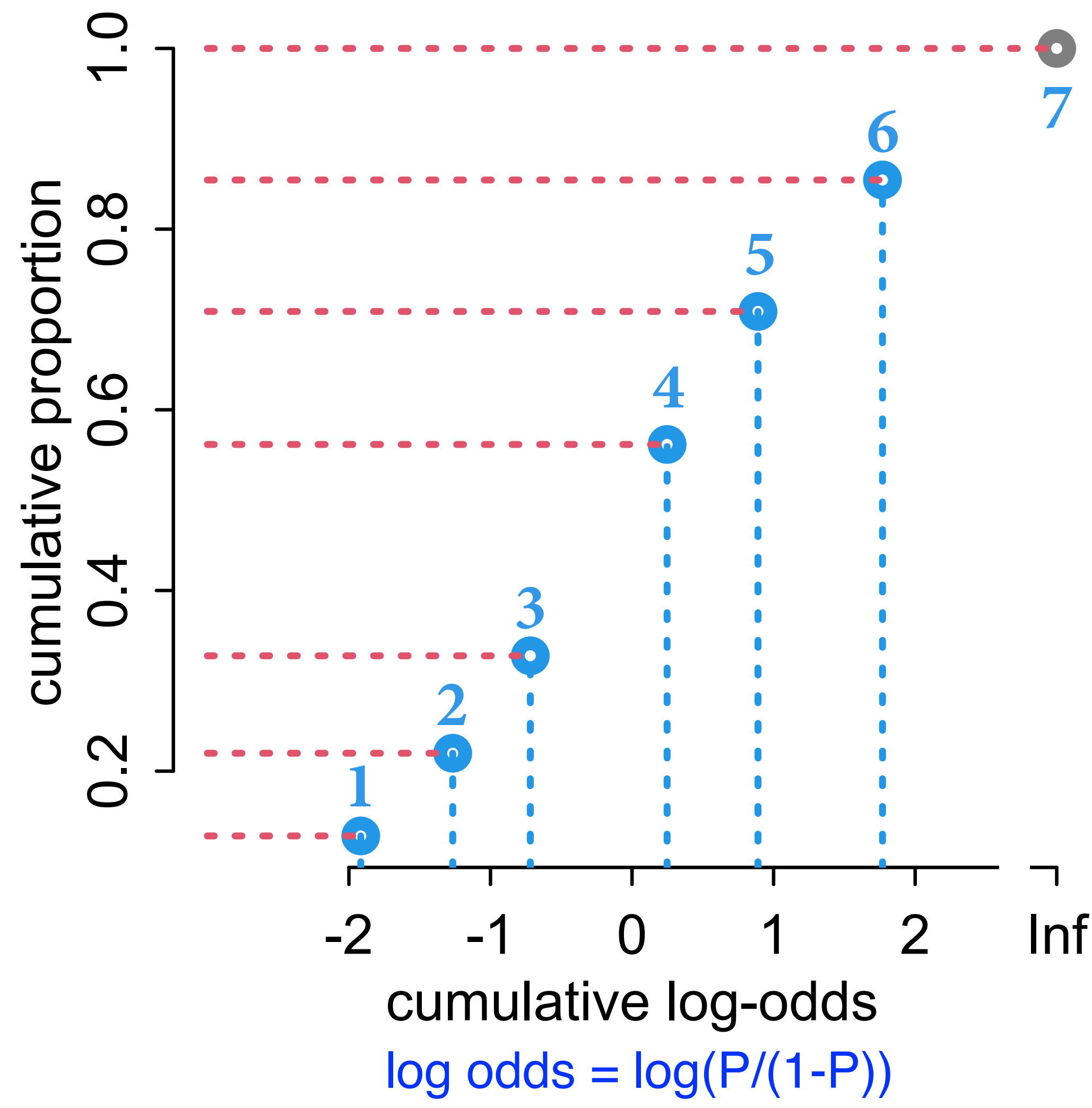
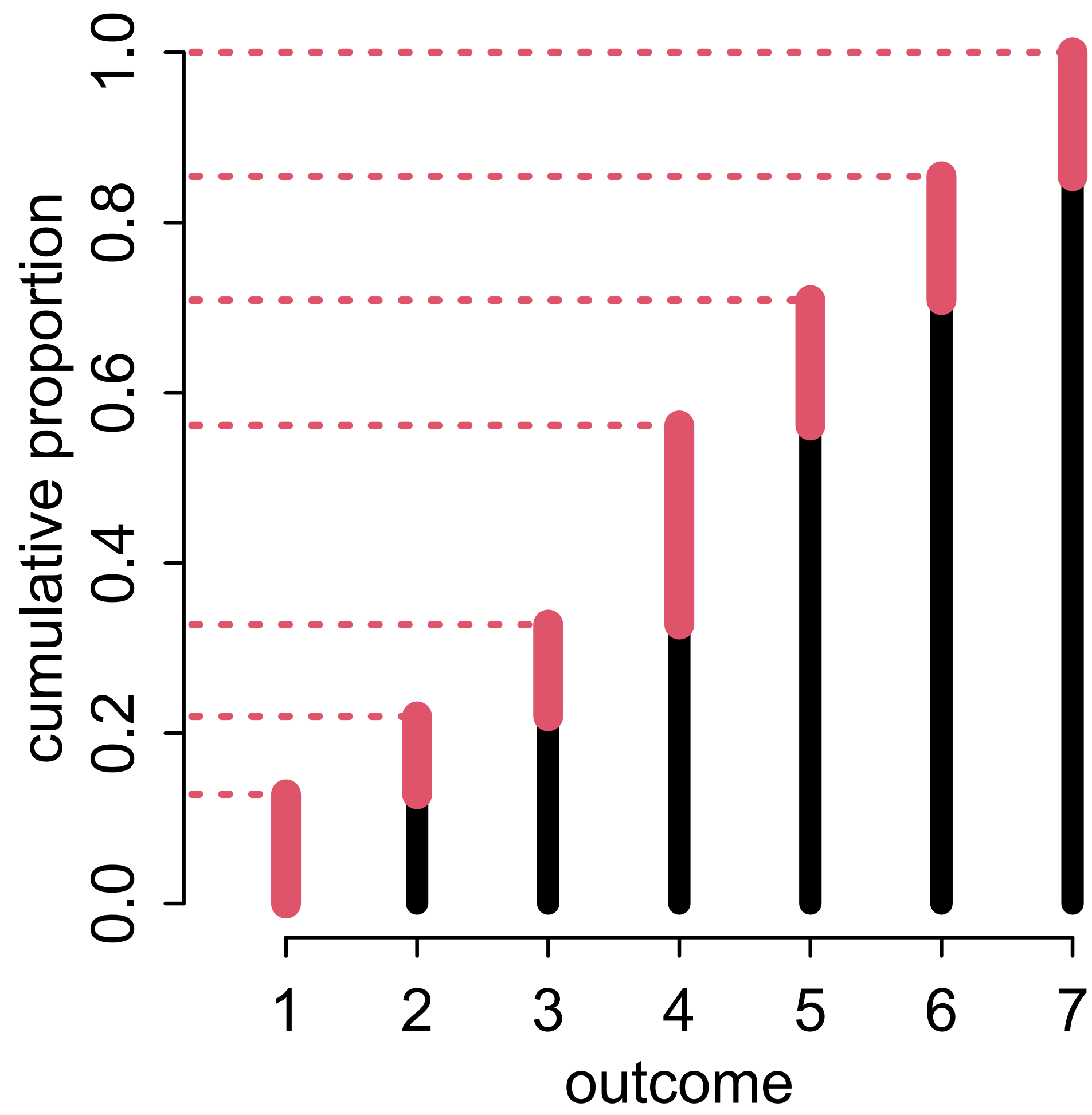


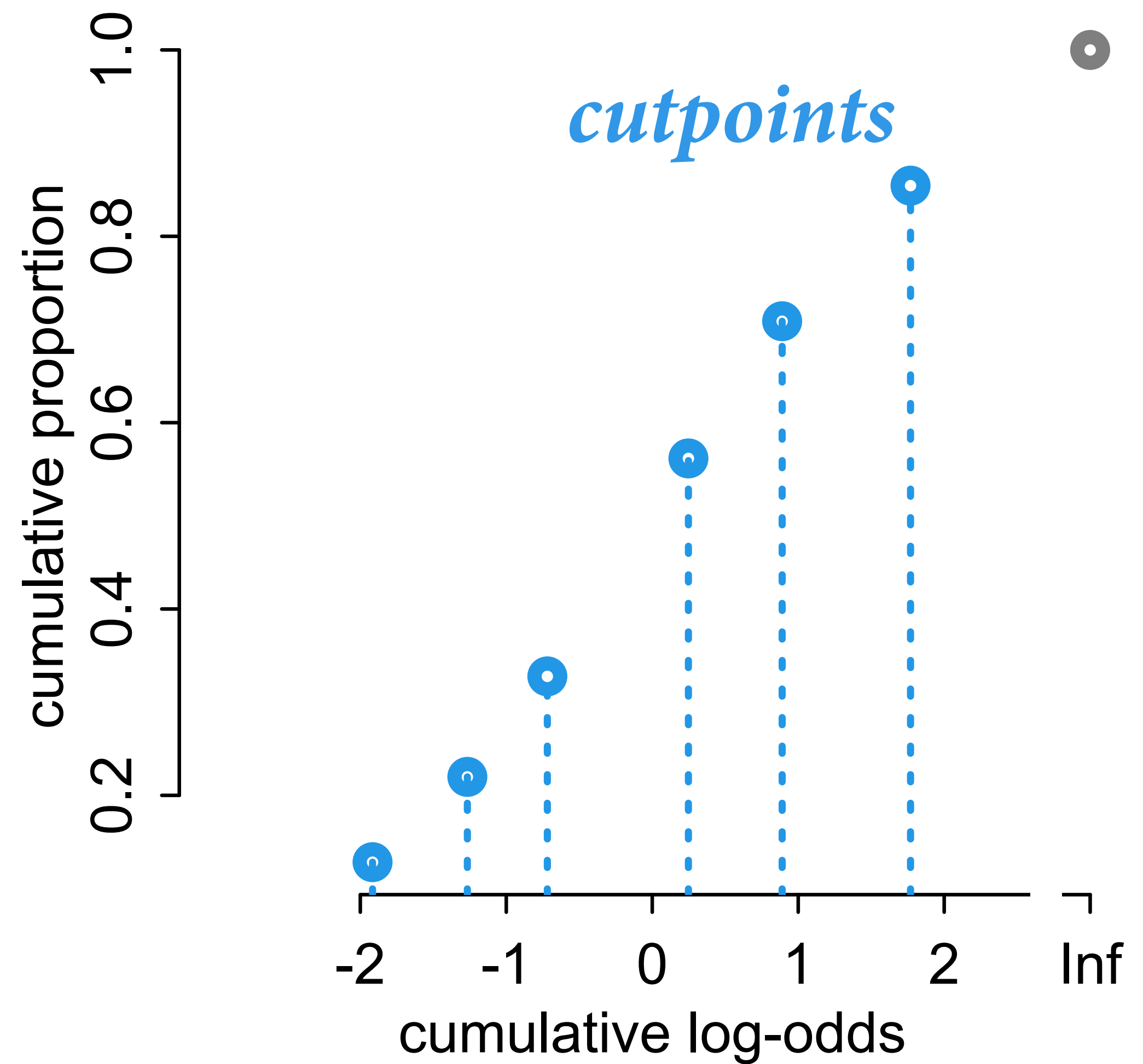
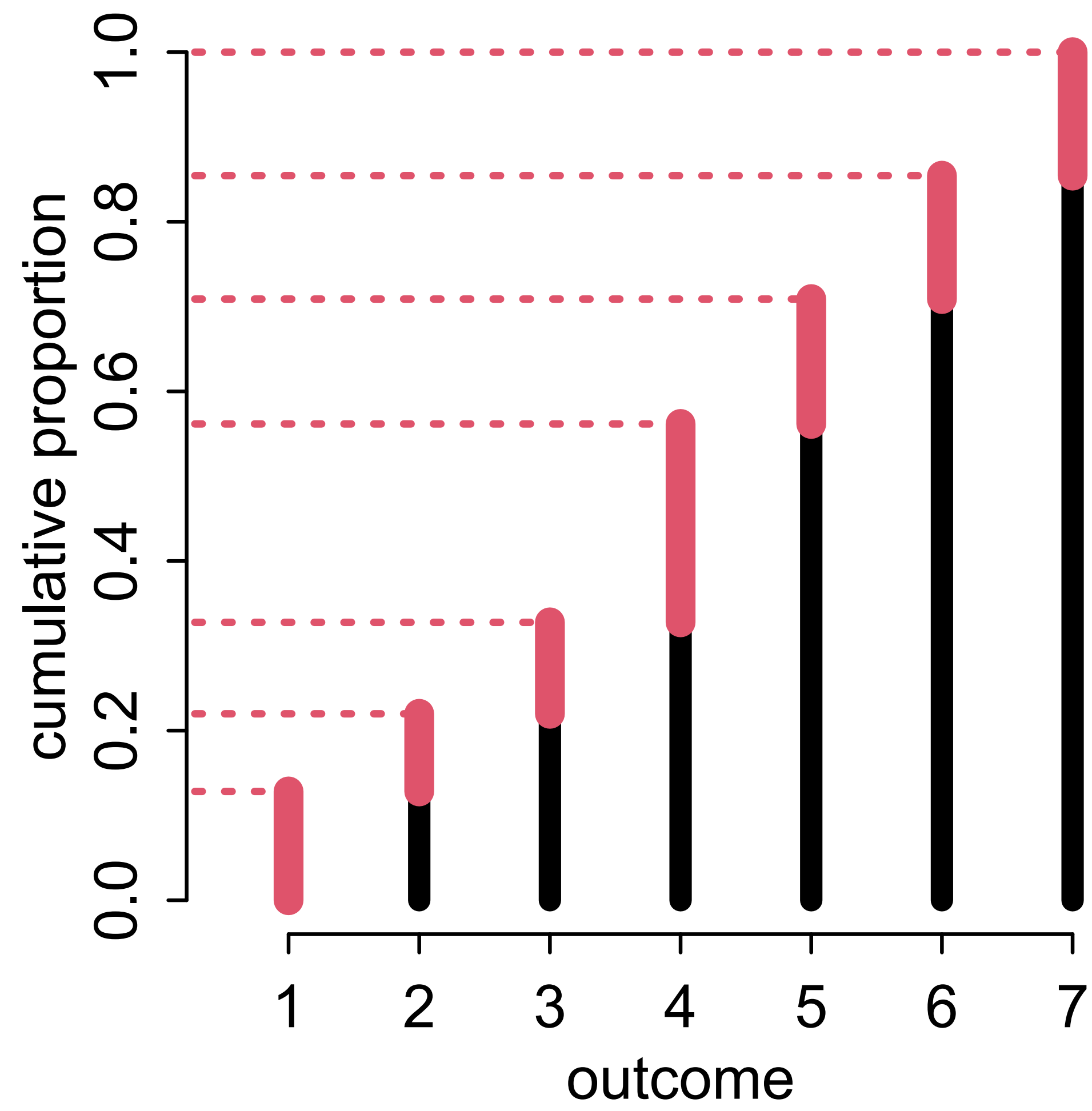


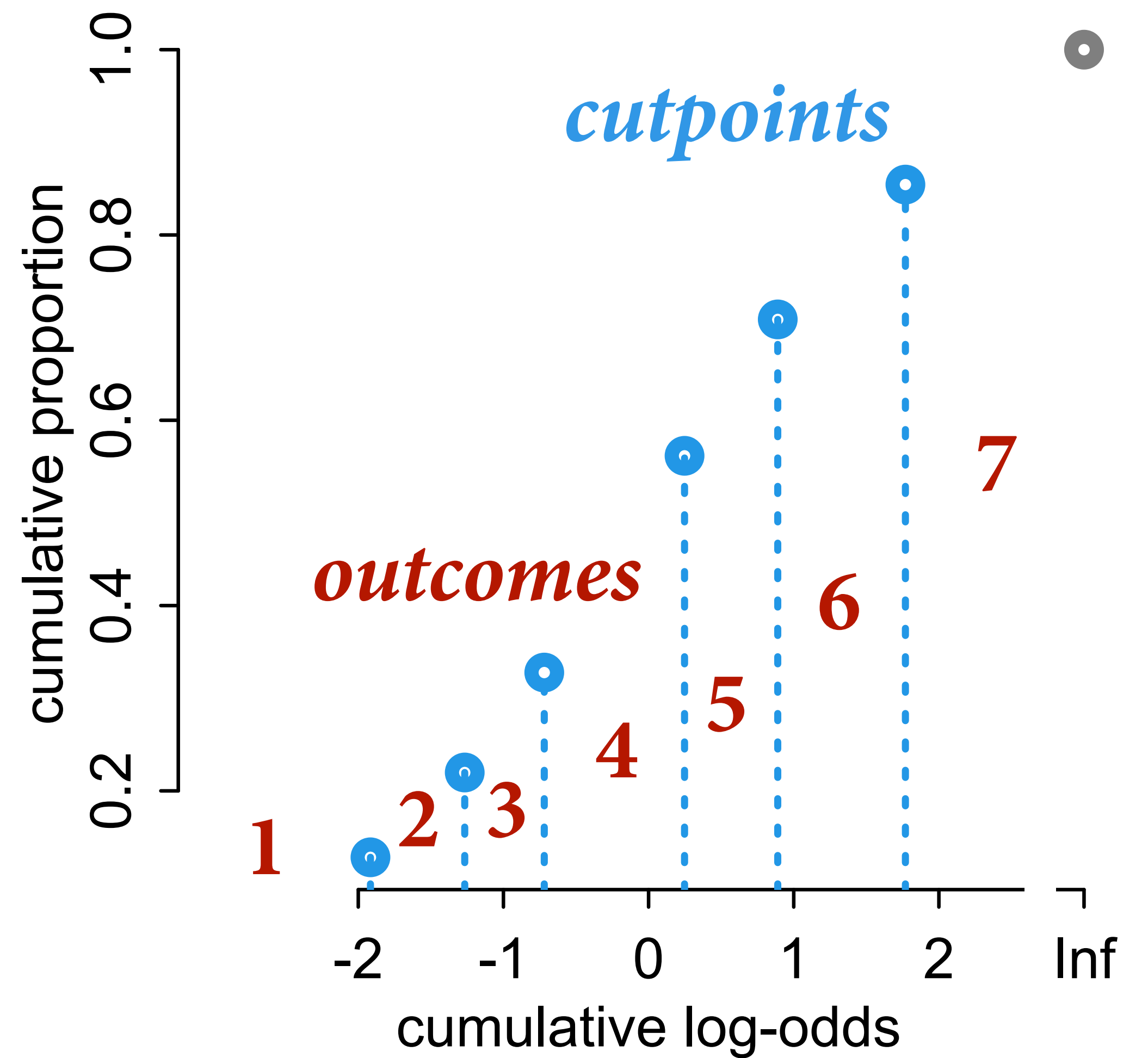
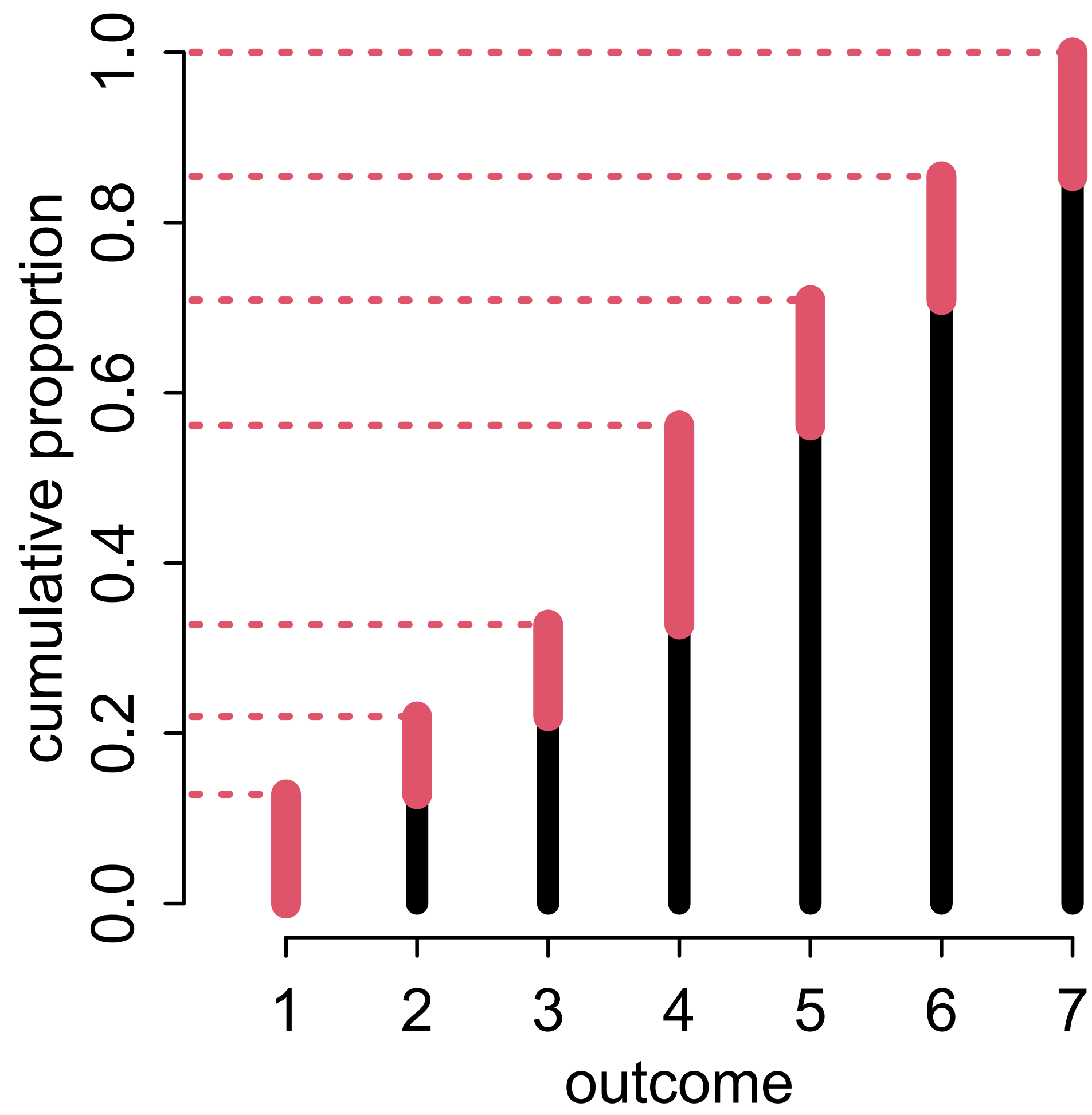
Ordered = Cumulative

Moving to a cumulative distribution allows us to maintain the ordering among categories



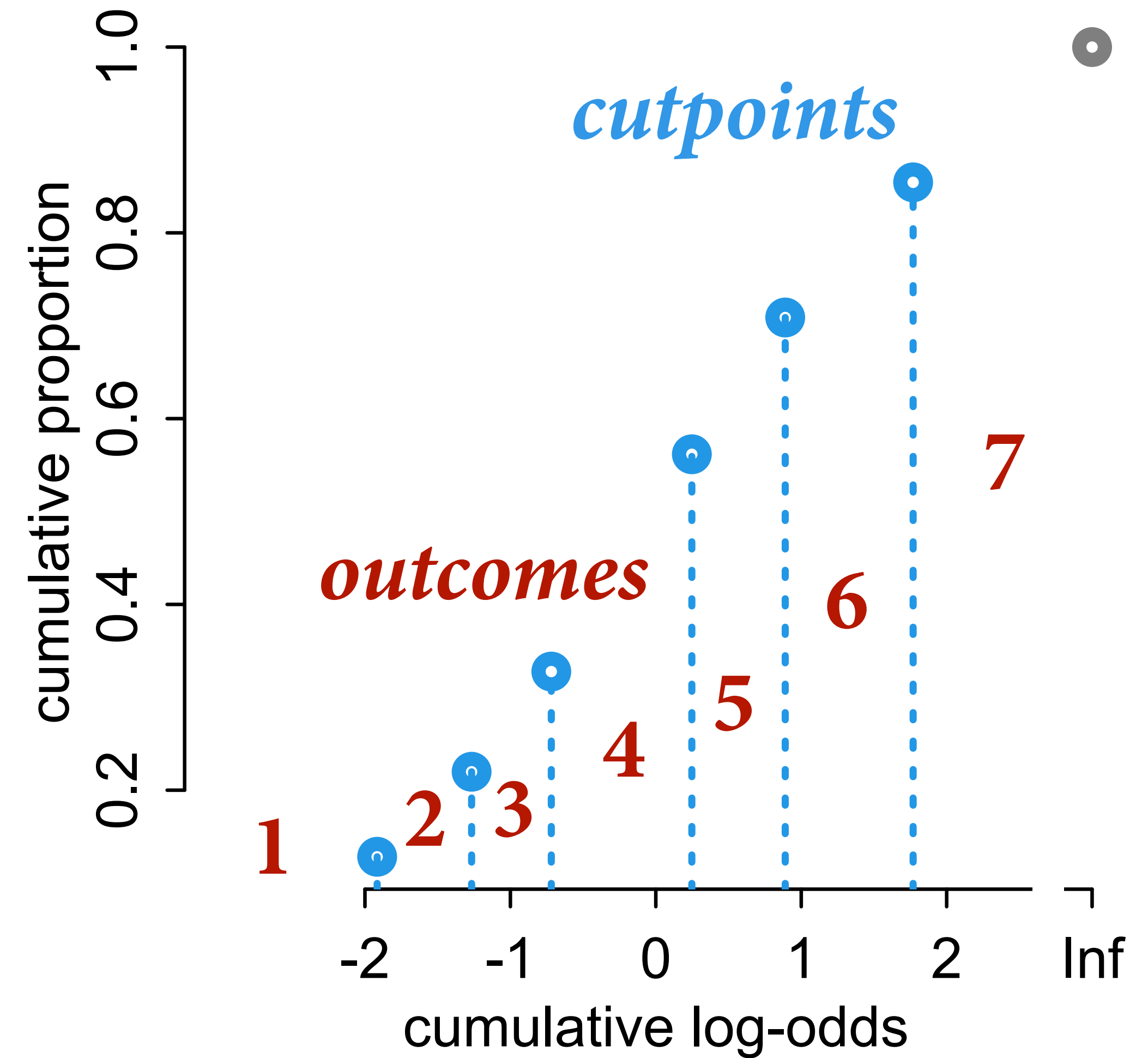




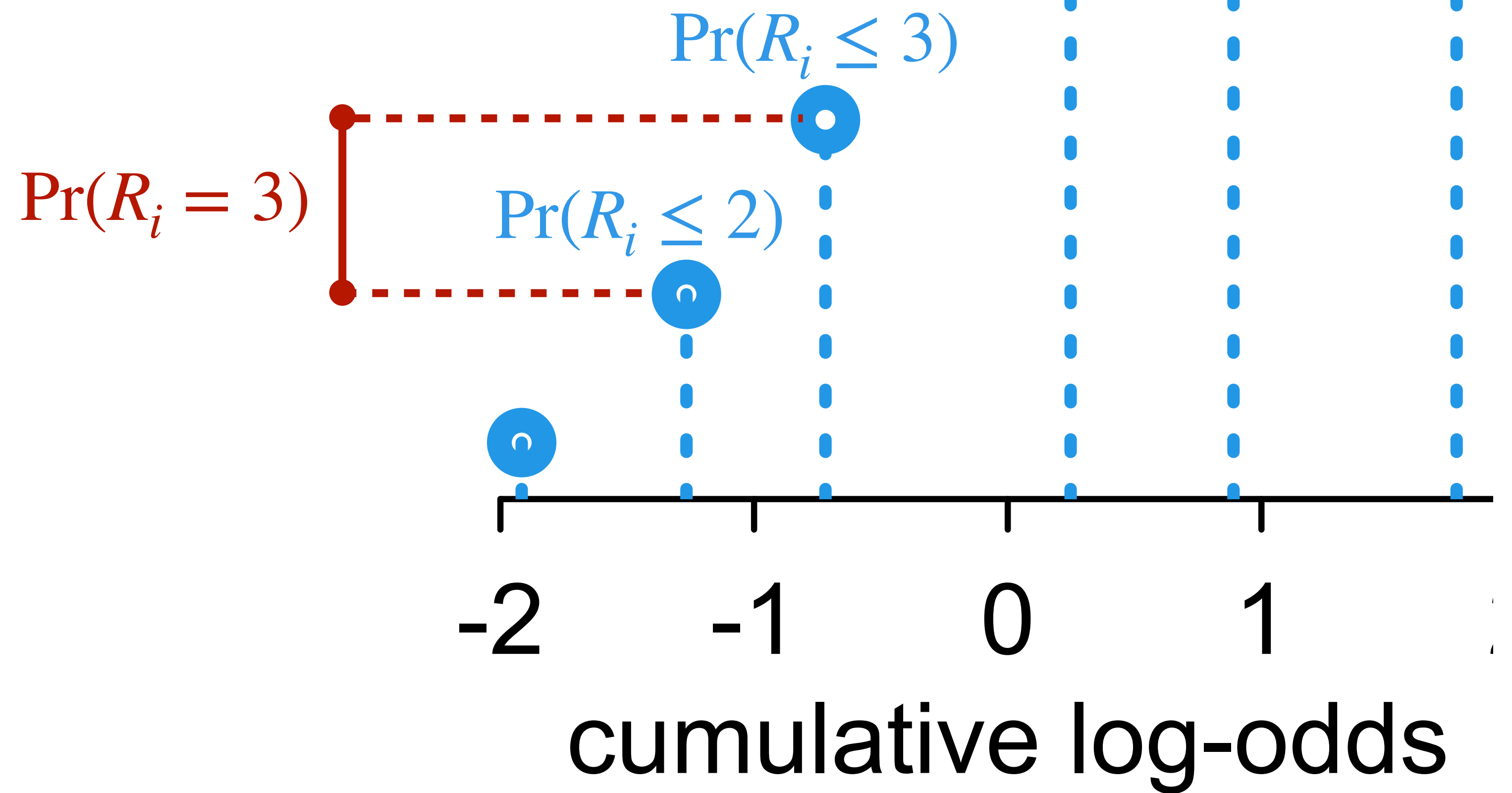


How to get back to discrete categories?

$$\Pr(R_i = k) = \Pr(R_i \leq k) - \Pr(R_i \leq k - 1)$$



$$\Pr(R_i = 3) = \Pr(R_i \leq 3) - \Pr(R_i \leq 2)$$

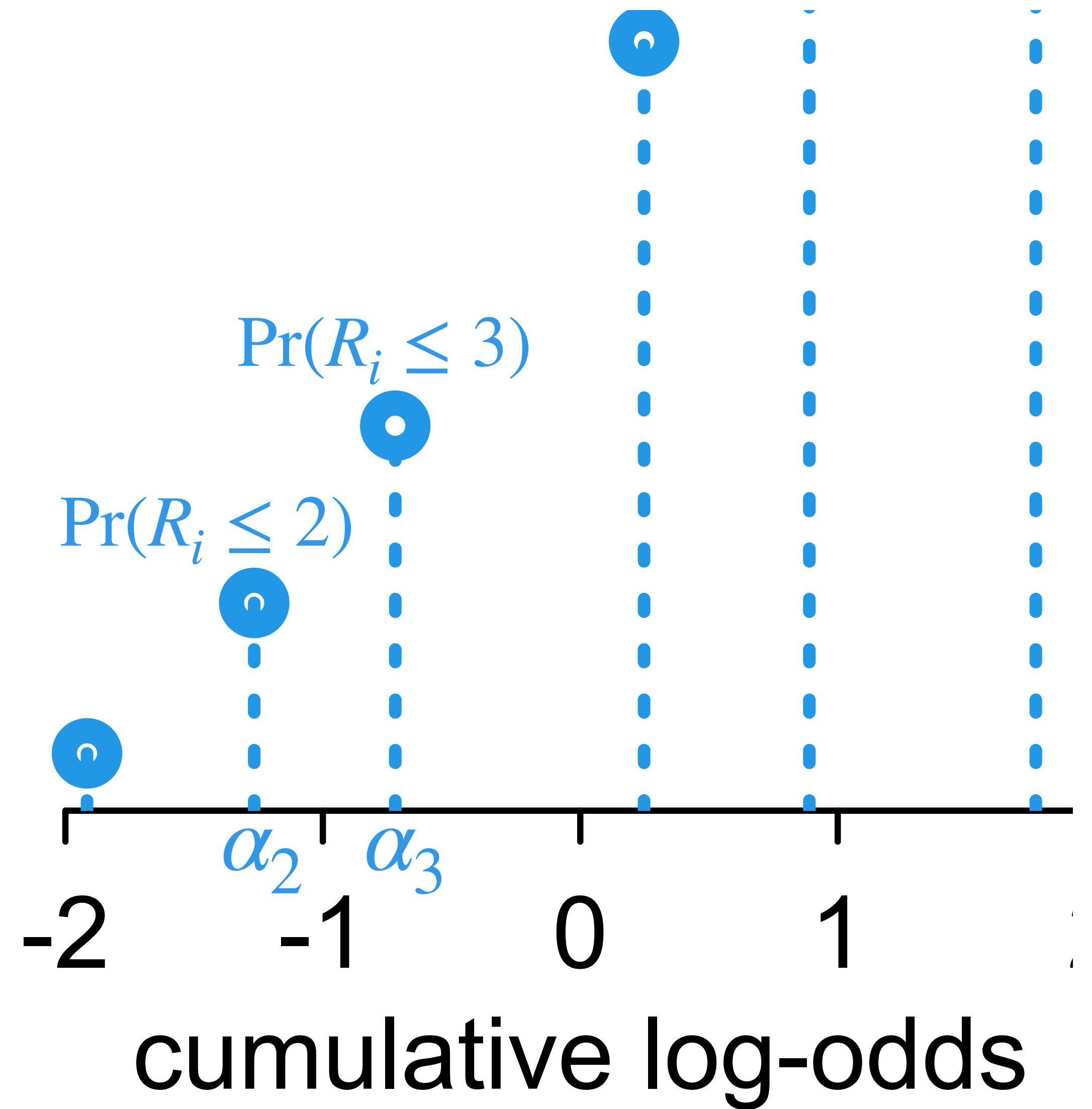


$$\Pr(R_i = 3) = \Pr(R_i \leq 3) - \Pr(R_i \leq 2)$$

$$\log \frac{\Pr(R_i \leq k)}{1 - \Pr(R_i \leq k)} = \alpha_k$$

cumulative log-odds

*cutpoint
(to estimate)*



Where's the GLM?

So far just estimating the histogram

How to make it a function of variables?

(1) Stratify cutpoints

E.g., different cut points for action, intention, and contact trials

(2) Offset each cutpoint by value of linear model ϕ_i

Where's the GLM?

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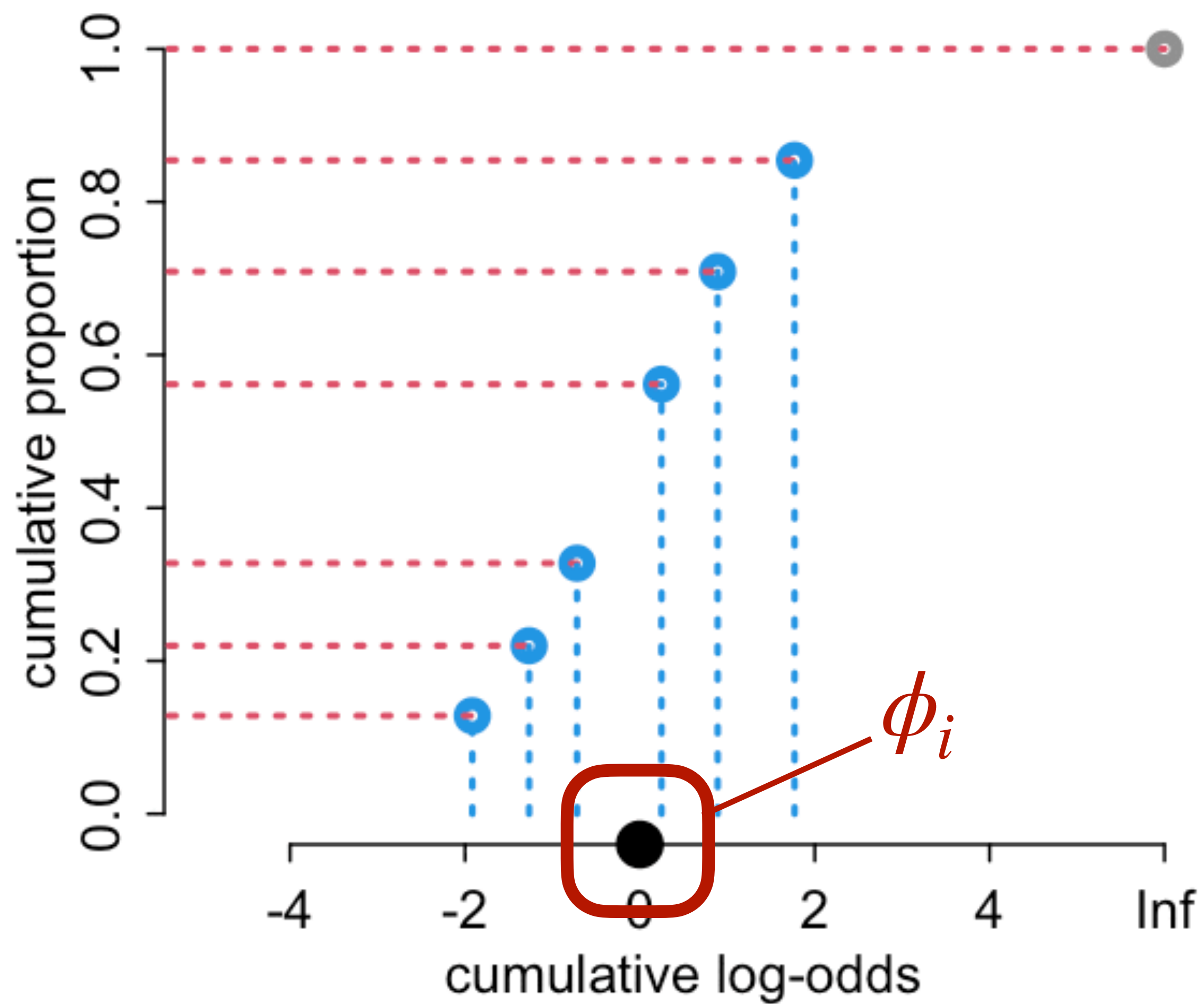
No intercept in this linear model b/c the cutpoint serves as an intercept



$$\phi_i = \beta x_i$$

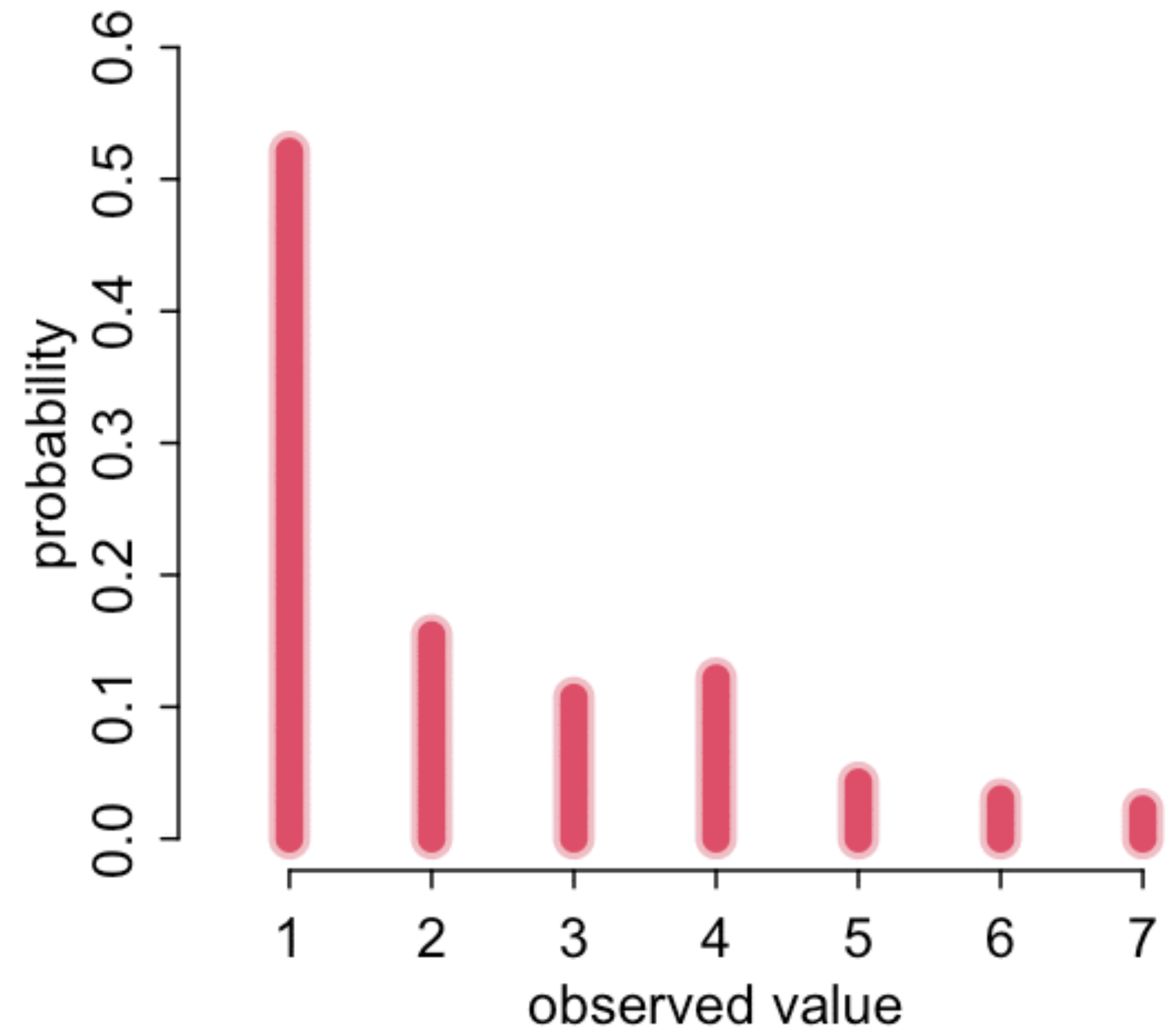
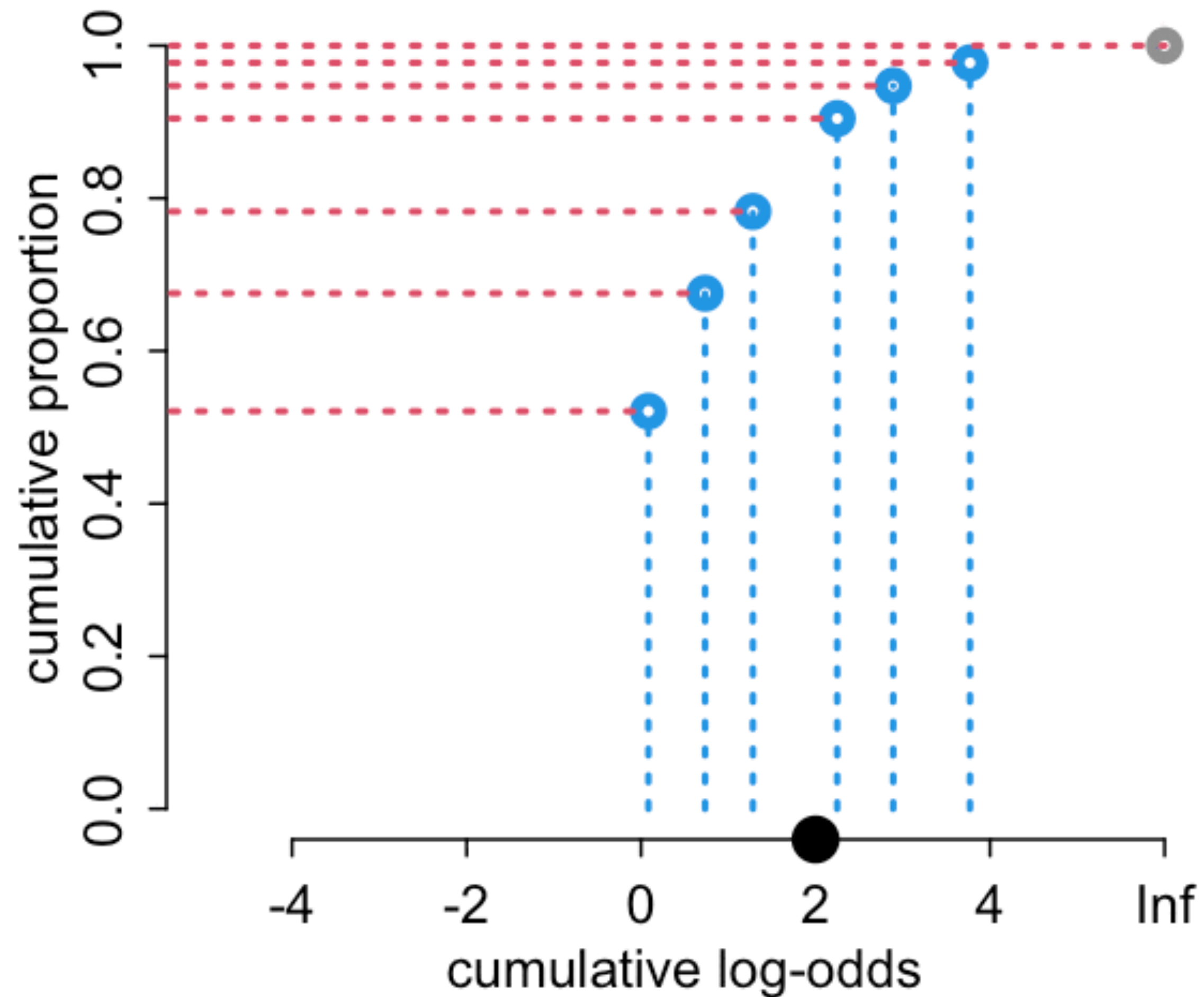
$$\log \frac{\Pr(R_i \leq k)}{1 - \Pr(R_i \leq k)} = \alpha_k + \phi_i$$

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

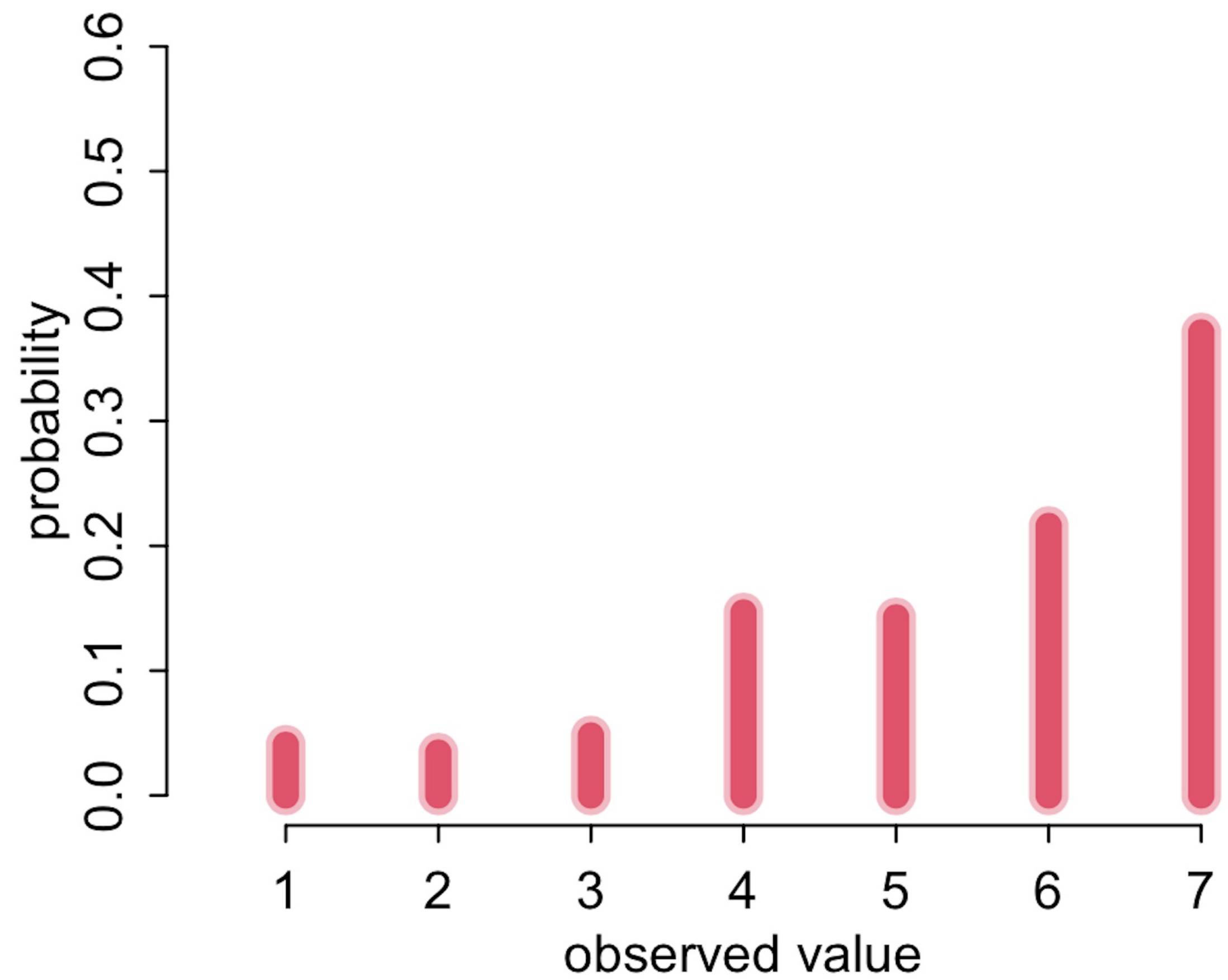
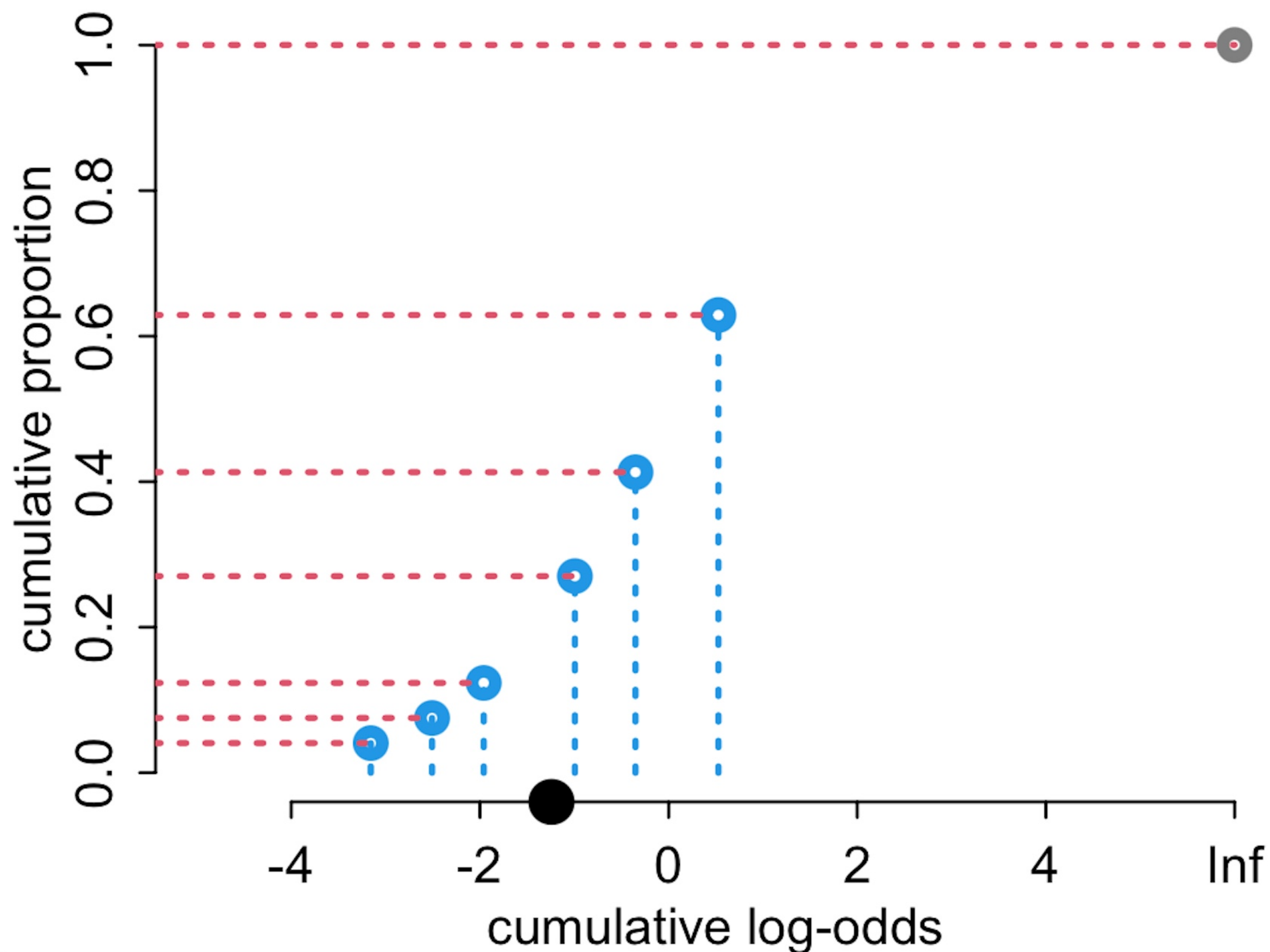


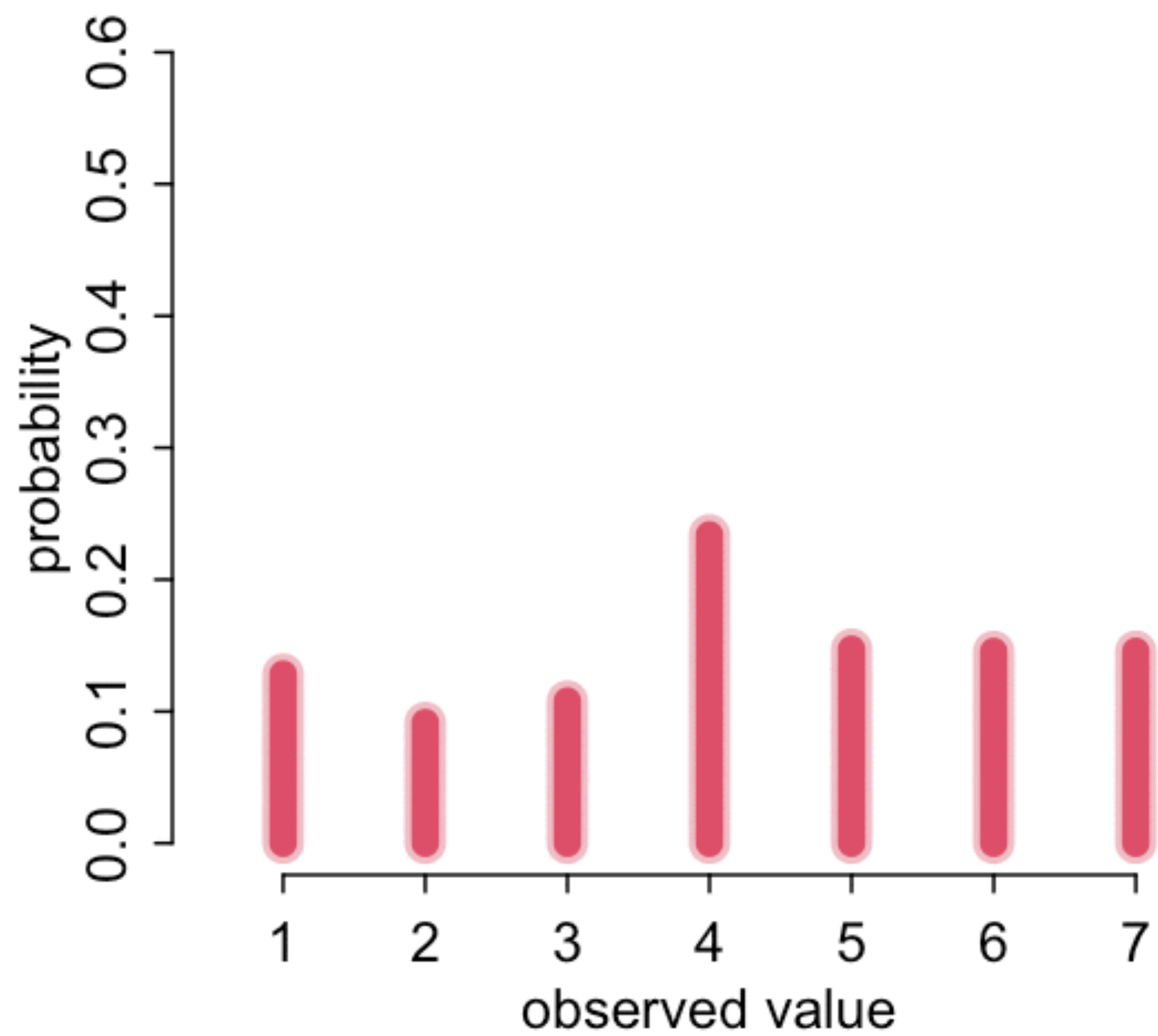
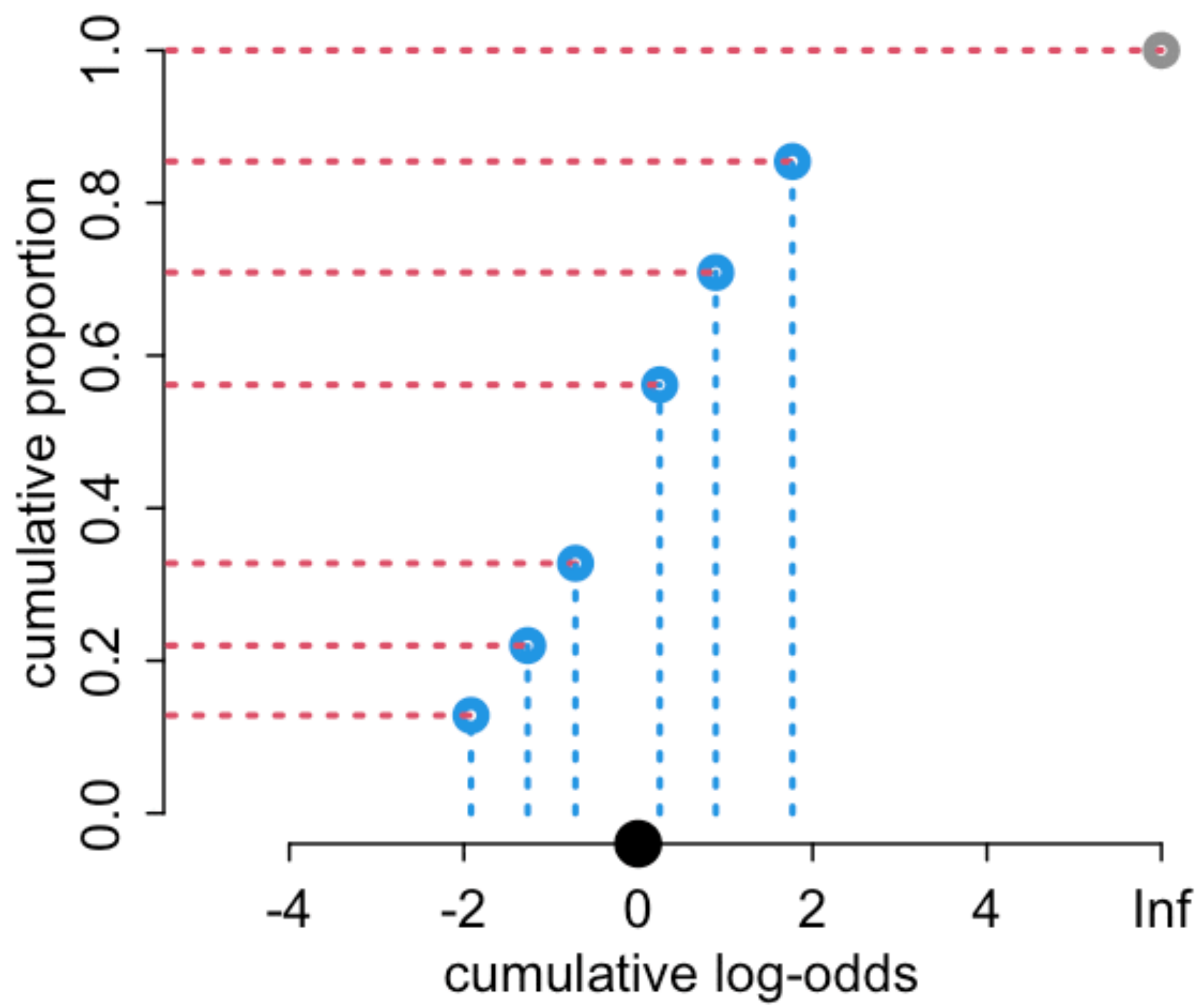
$$\log \frac{\Pr(R_i \leq k)}{1 - \Pr(R_i \leq k)} = \alpha_k + \phi_i$$

If adding phi to cutpoints, then larger phi give more low responses.
Some software subtracts phi so it will have the opposite effect.
check

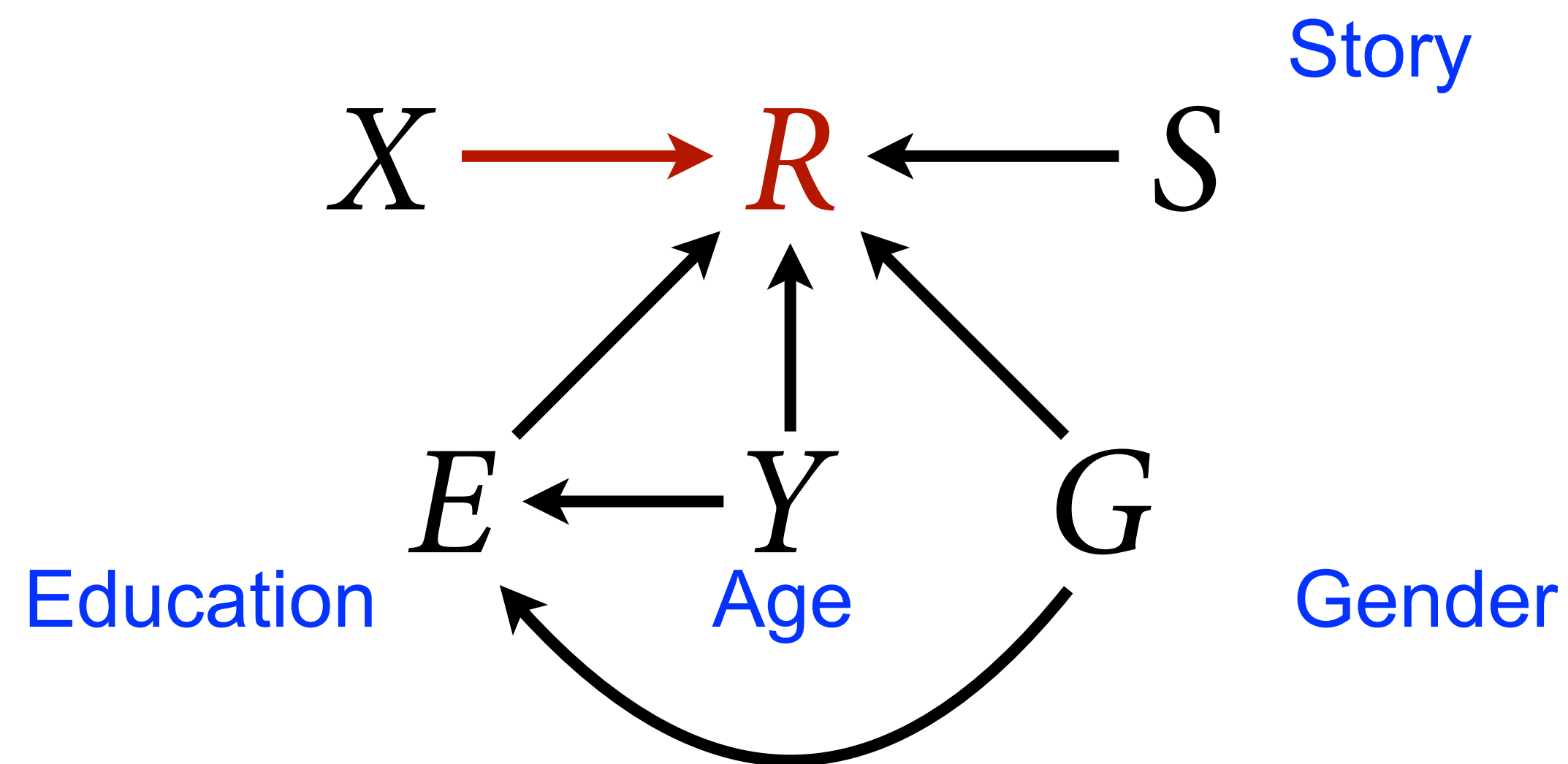


Moving the cutpoints to the left puts more probability on larger scale values (e.g., 7 in this case)





Start off easy:



$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_A A_i + \beta_C C_i + \beta_I I_i$$

$$\beta_- \sim \text{Normal}(0, 0.5)$$

$$\alpha_j \sim \text{Normal}(0, 1)$$

```

data(Trolley)
d <- Trolley
dat <- list(
  R = d$response,
  A = d$action,
  I = d$intention,
  C = d$contact
)

mRX <- ulam(
  alist(
    R ~ dordlogit(phi,alpha),
    phi <- bA*A + bI*I + bC*C,
    c(bA,bI,bC) ~ normal(0,0.5),
    alpha ~ normal(0,1)
  ) , data=dat , chains=4 , cores=4 )

```

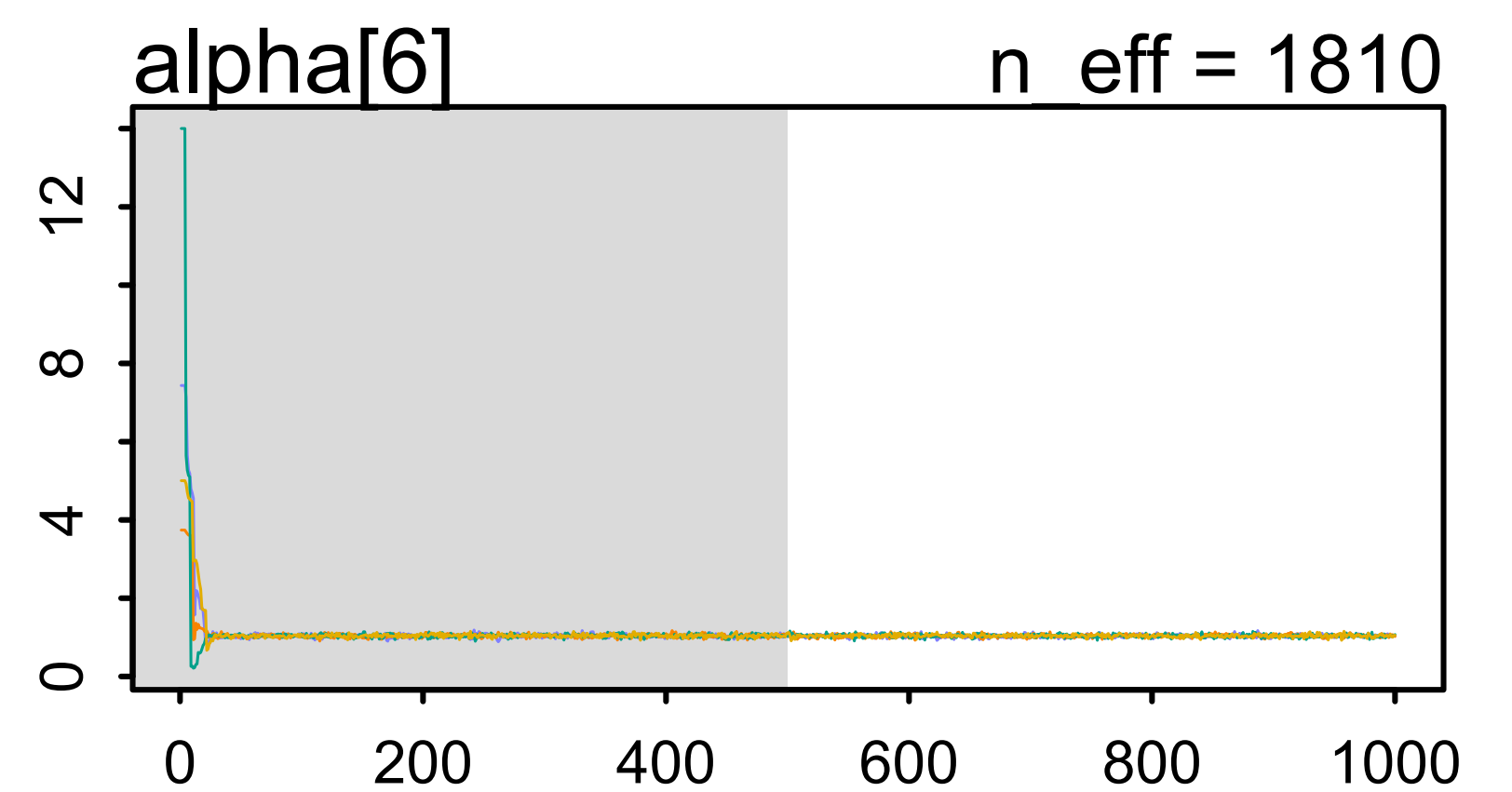
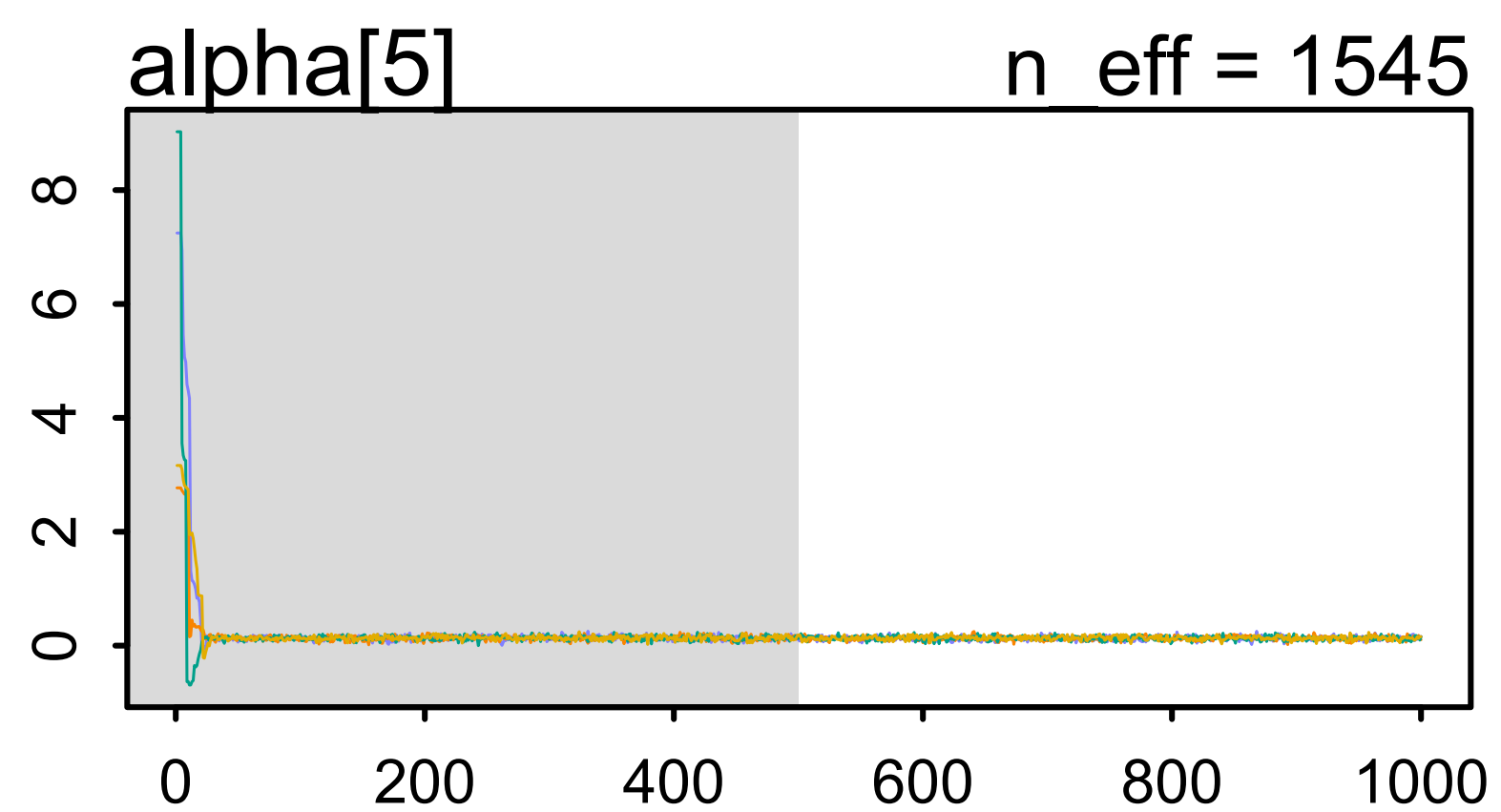
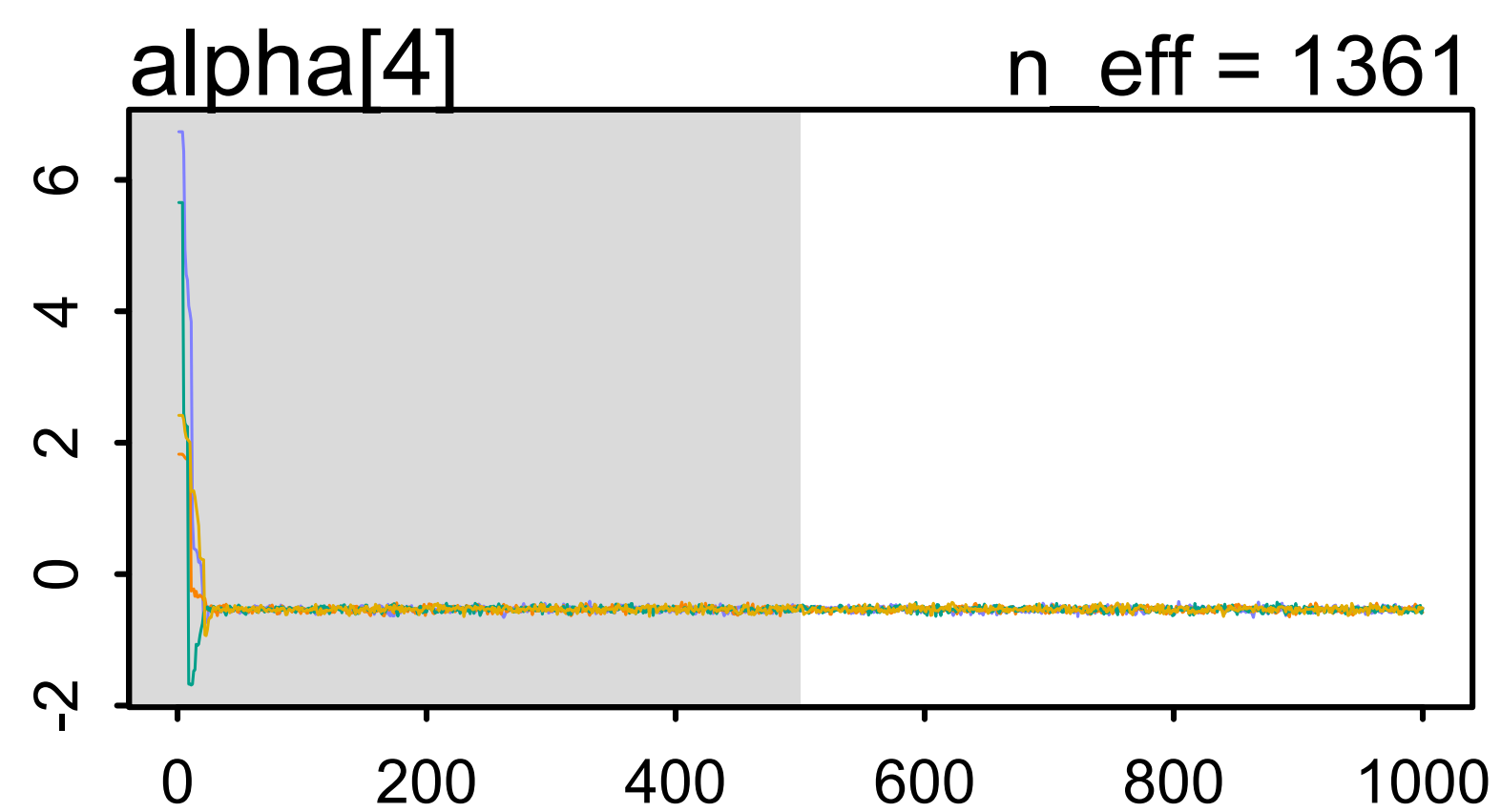
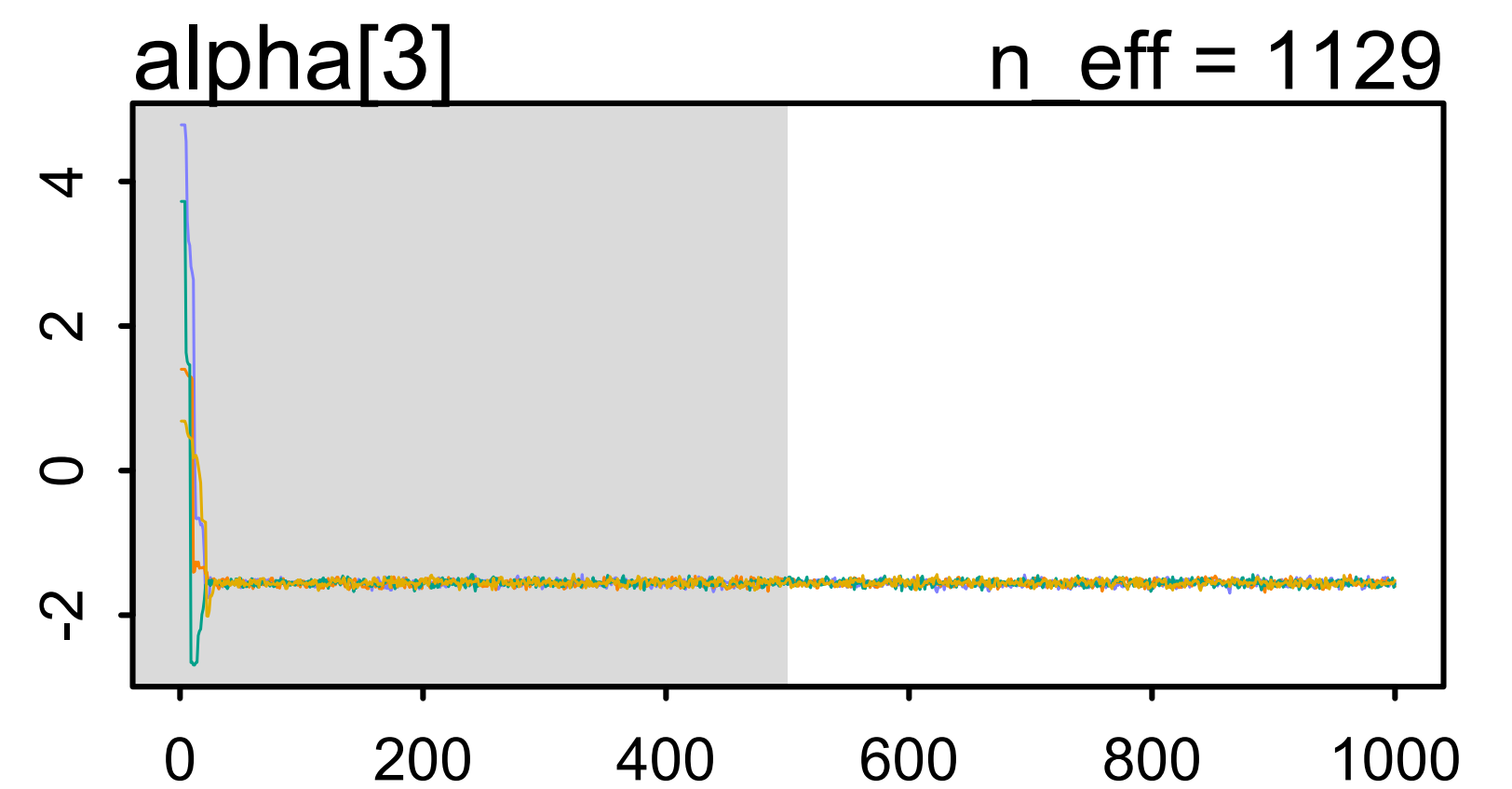
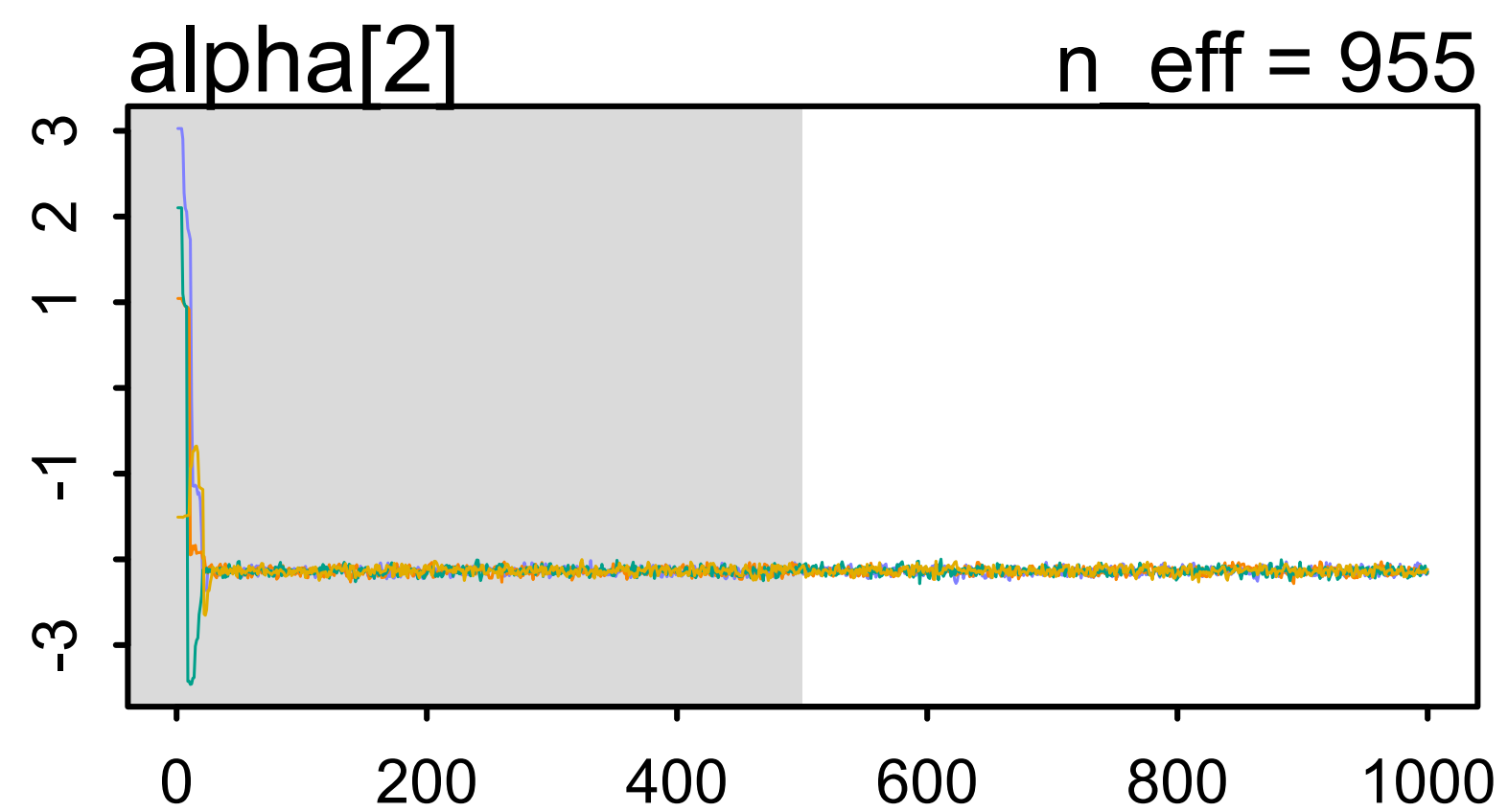
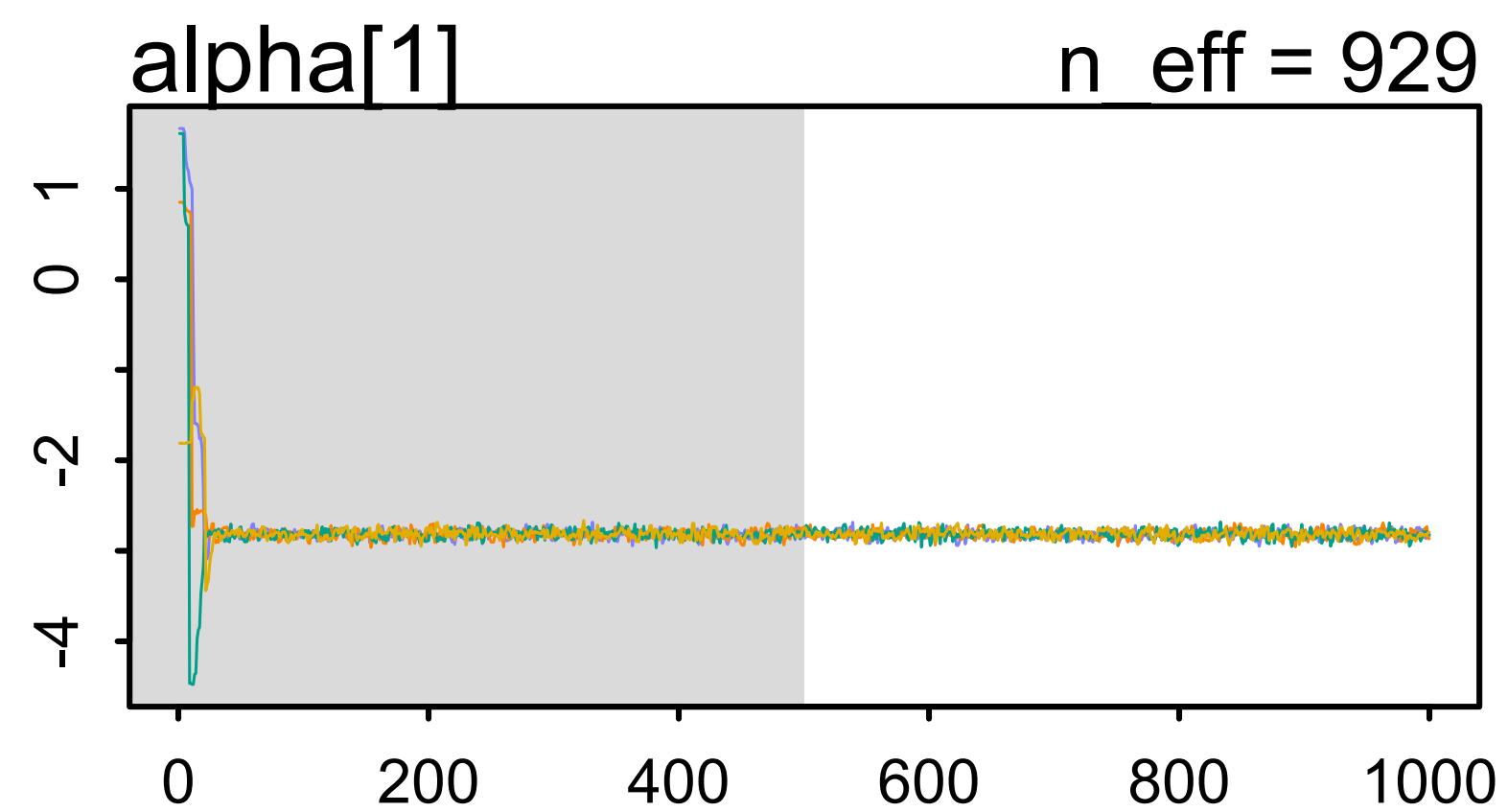
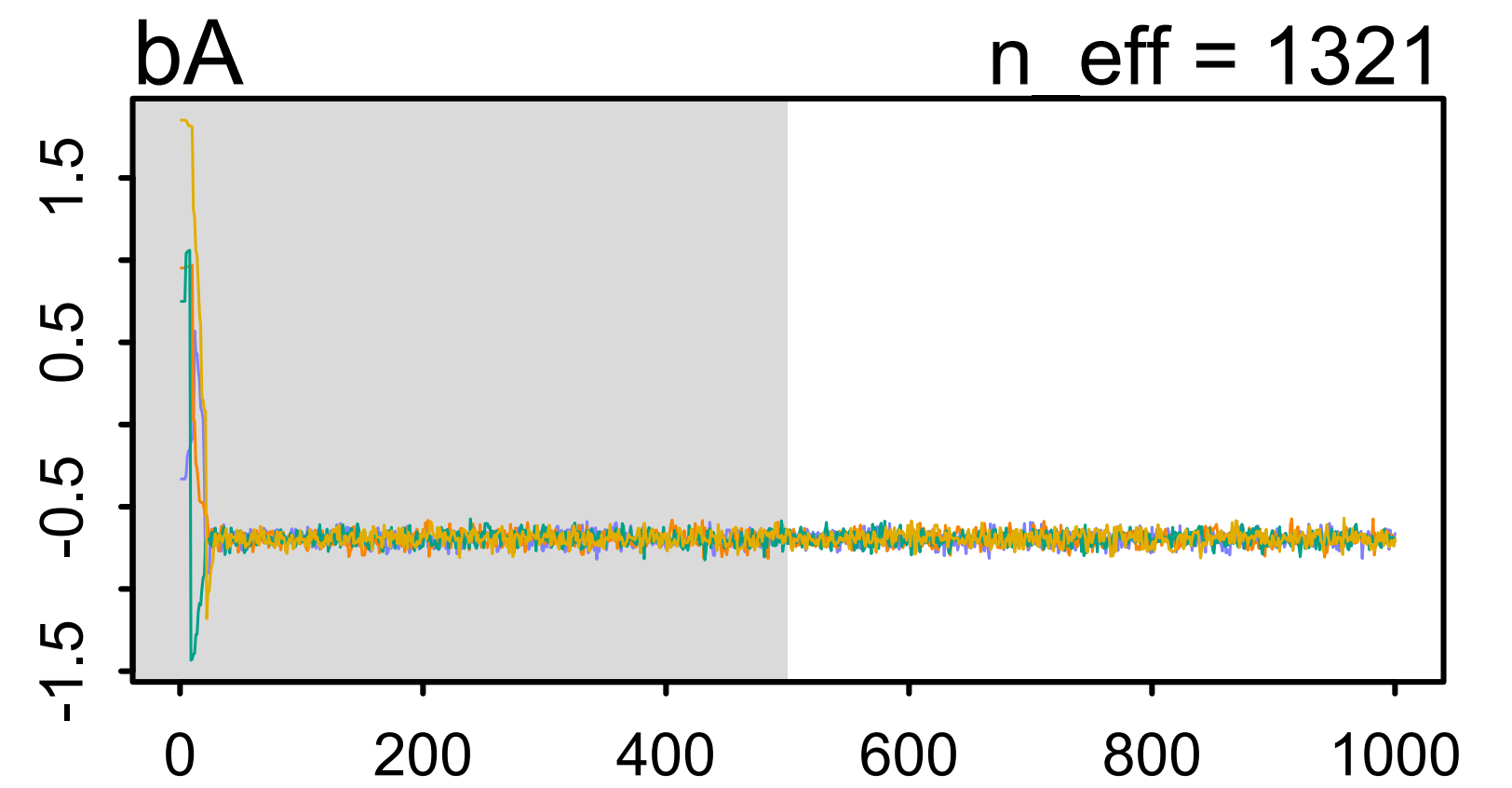
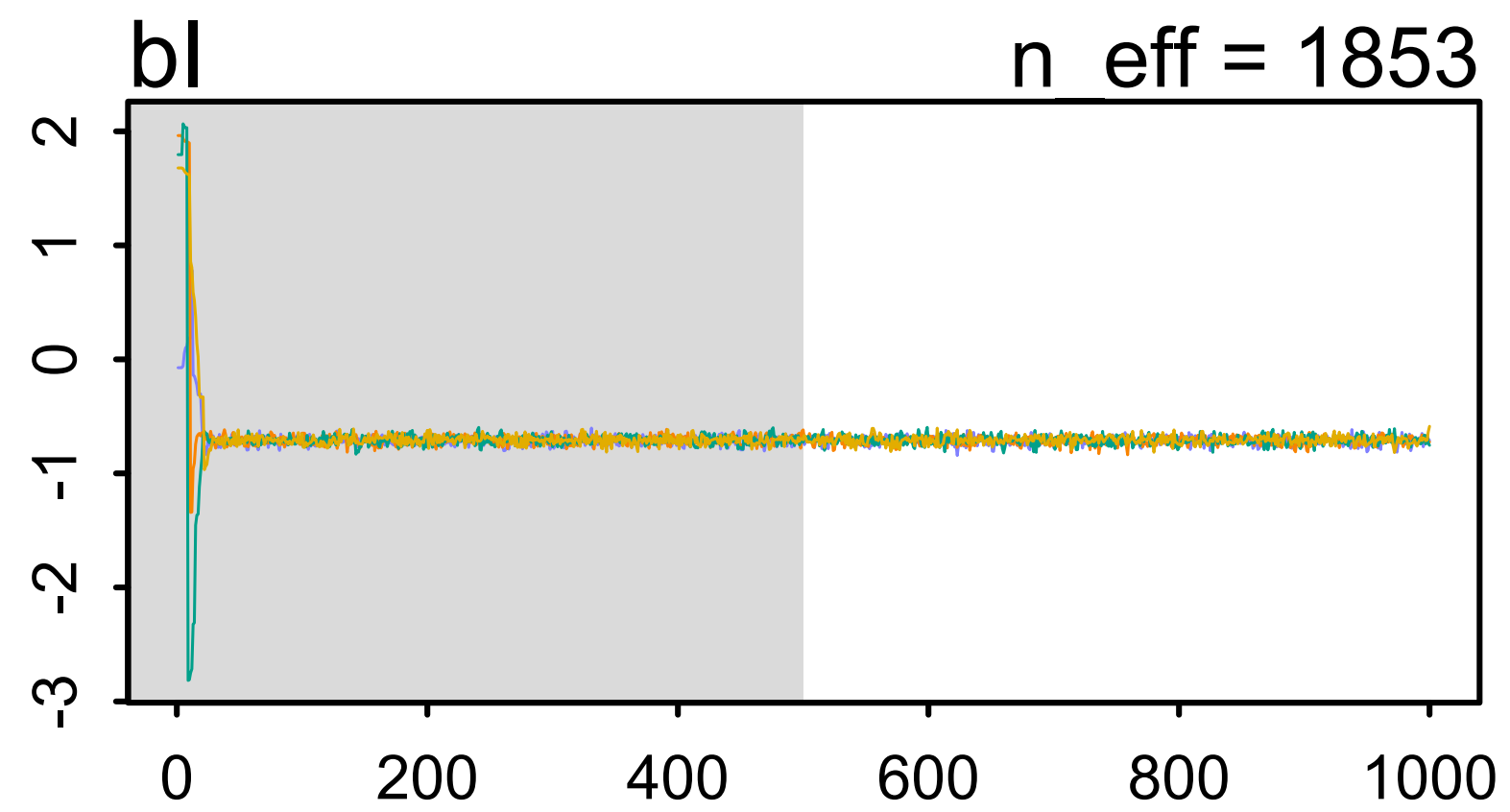
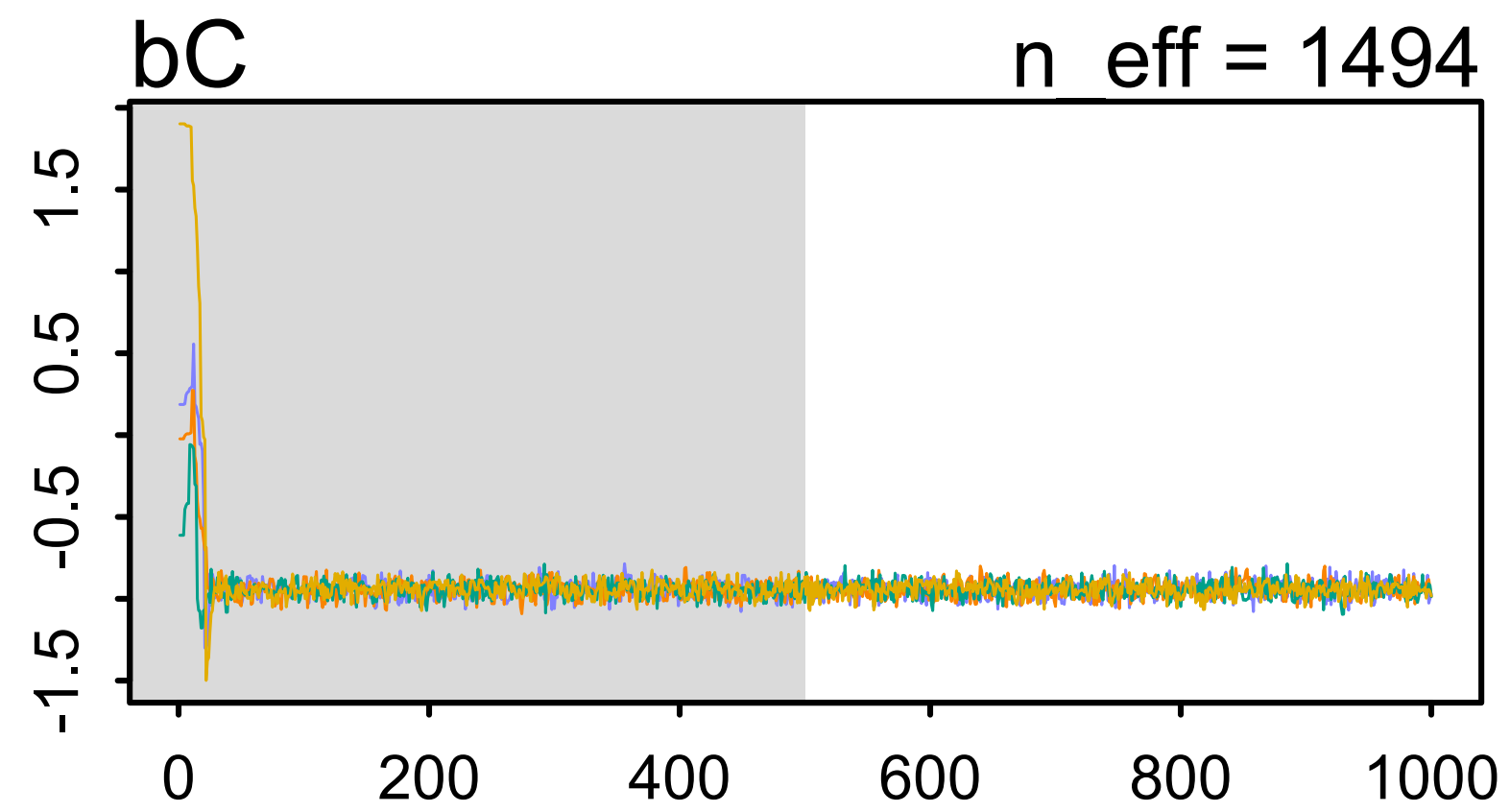
$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

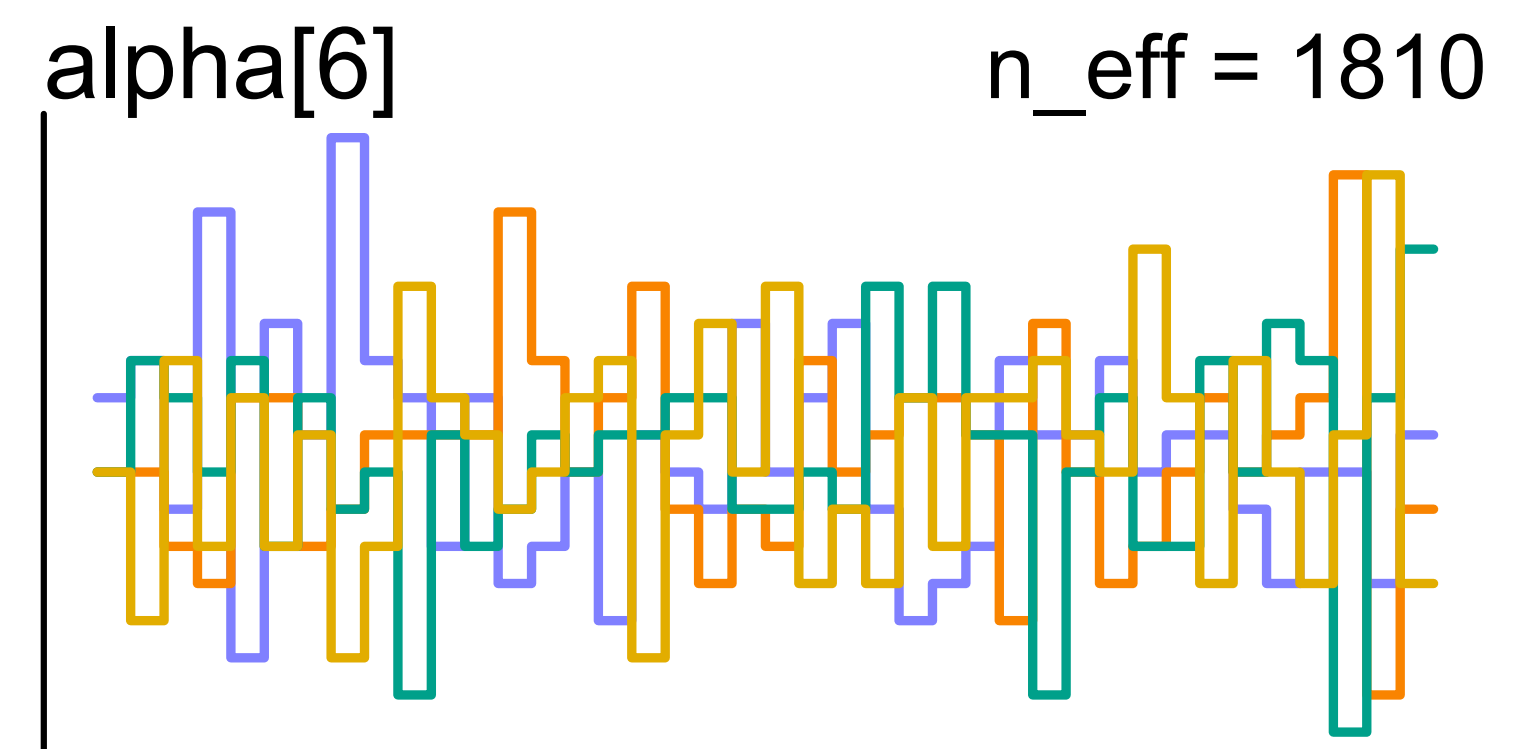
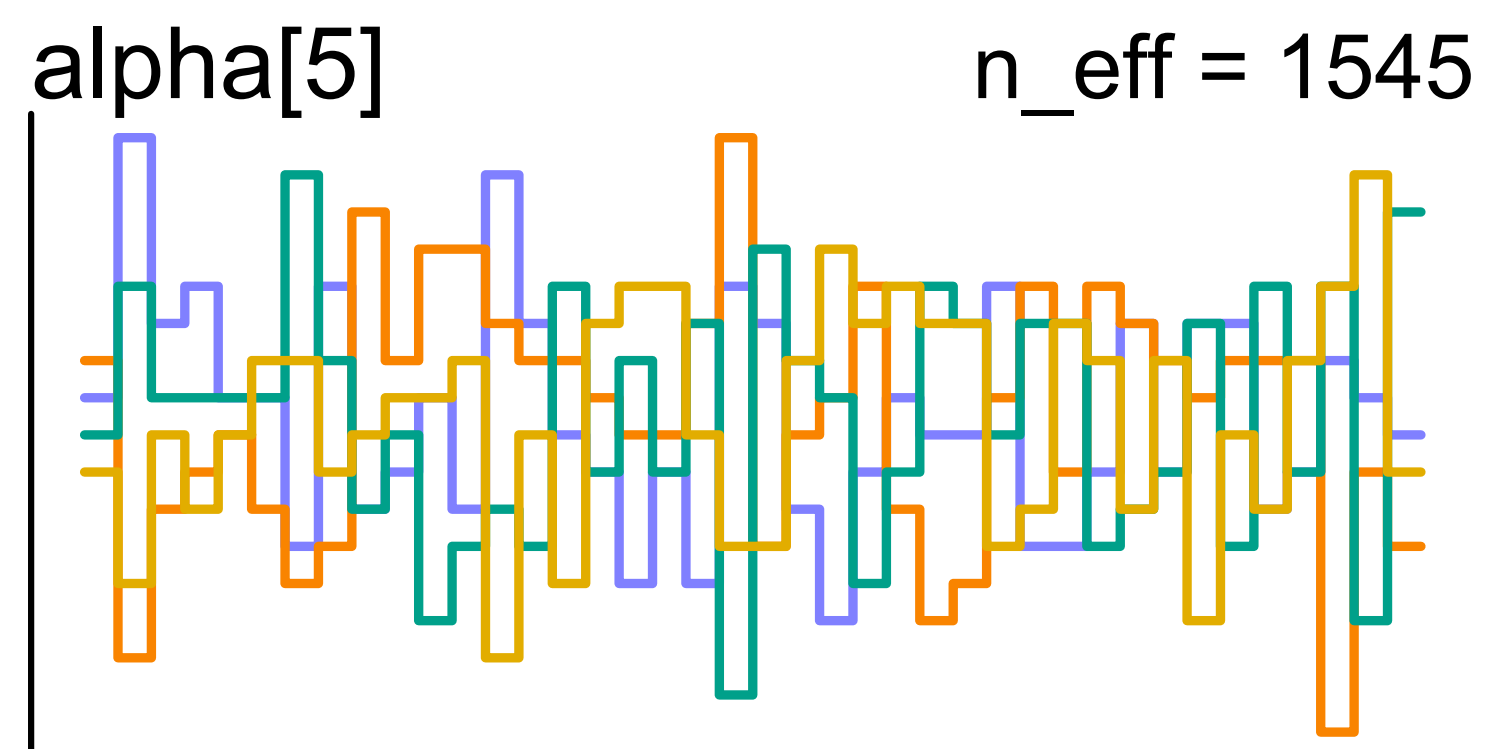
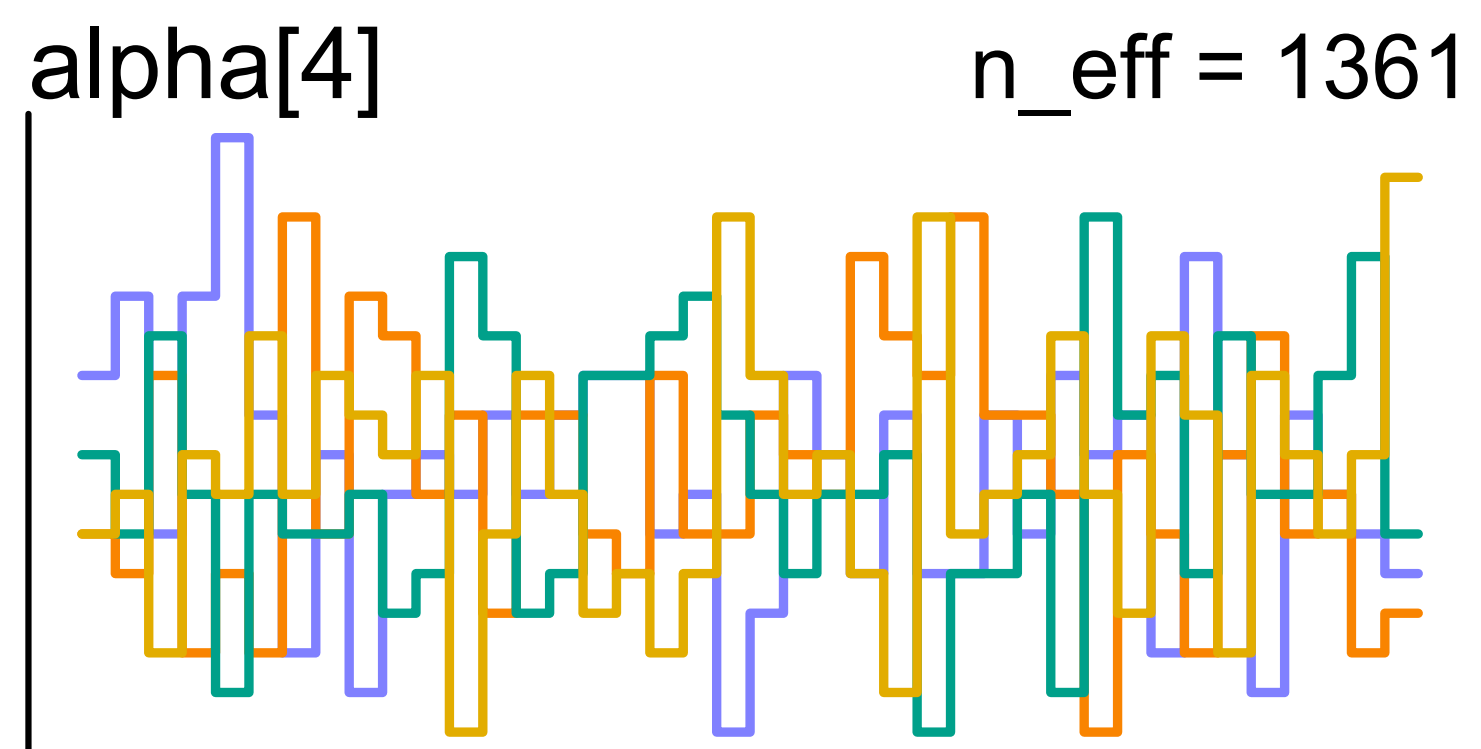
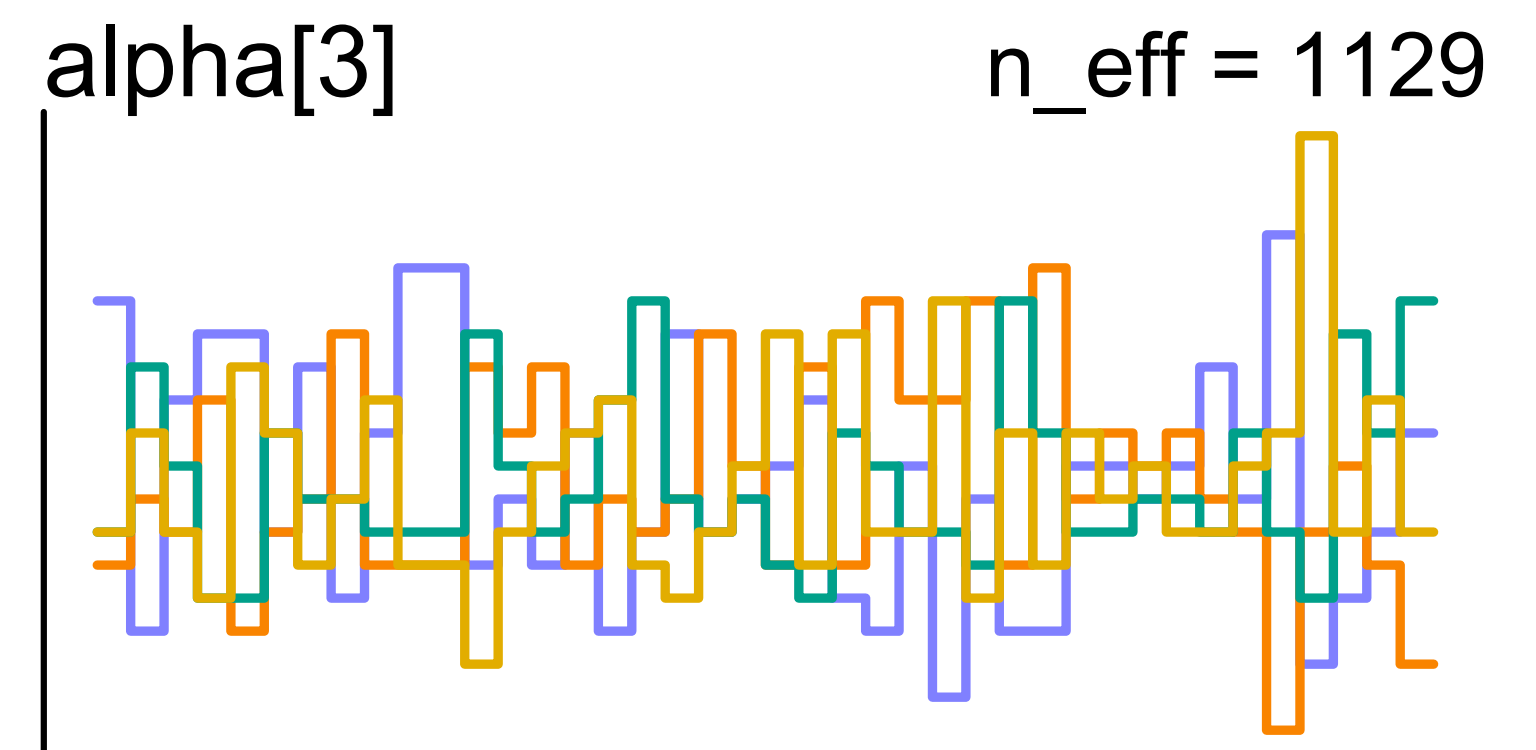
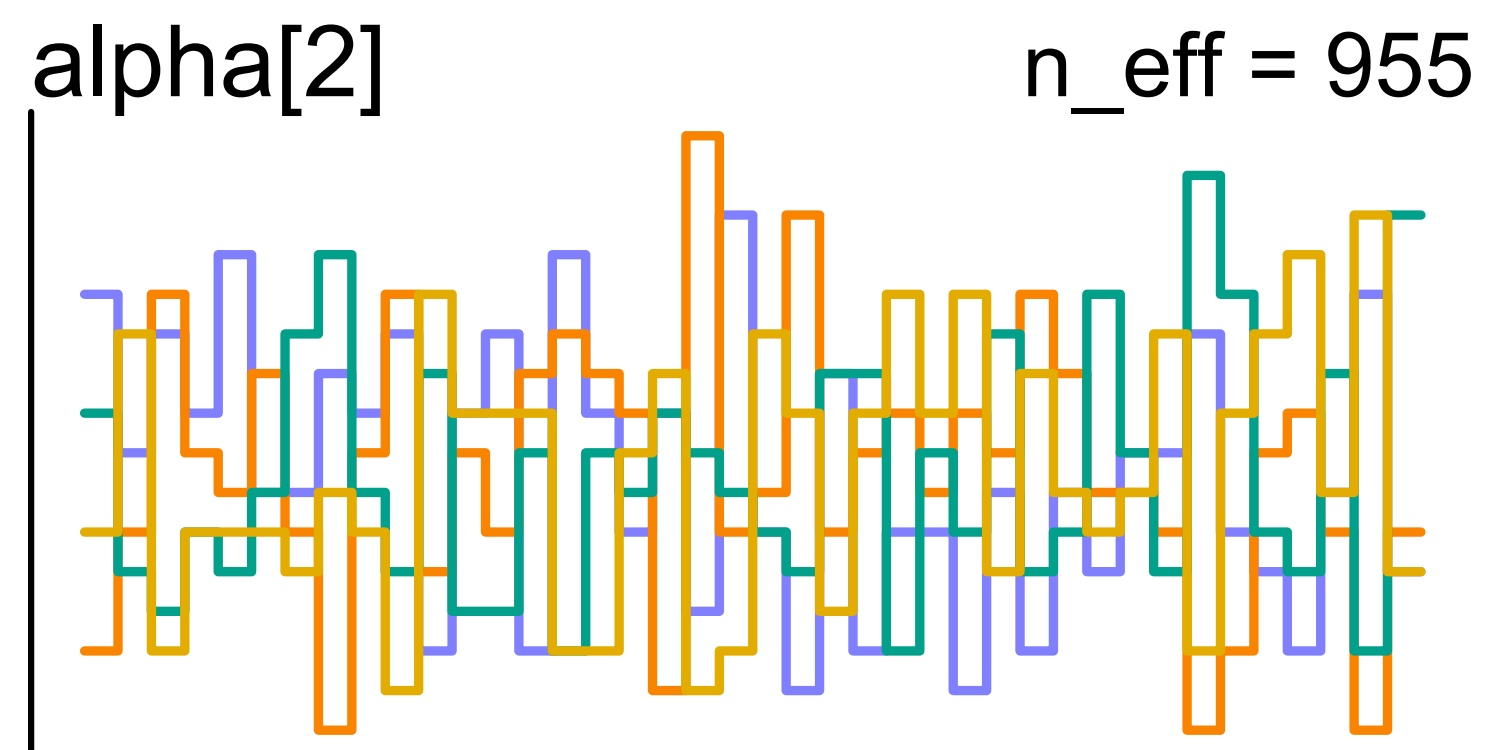
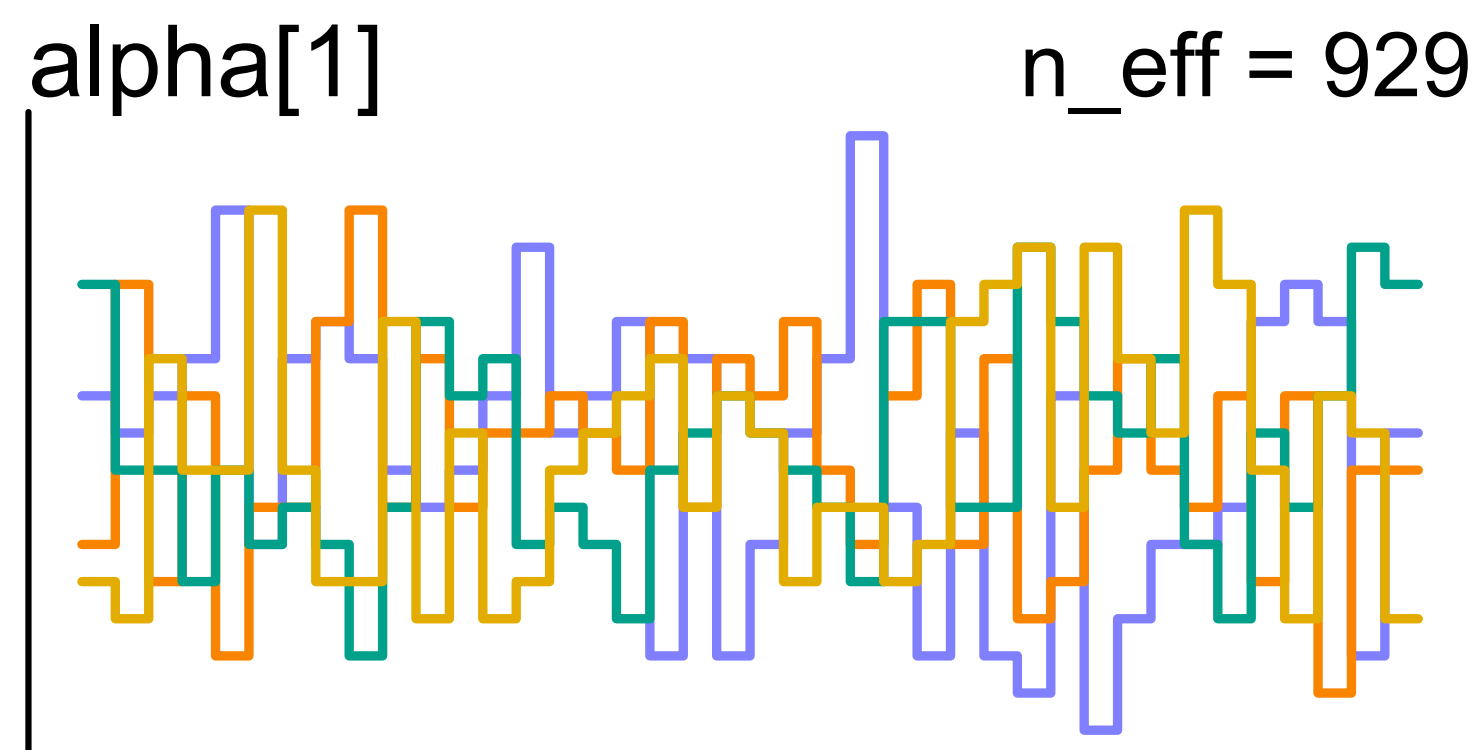
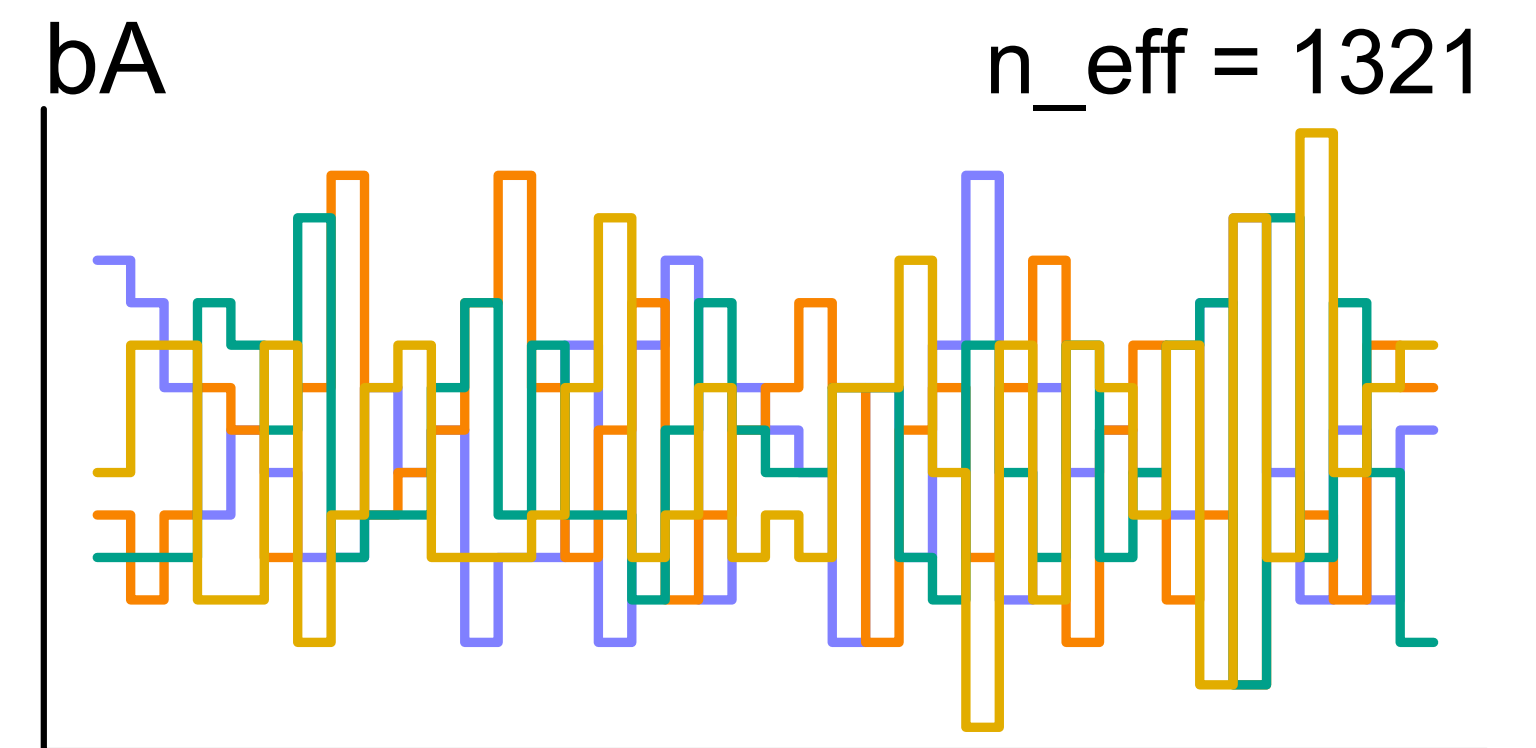
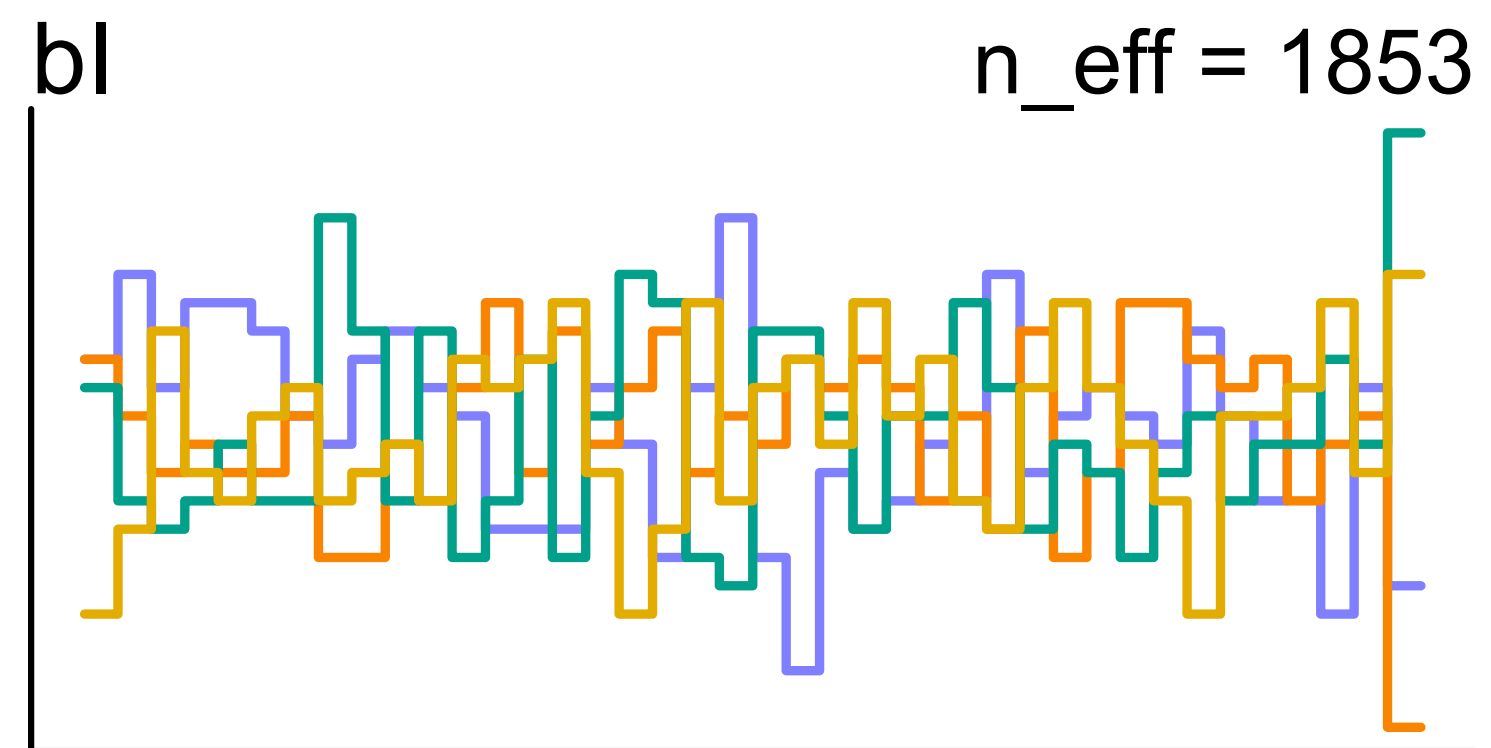
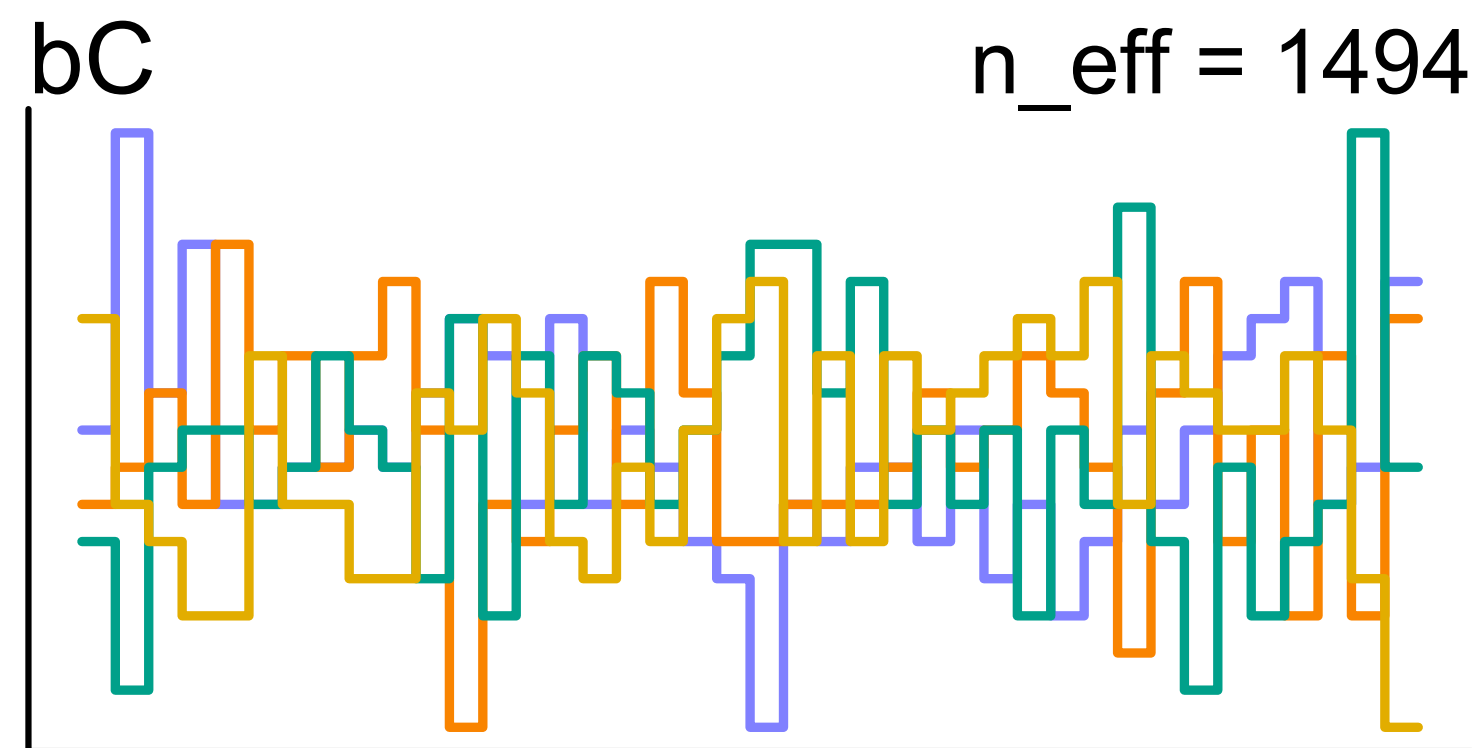
$$\phi_i = \beta_A A_i + \beta_C C_i + \beta_I I_i$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\alpha_j \sim \text{Normal}(0,1)$$

Phi tests how story type changes ratings. *Phi is subtracted in ulam






```

data(Trolley)
d <- Trolley
dat <- list(
  R = d$response,
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  alist(
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    c(bA,bI,bC) ~ normal(0,0.5),
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  ) , data=dat , chains=4 , cores=4 )

```

```

> precis(mRX,2)

```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
bC	-0.94	0.05	-1.02	-0.87	1494	1
bI	-0.71	0.04	-0.77	-0.65	1853	1
bA	-0.69	0.04	-0.76	-0.63	1321	1
alpha[1]	-2.82	0.05	-2.89	-2.74	929	1
alpha[2]	-2.14	0.04	-2.20	-2.07	955	1
alpha[3]	-1.56	0.04	-1.62	-1.49	1129	1
alpha[4]	-0.54	0.04	-0.59	-0.48	1361	1
alpha[5]	0.13	0.04	0.07	0.19	1545	1
alpha[6]	1.04	0.04	0.97	1.10	1810	1

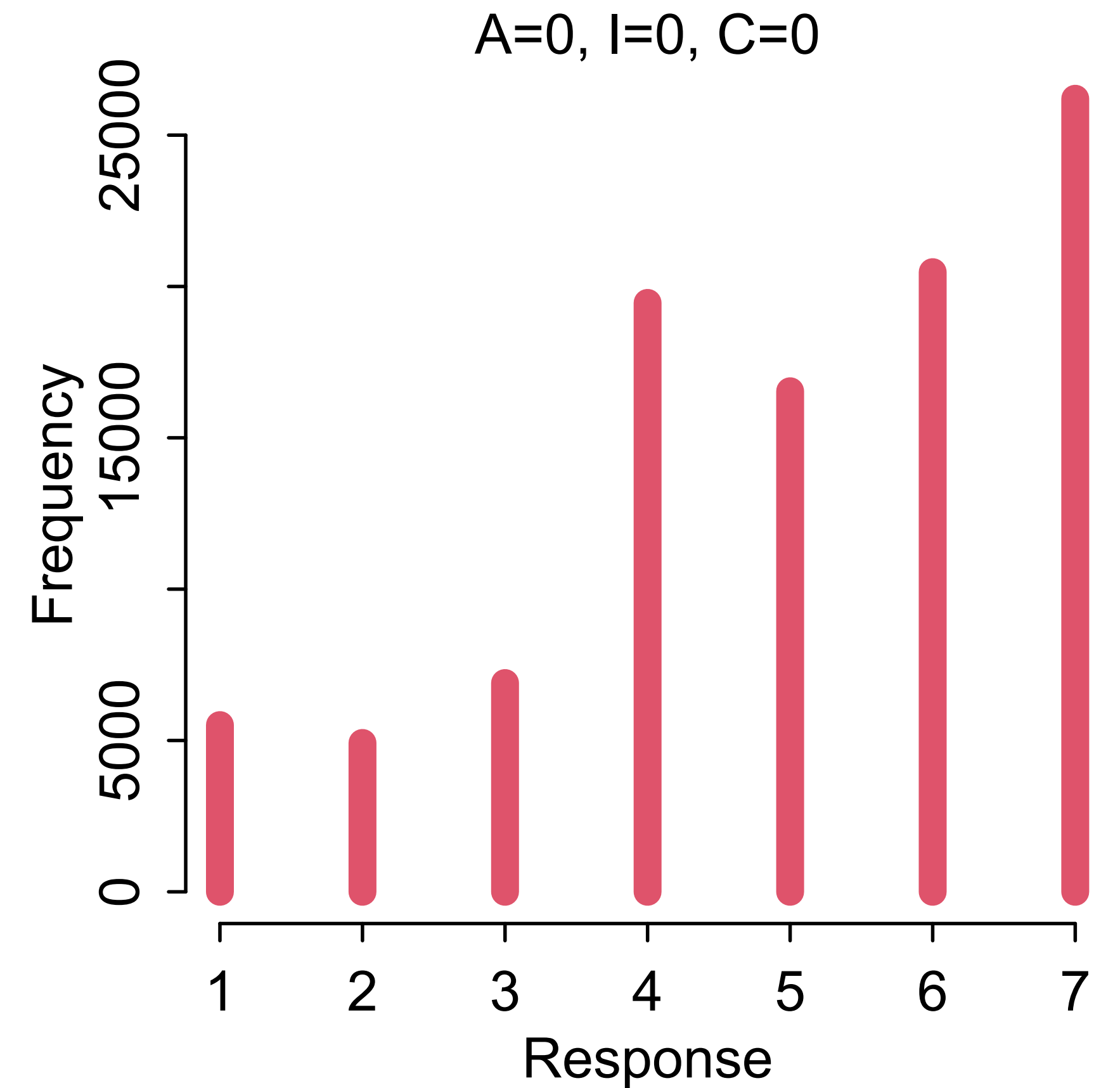
```

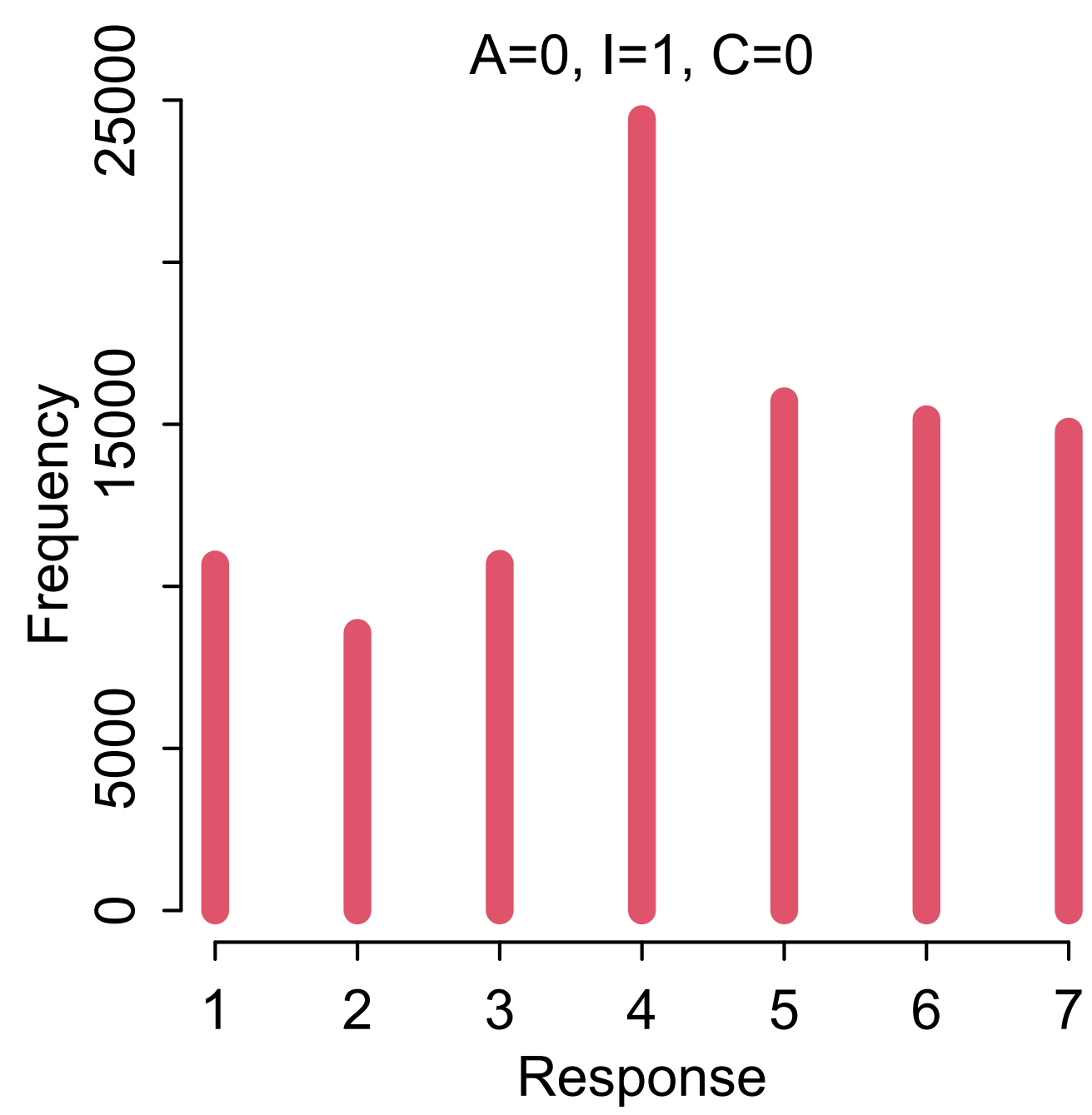
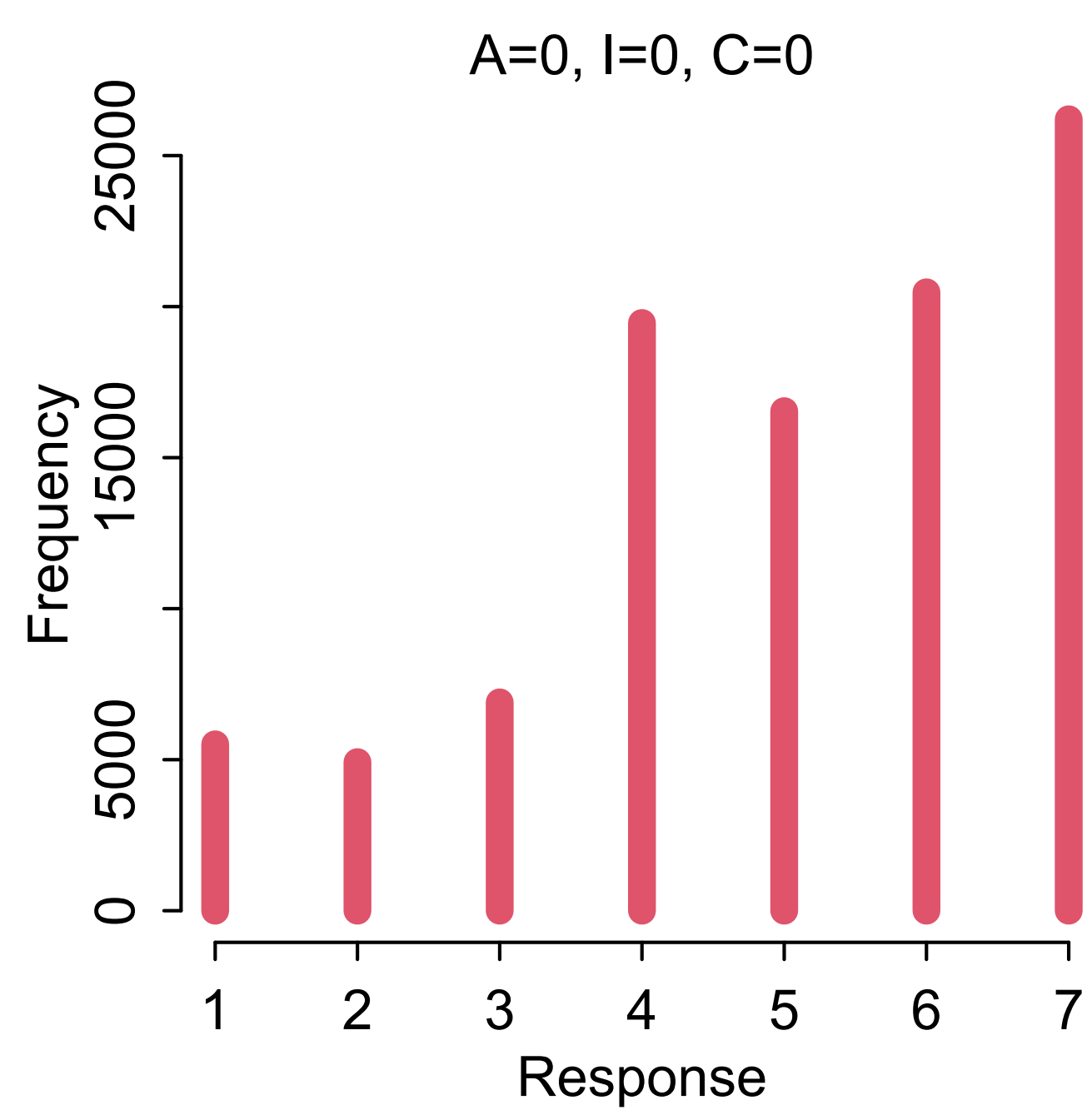
# plot predictive distributions for each treatment

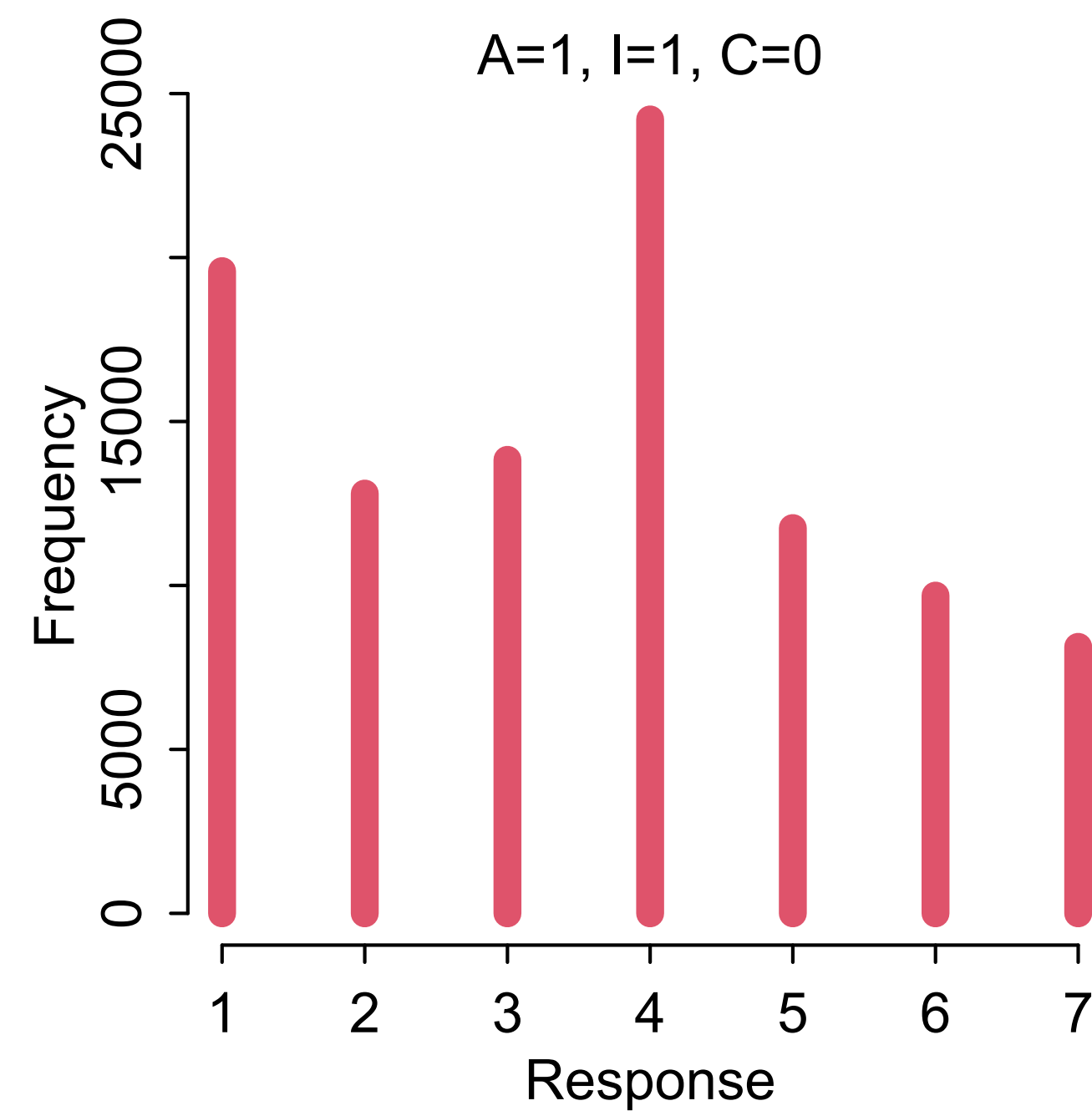
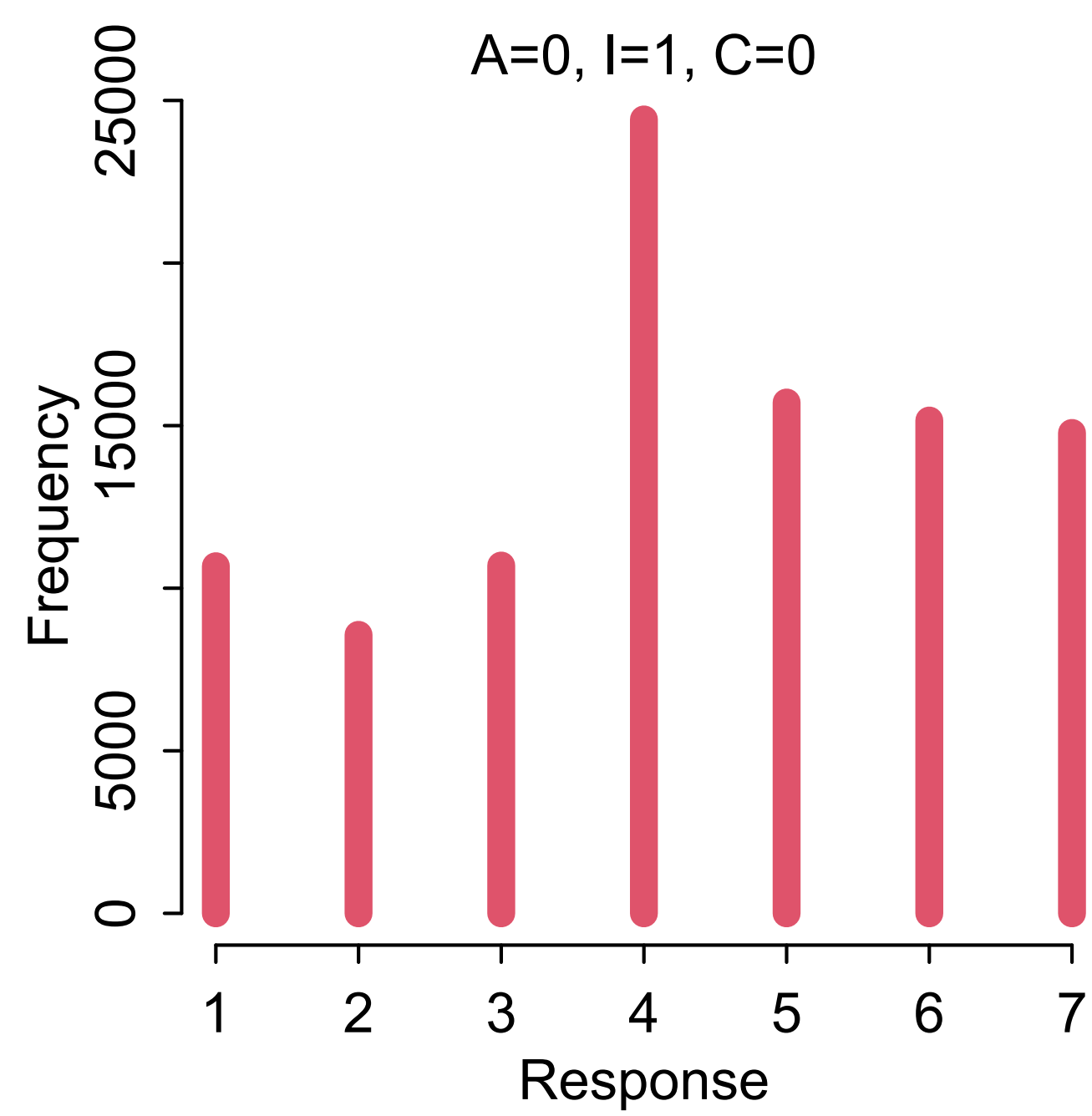
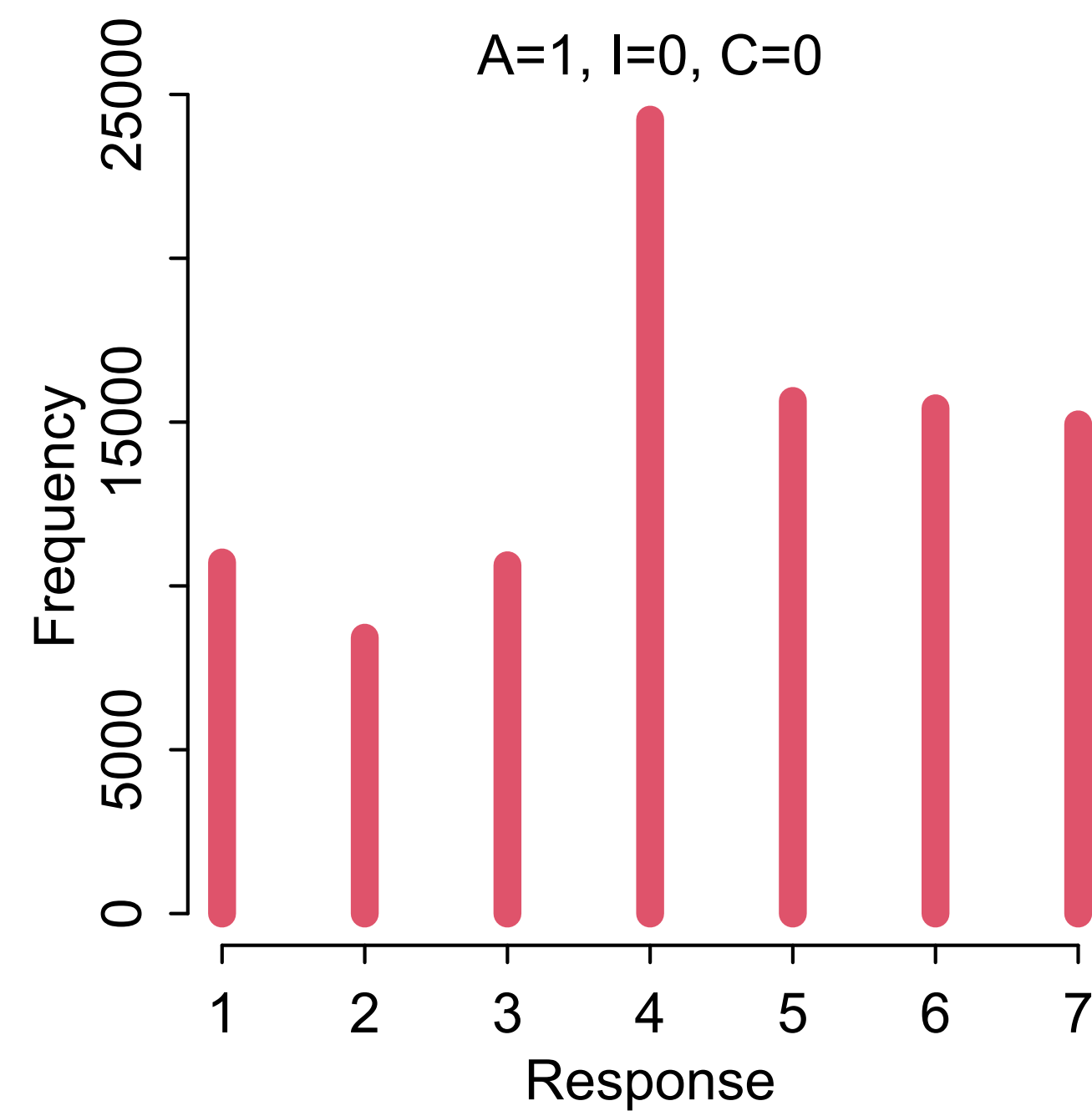
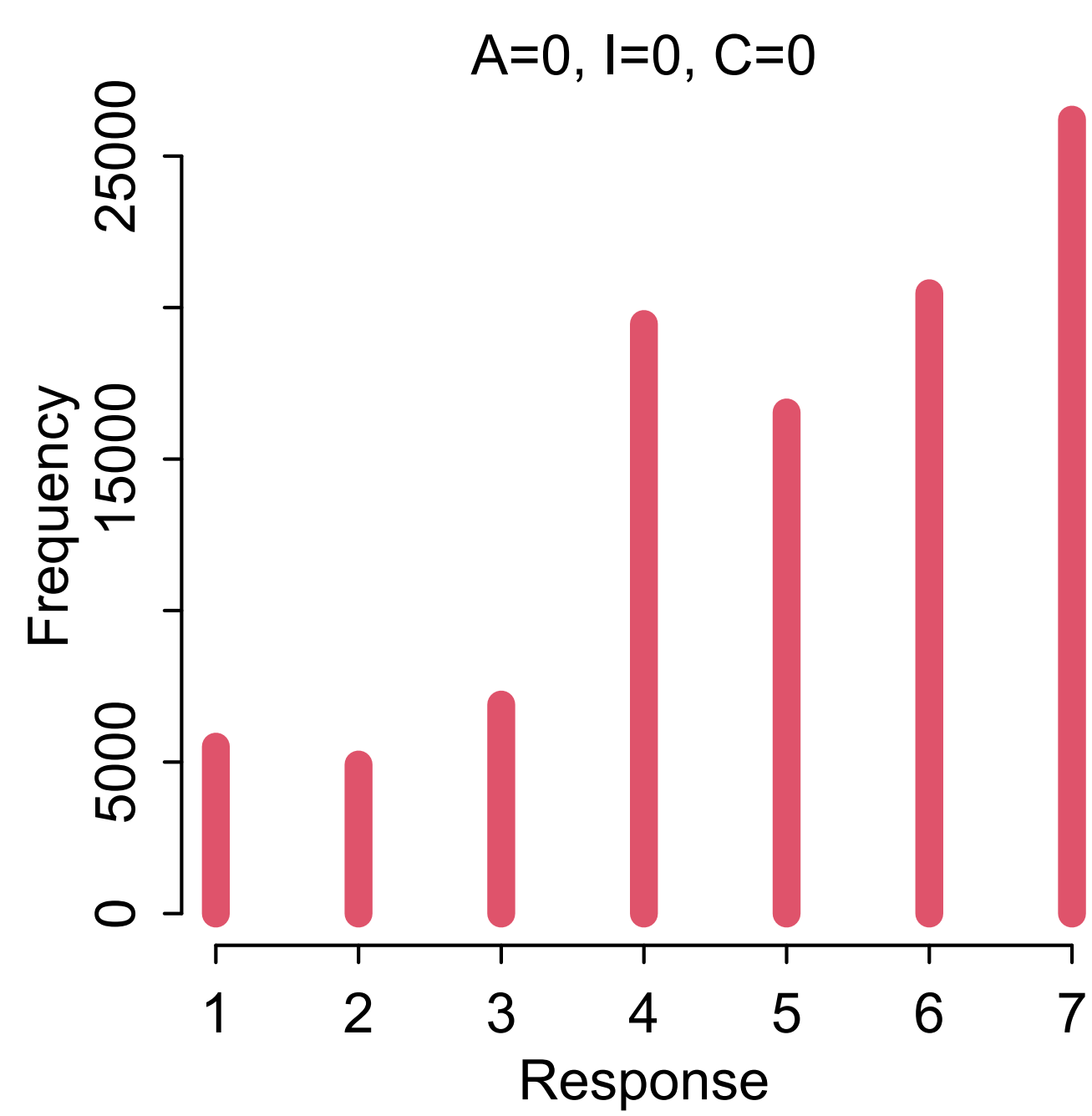
vals <- c(0,0,0)
Rsim <- mcreplicate( 100 ,
sim(mRX,data=list(A=vals[1],I=vals[2],C=vals[3])) ,
mc.cores=6 )

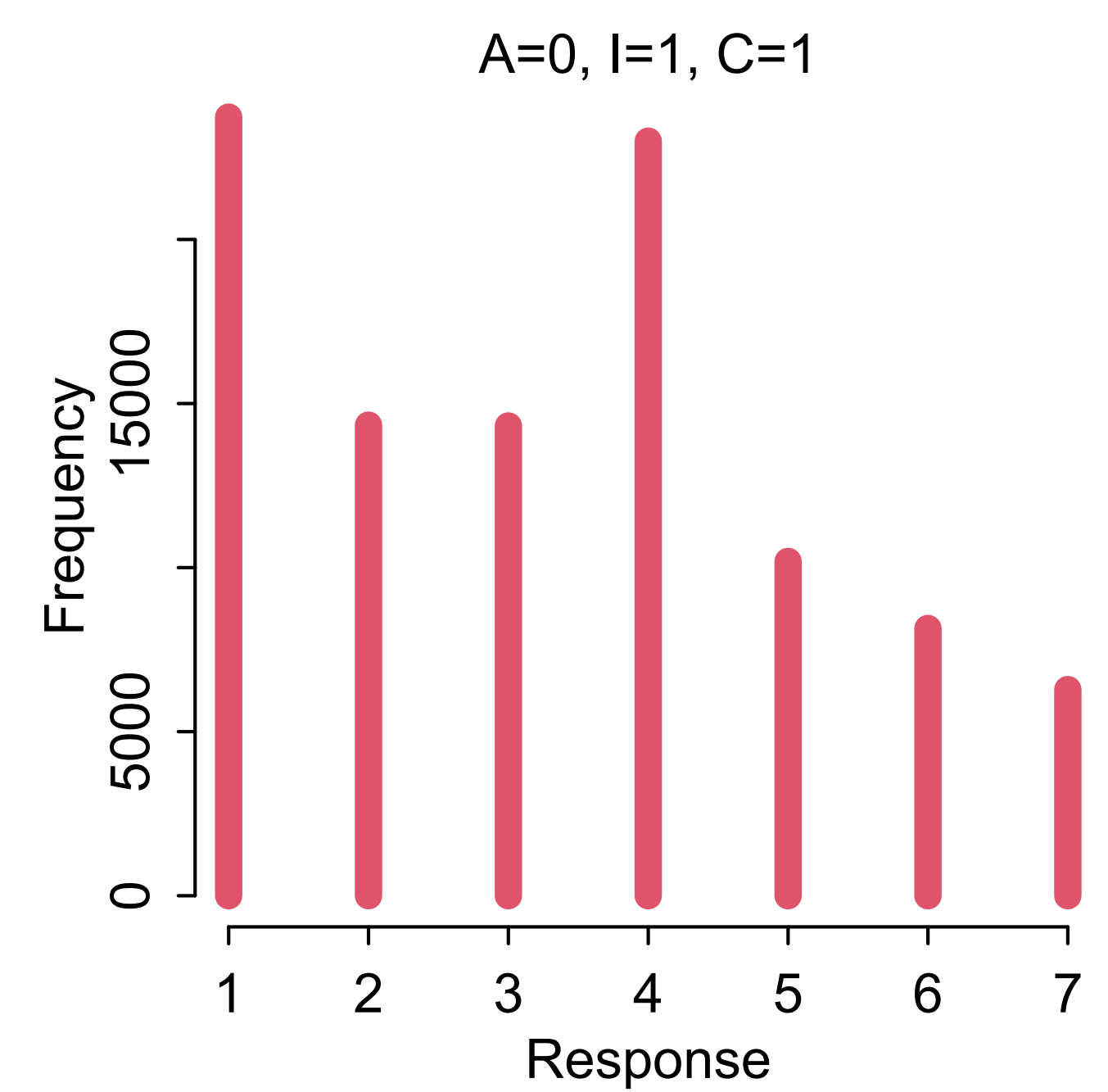
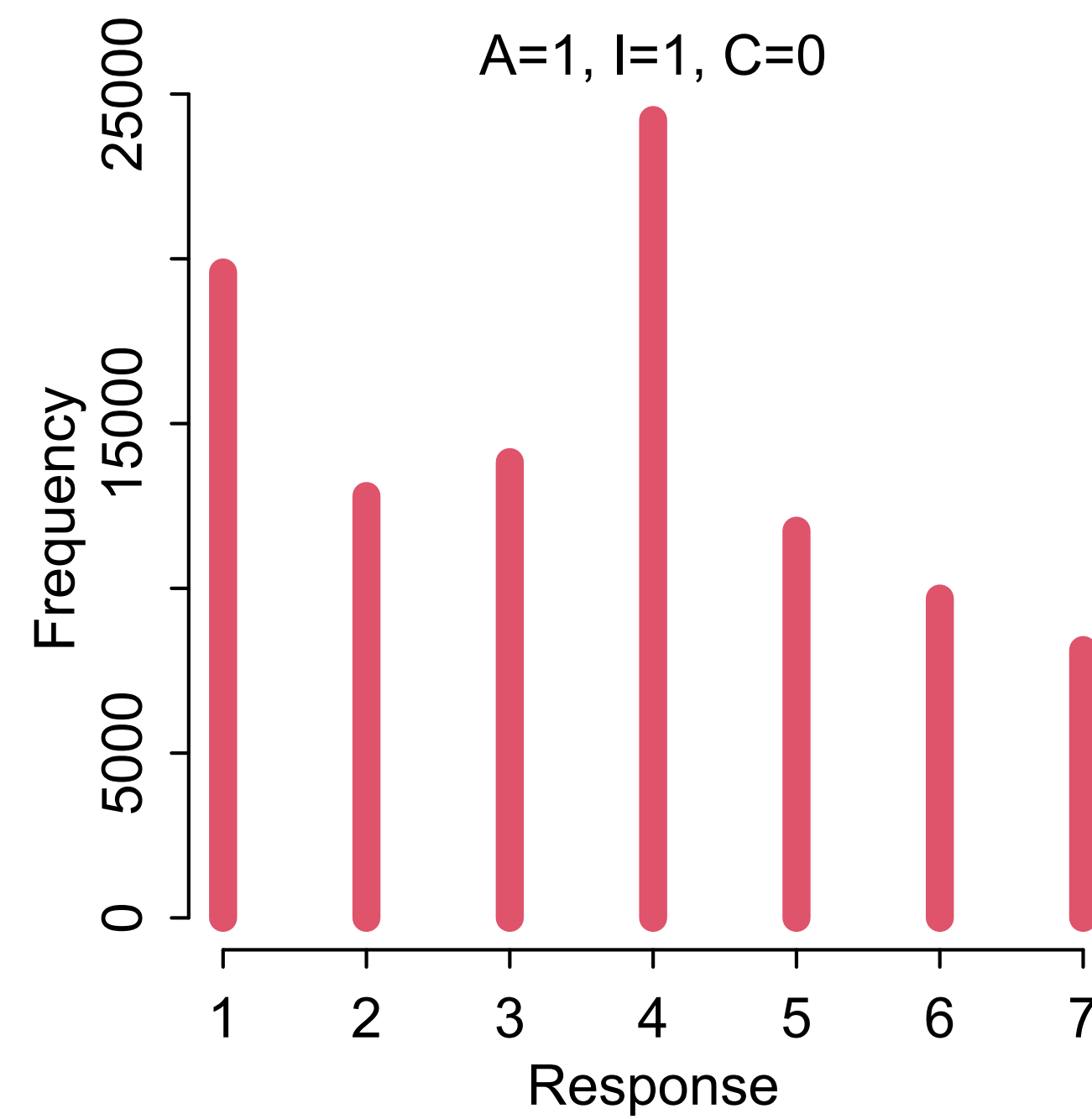
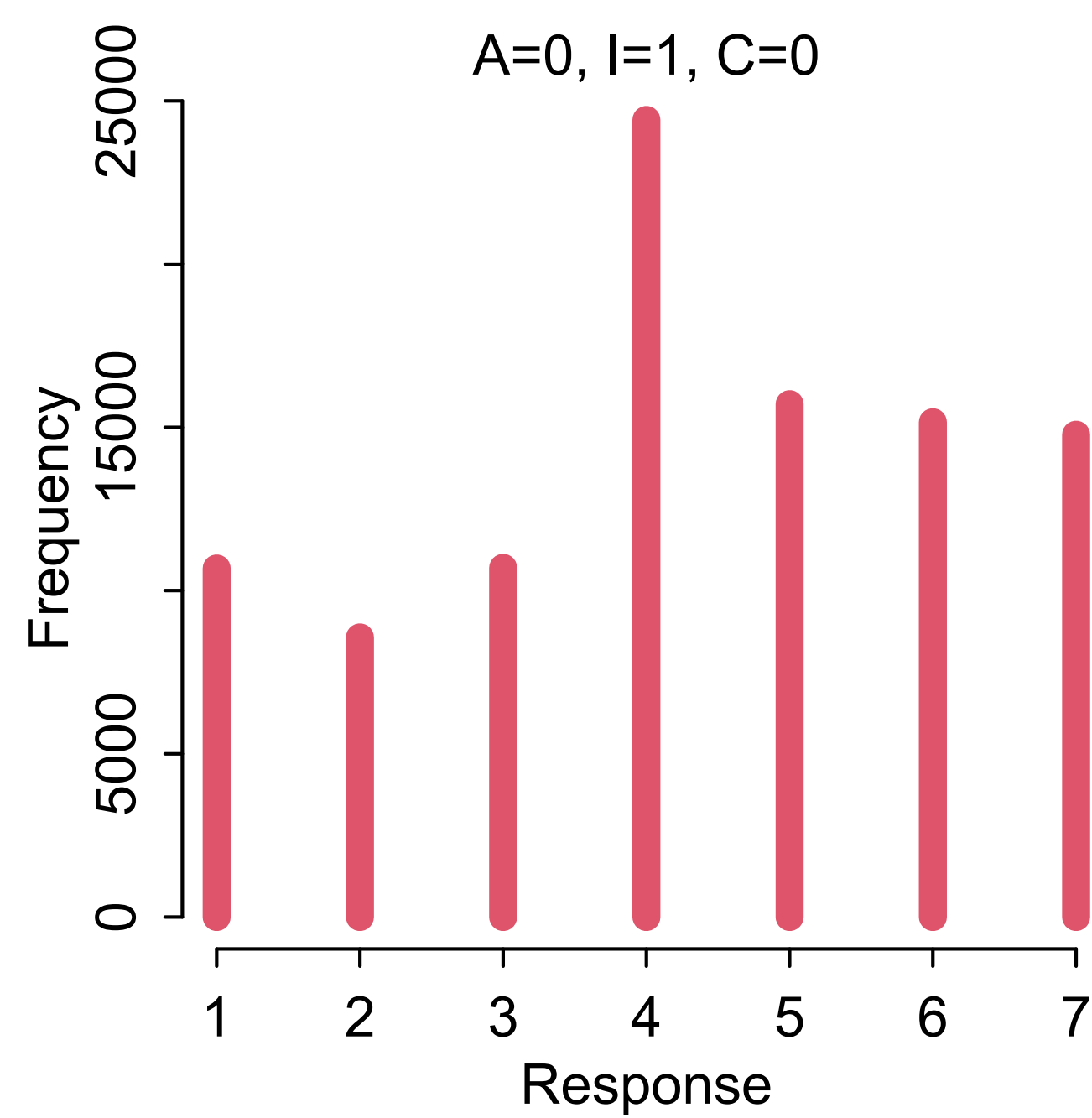
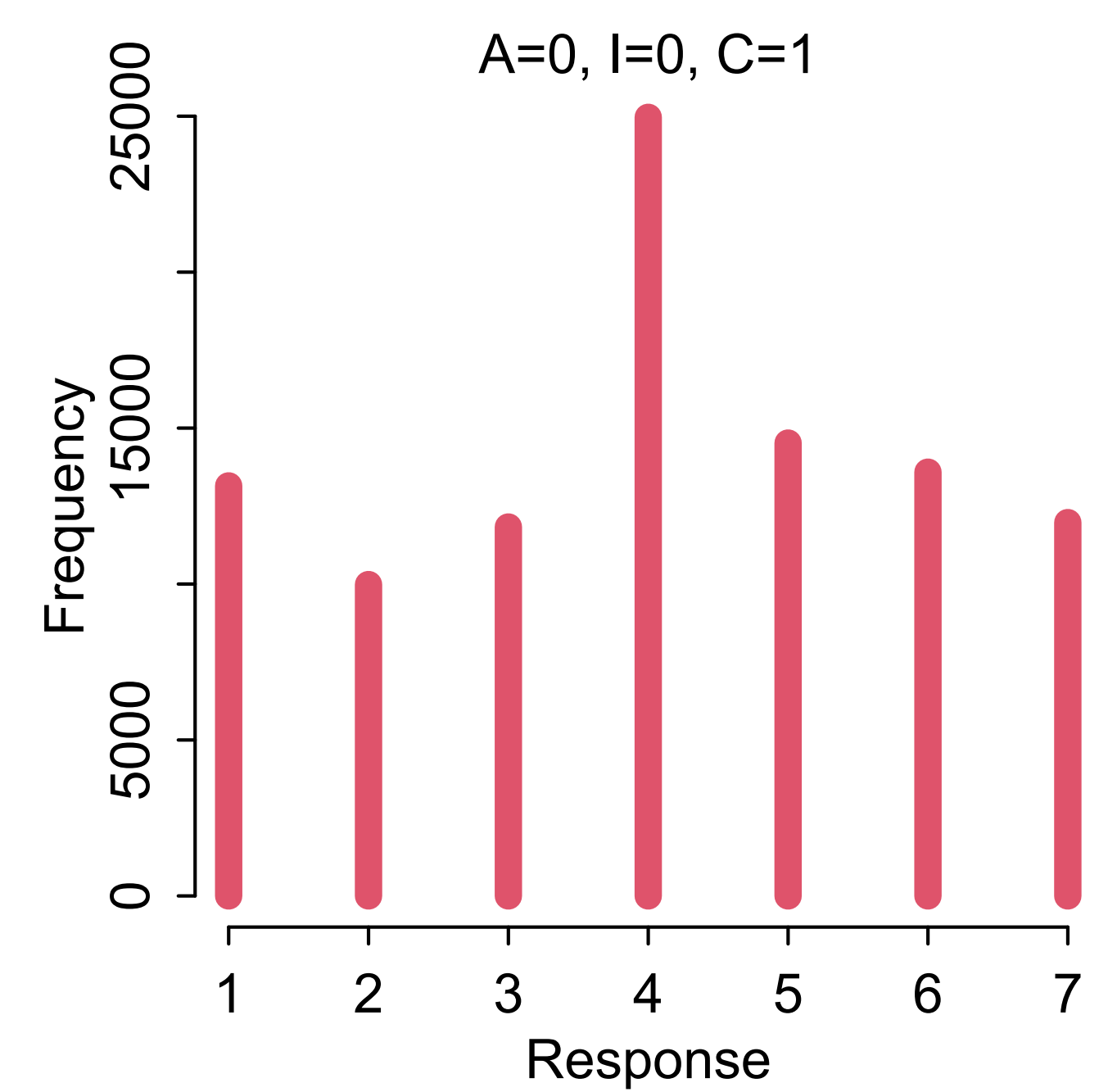
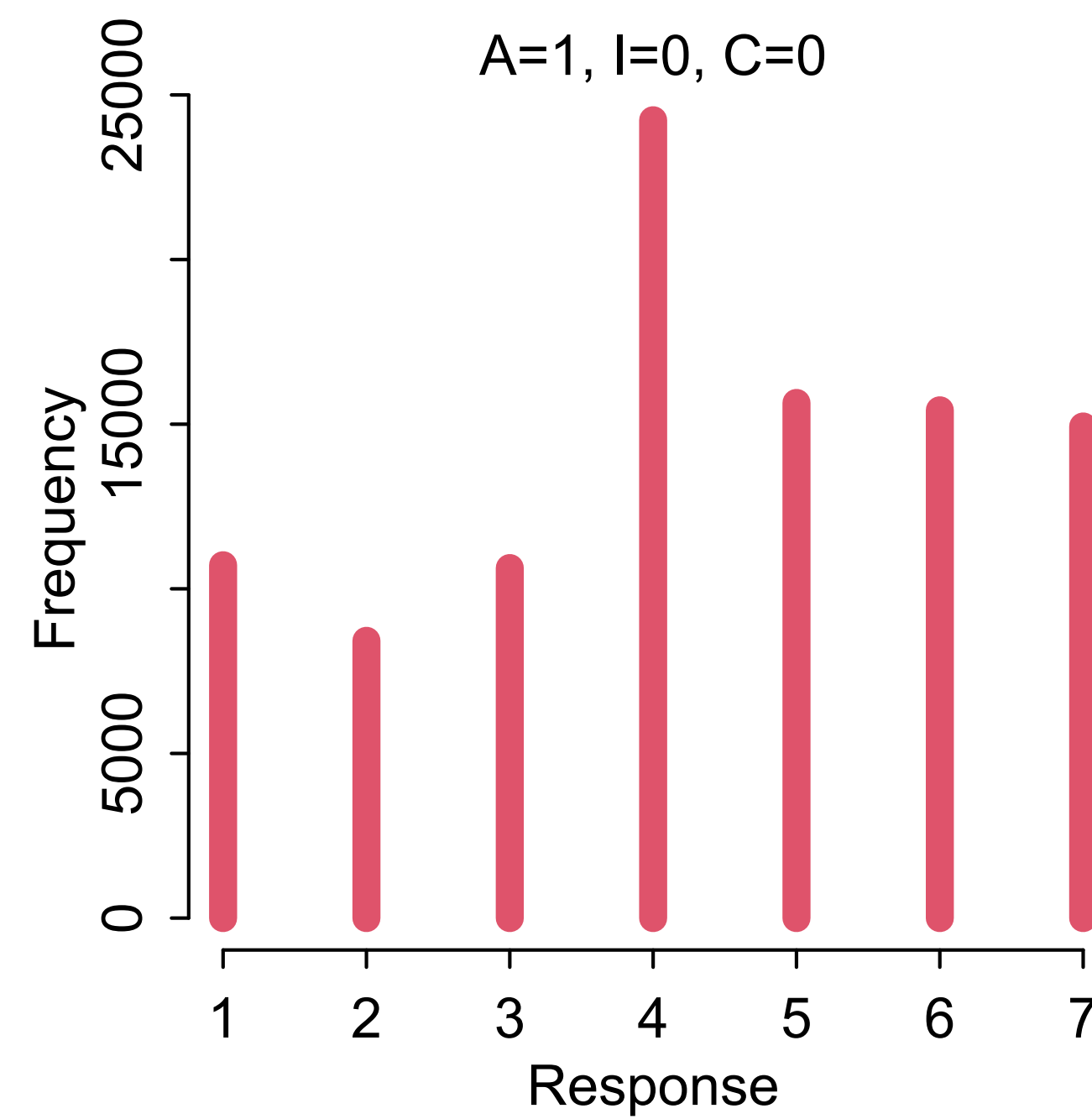
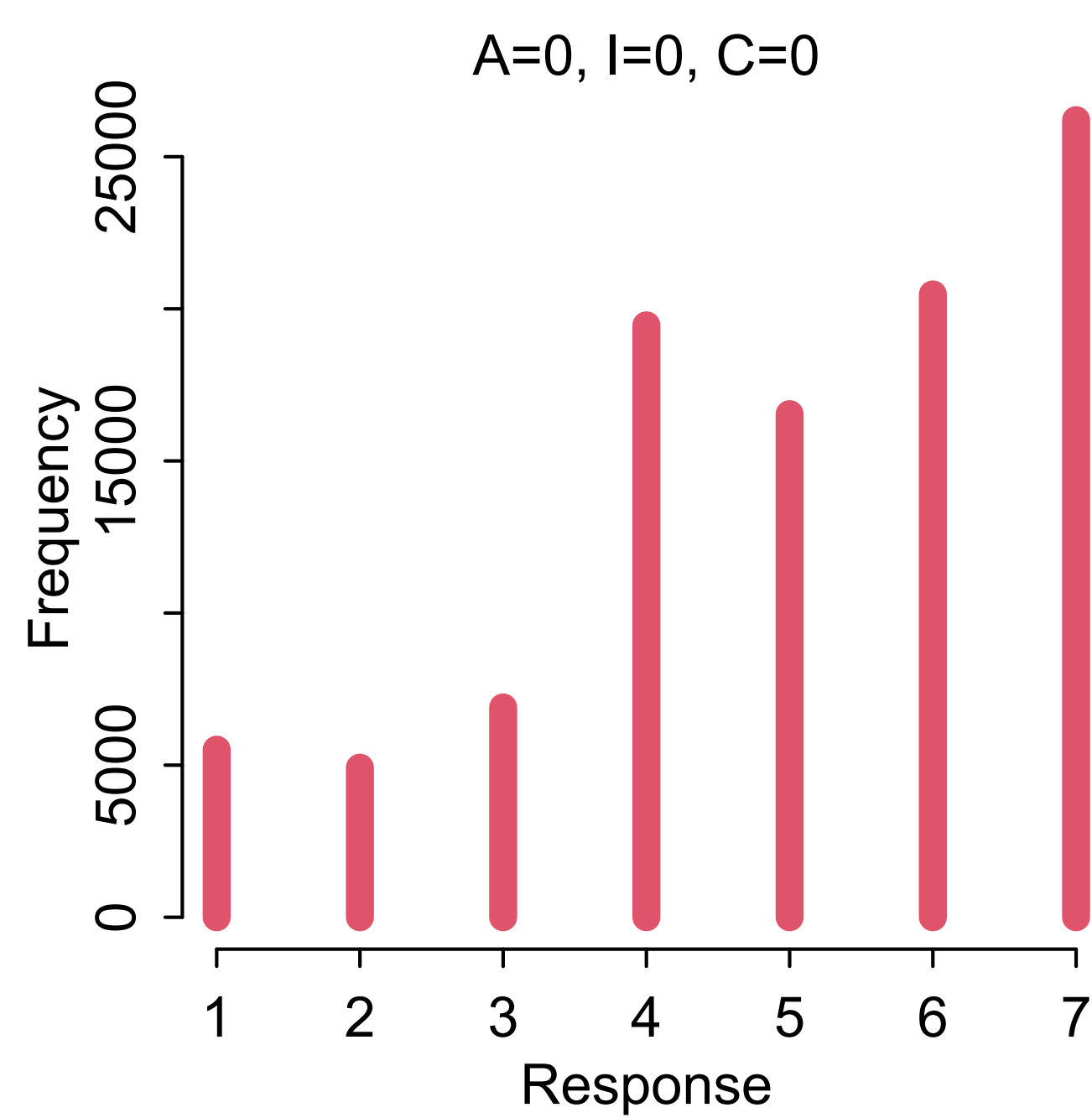
simplehist(as.vector(Rsim),lwd=8,col=2,xlab="Response")
mtext(concat("A=",vals[1],", I=",vals[2],",
C=",vals[3]))

```



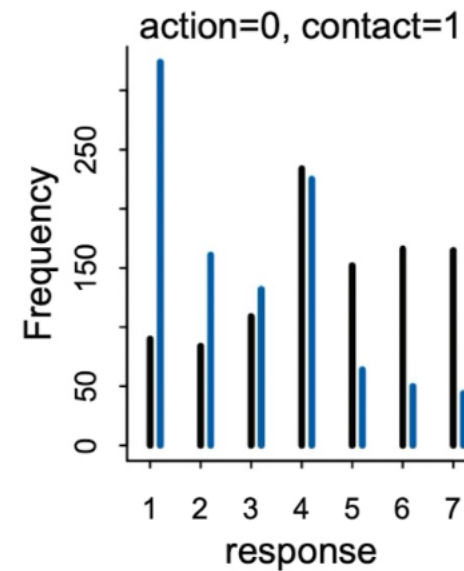
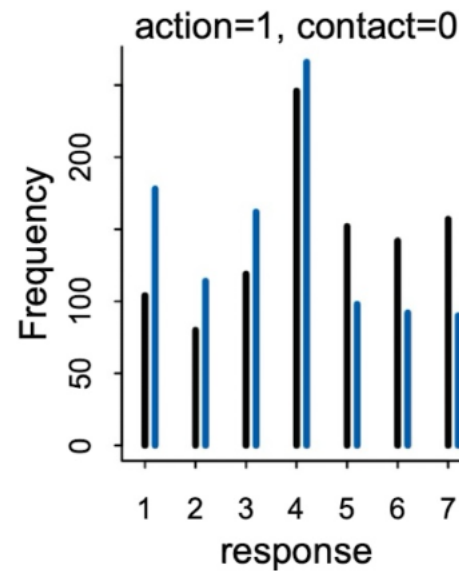
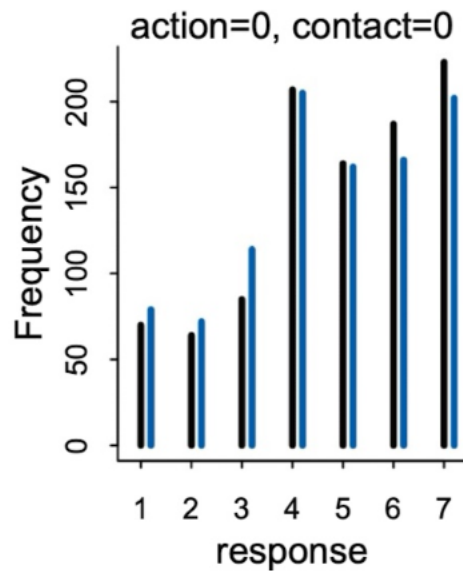






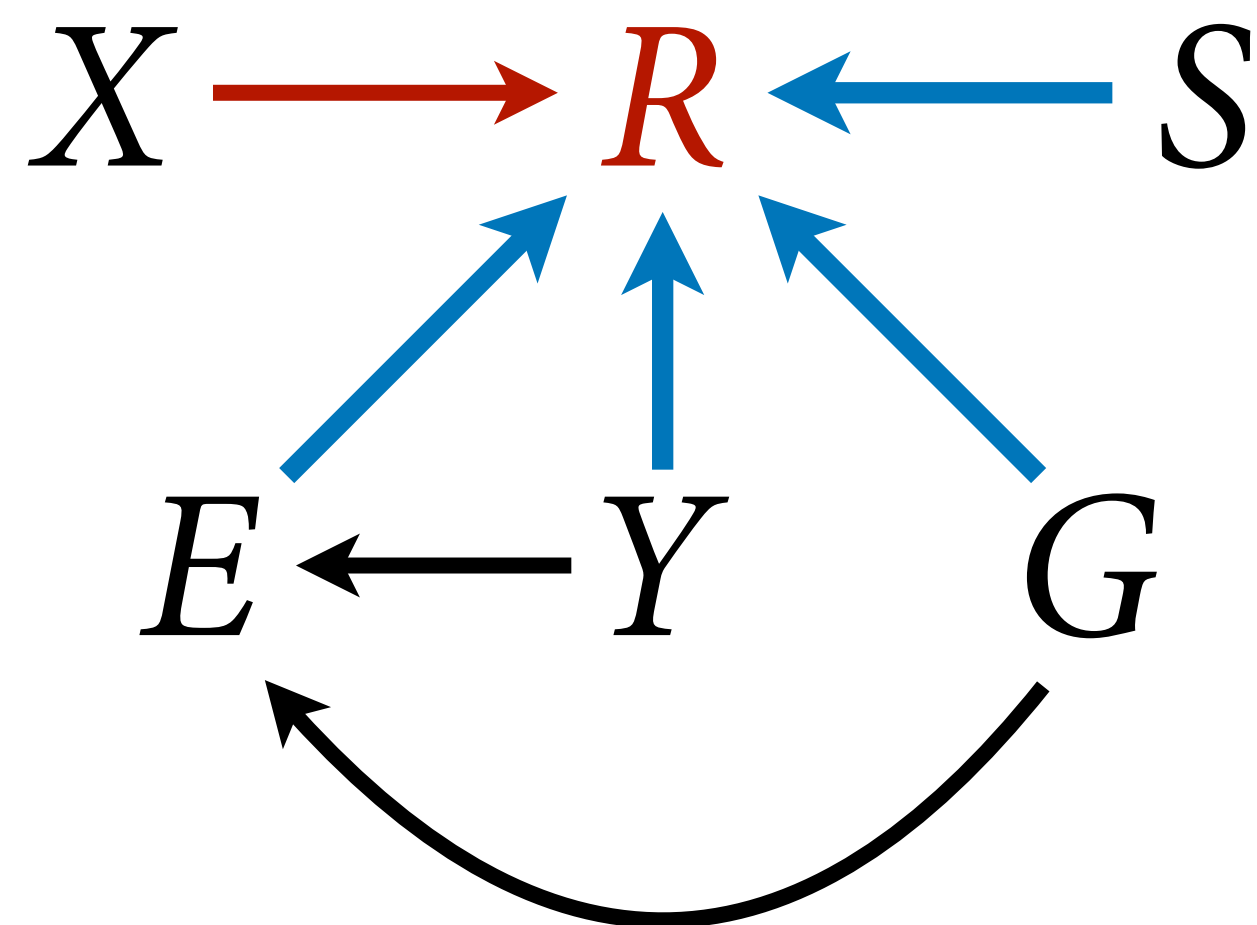
Intention = 1

Intention = 0



PAUSE

What about the competing causes?



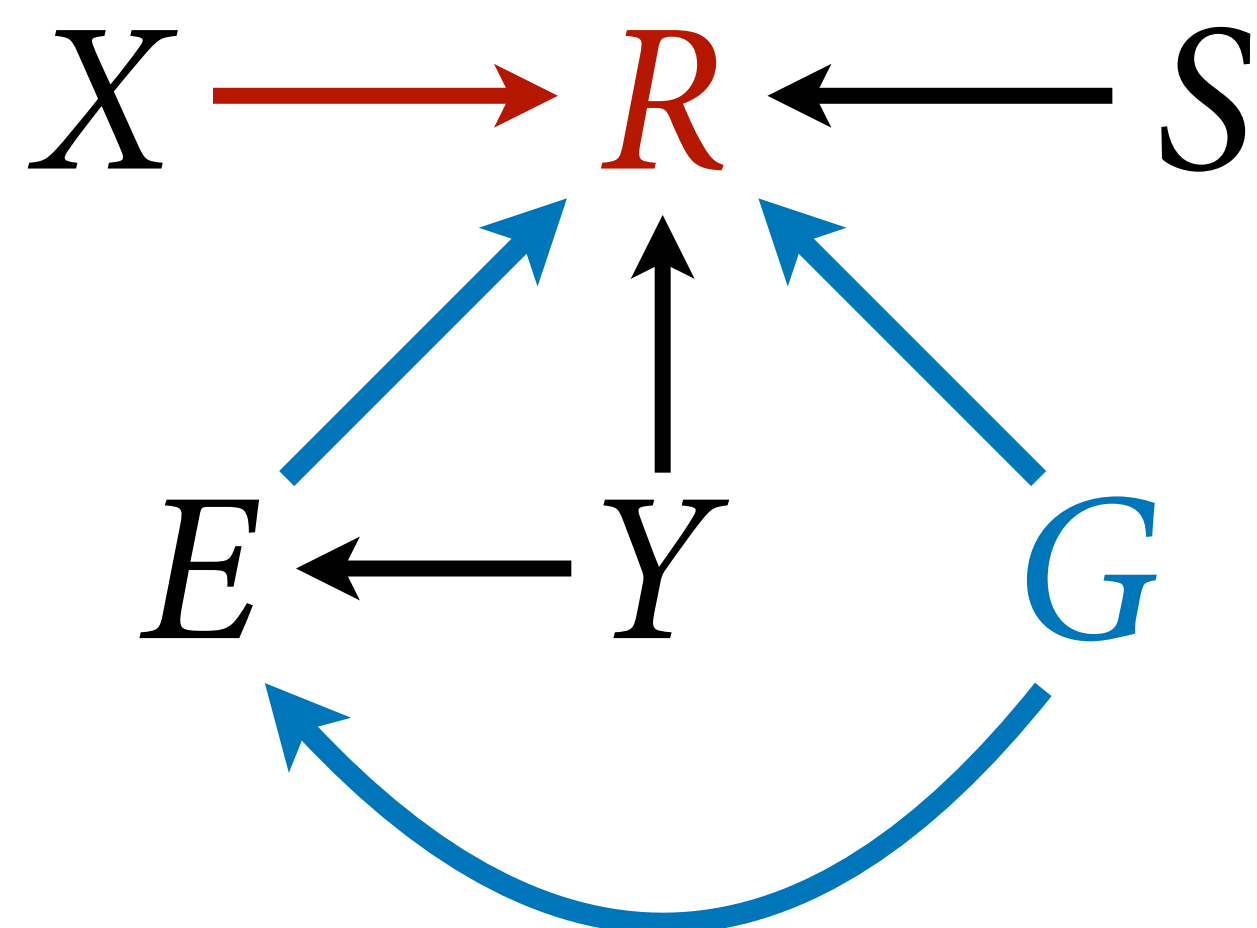
$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_A A_i + \beta_C C_i + \beta_I I_i$$

$$\beta \sim \text{Normal}(0, 0.5)$$

$$\alpha_j \sim \text{Normal}(0, 1)$$

Total effect of **gender**:



$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{A,G[i]}A_i + \beta_{C,G[i]}C_i + \beta_{I,G[i]}I_i$$

$$\beta_{_} \sim \text{Normal}(0,0.5)$$

$$\alpha_j \sim \text{Normal}(0,1)$$


```

# total effect of gender
dat$G <- ifelse(d$male==1,2,1)
mRXG <- ulam(
  alist(
    R ~ dordlogit(phi,alpha),
    phi <- bA[G]*A + bI[G]*I + bC[G]*C,
    bA[G] ~ normal(0,0.5),
    bI[G] ~ normal(0,0.5),
    bC[G] ~ normal(0,0.5),
    alpha ~ normal(0,1)
  ) , data=dat , chains=4 , cores=4 )

```

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{A,G[i]}A_i + \beta_{C,G[i]}C_i + \beta_{I,G[i]}I_i$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\alpha_j \sim \text{Normal}(0,1)$$

```

# total effect of gender
dat$G <- ifelse(d$male==1,2,1)
mRXG <- ulam(
  alist(
    R ~ dordlogit(phi,alpha),
    phi <- bA[G]*A + bI[G]*I + bC[G]*C,
    bA[G] ~ normal(0,0.5),
    bI[G] ~ normal(0,0.5),
    bC[G] ~ normal(0,0.5),
    alpha ~ normal(0,1)
  ) , data=dat , chains=4 , cores=4 )

```

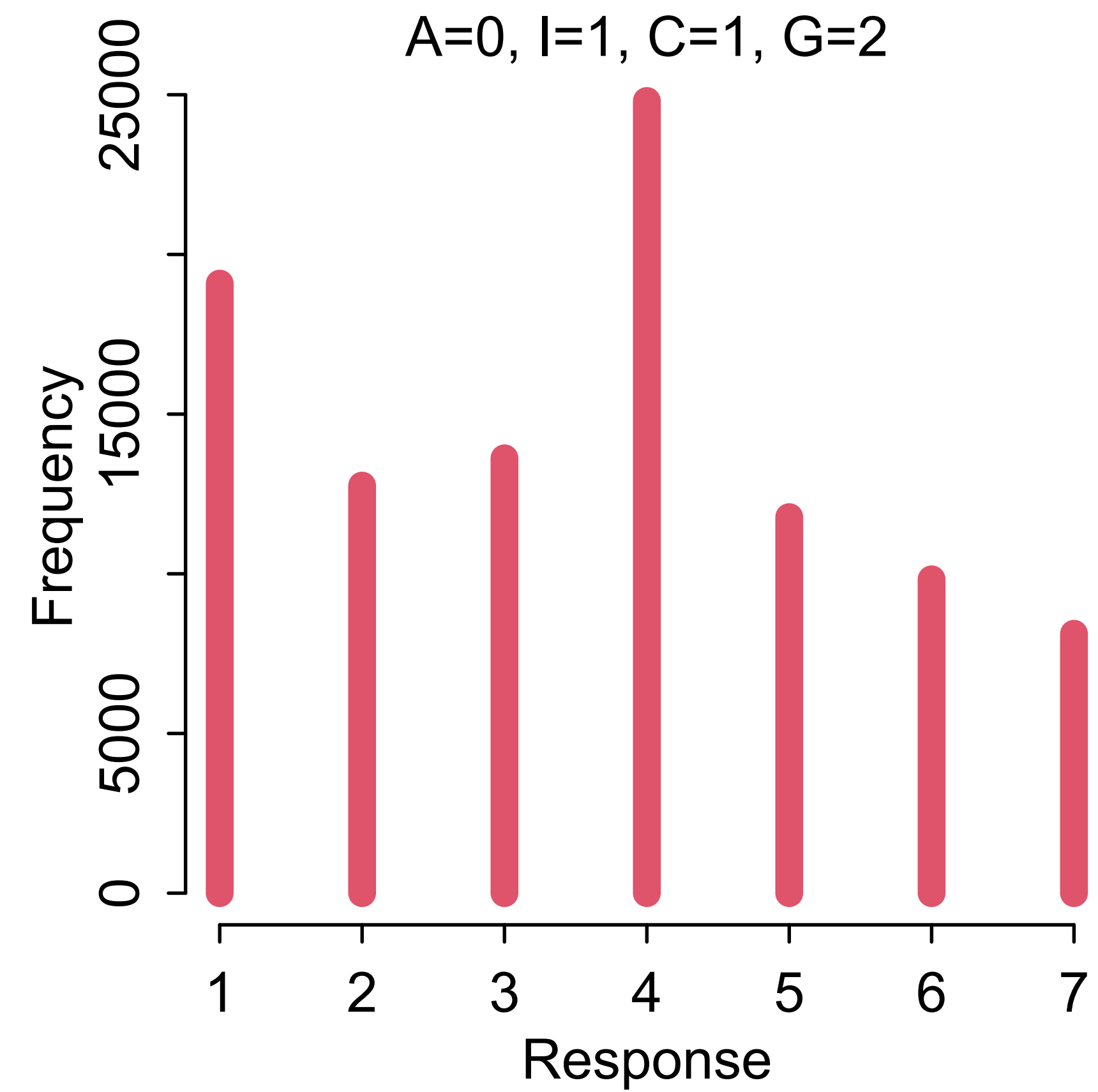
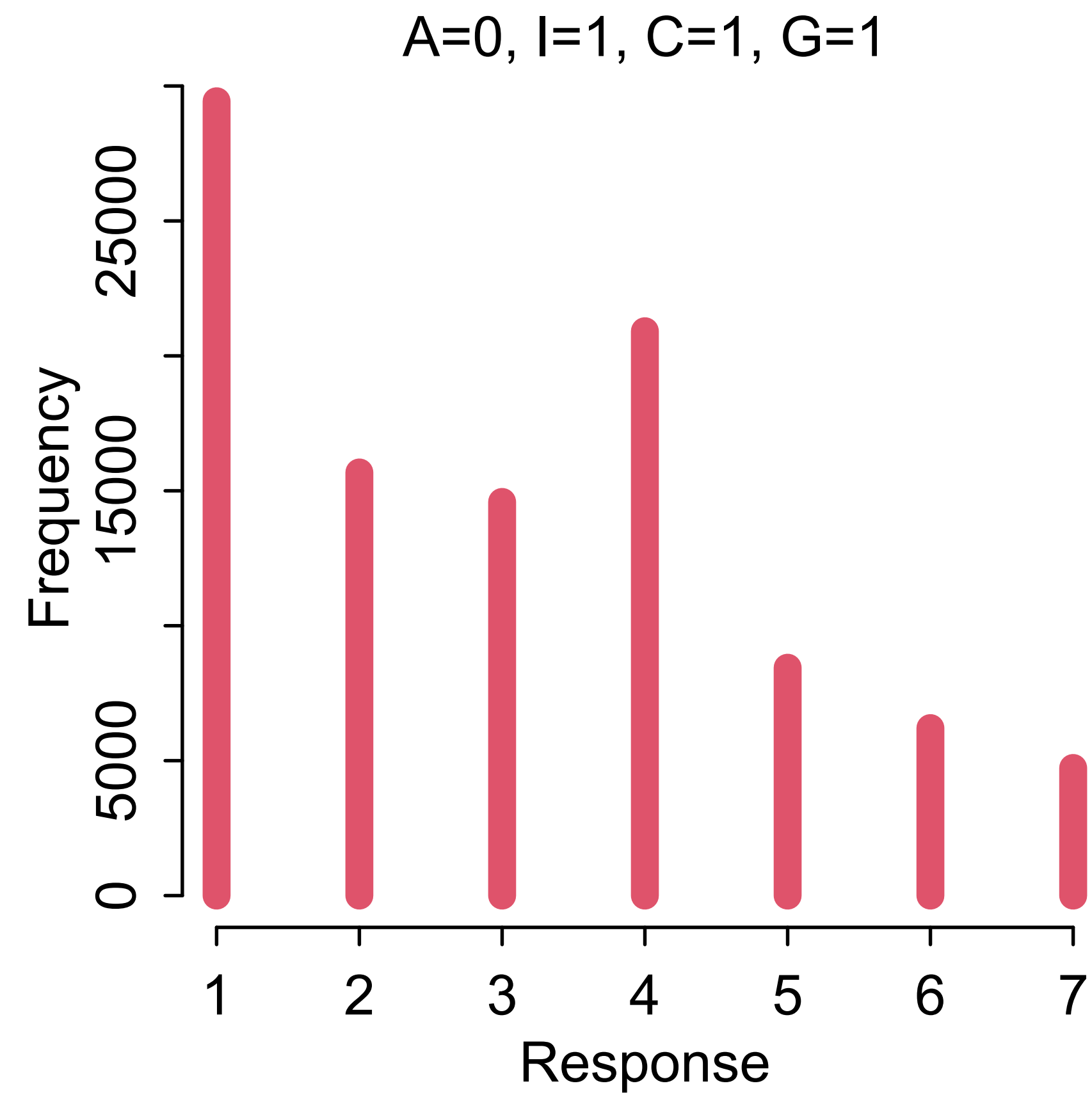
```

> precis(mRXG,2)

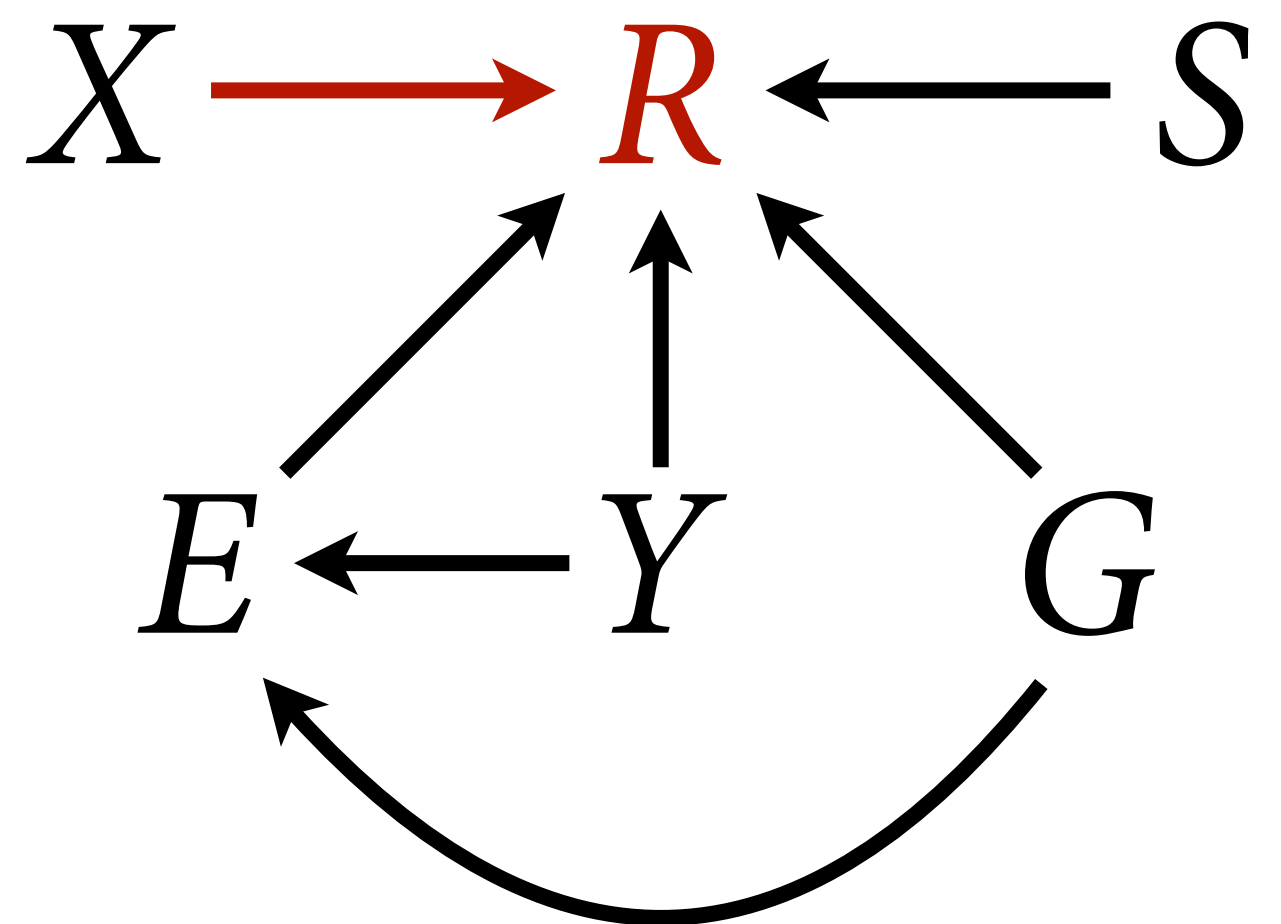
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
bA[1]	-0.88	0.05	-0.96	-0.80	1858	1.00
bA[2]	-0.53	0.05	-0.61	-0.45	1724	1.00
bI[1]	-0.90	0.05	-0.97	-0.82	2189	1.00
bI[2]	-0.55	0.05	-0.63	-0.48	2382	1.00
bC[1]	-1.06	0.07	-1.17	-0.95	2298	1.00
bC[2]	-0.84	0.06	-0.94	-0.74	2000	1.00
alpha[1]	-2.83	0.05	-2.90	-2.75	1054	1.01
alpha[2]	-2.15	0.04	-2.21	-2.08	1104	1.00
alpha[3]	-1.56	0.04	-1.62	-1.50	1076	1.00
alpha[4]	-0.53	0.04	-0.59	-0.47	1080	1.00
alpha[5]	0.14	0.04	0.09	0.20	1216	1.00
alpha[6]	1.06	0.04	1.00	1.12	1532	1.00

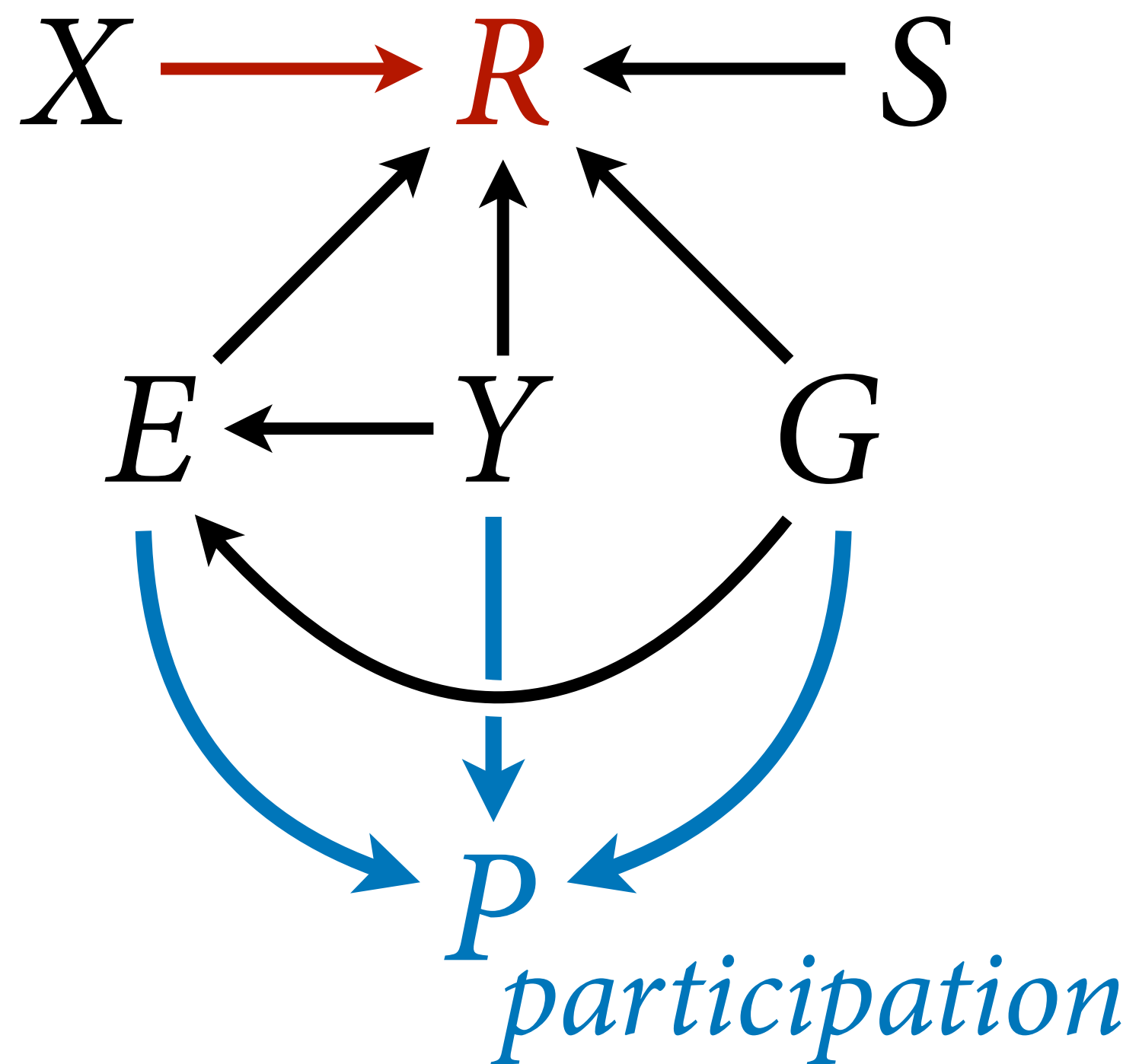
$G[i], i$



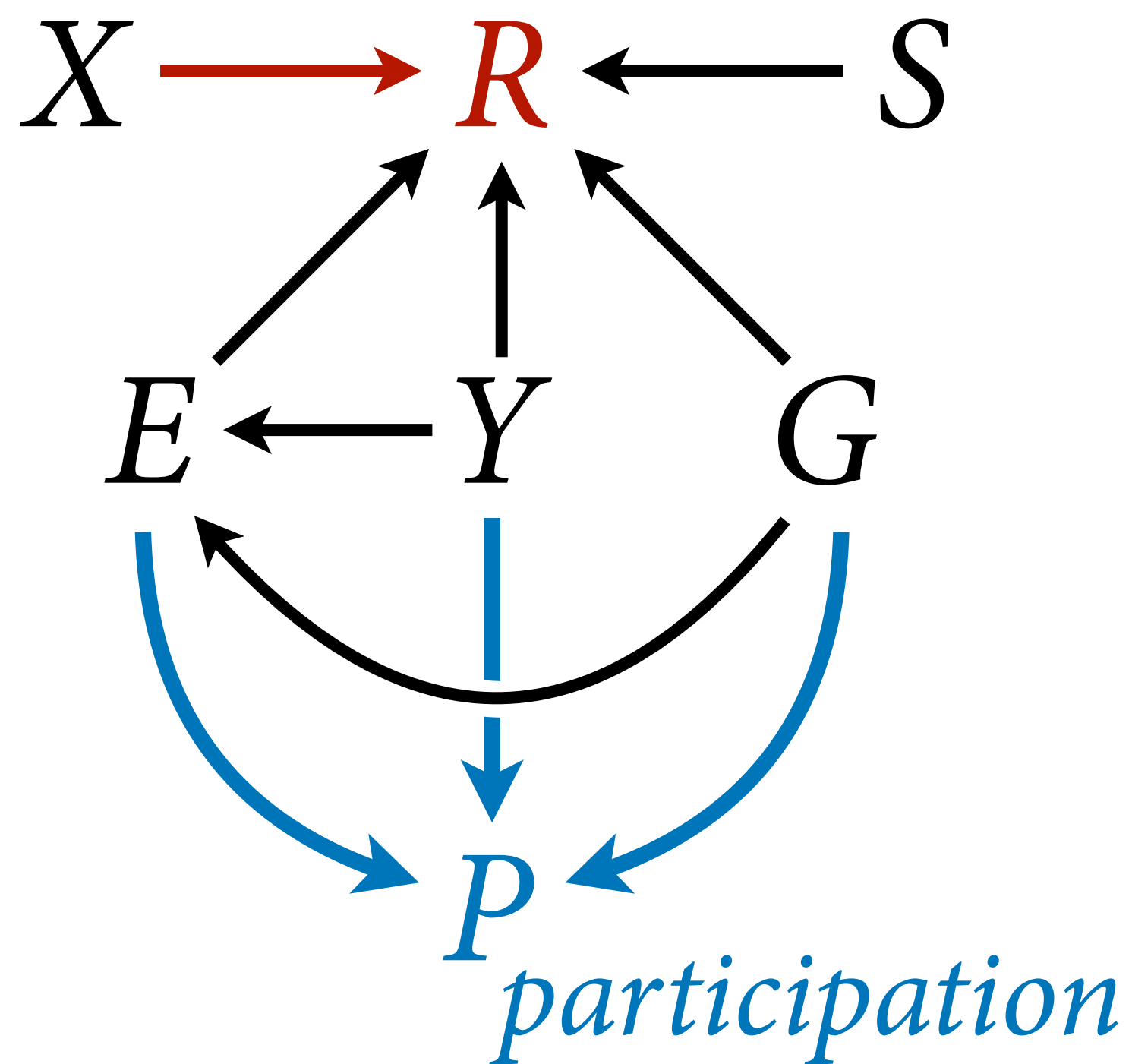
Hang on! This is a voluntary sample



Hang on! This is a **voluntary** sample



Hang on! This is a **voluntary** sample



Conditioning on P makes E, Y, G covary in sample

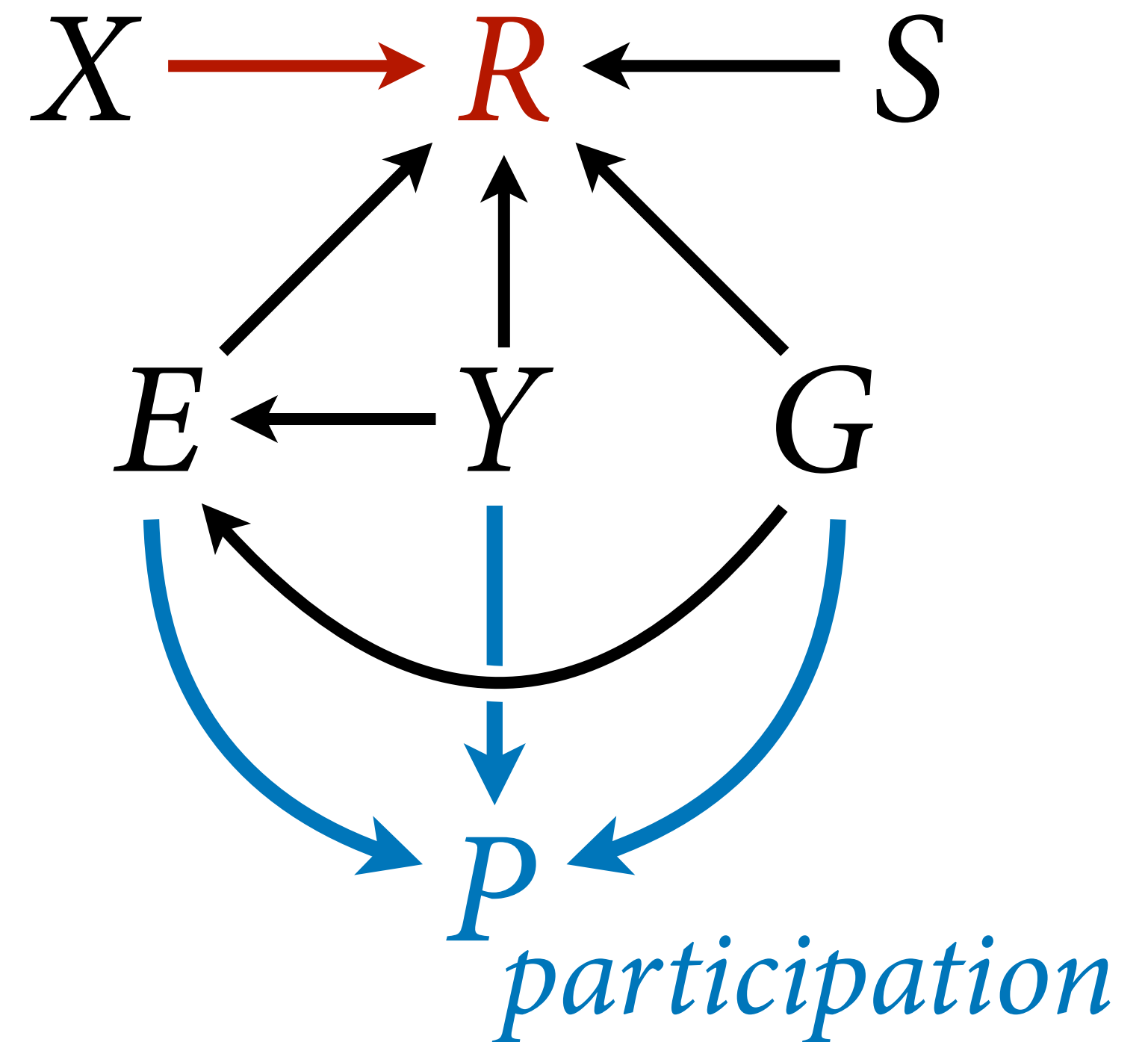
Endogenous selection

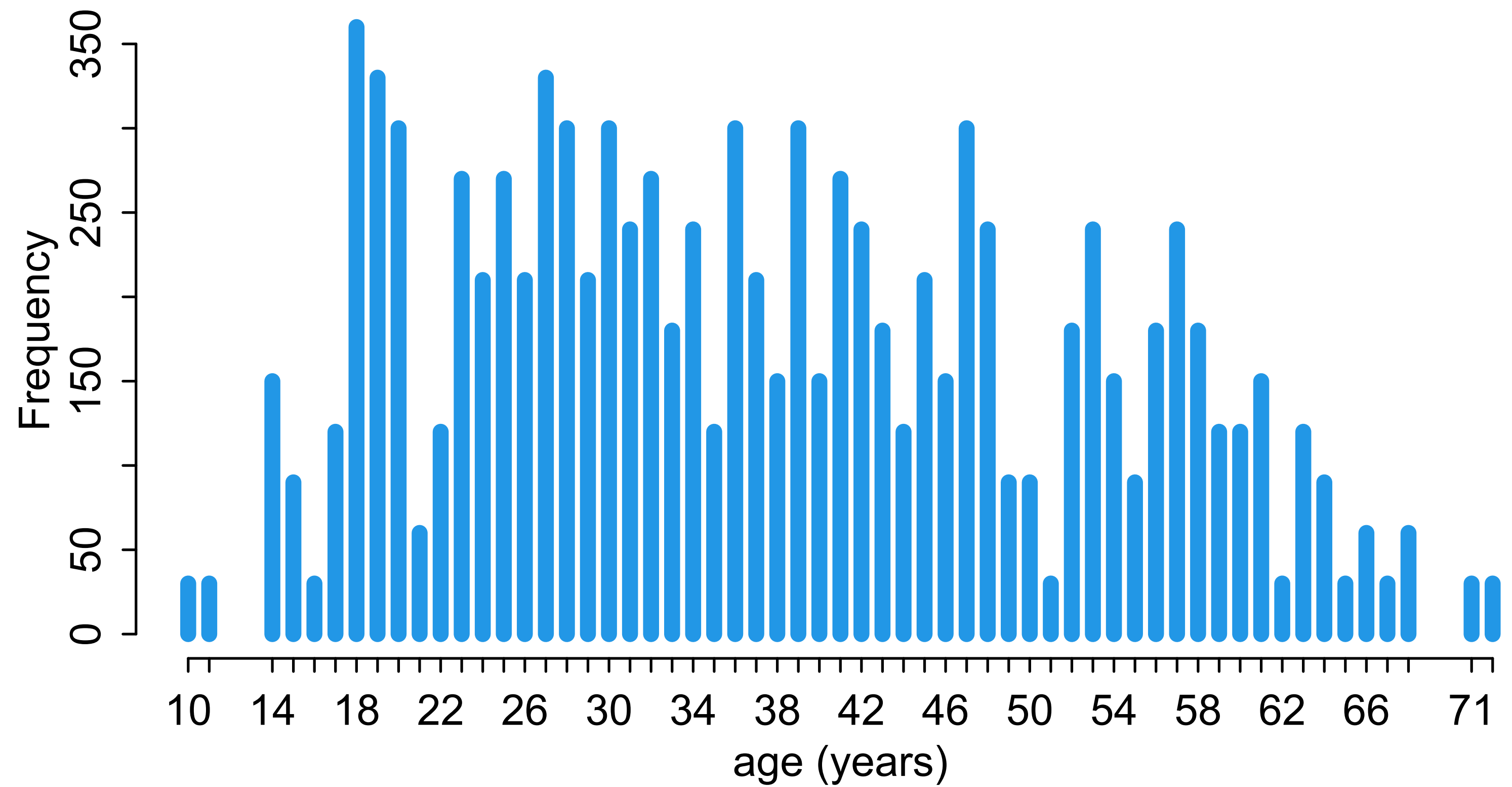
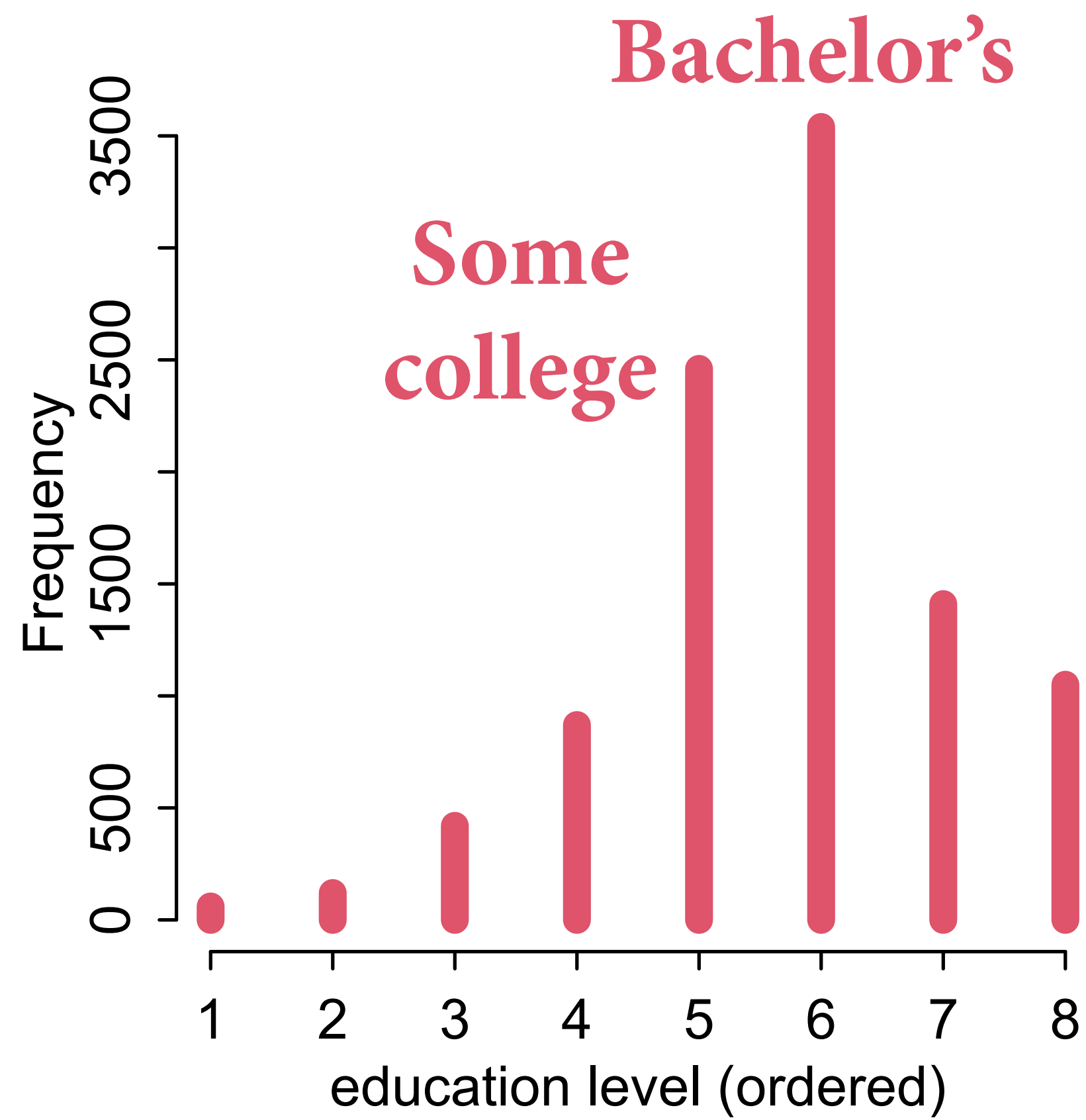
Sample is selected on a collider

Induces misleading associations among variables

Not possible here to estimate total effect of G , BUT can get direct effect

Need to stratify by E and Y and G





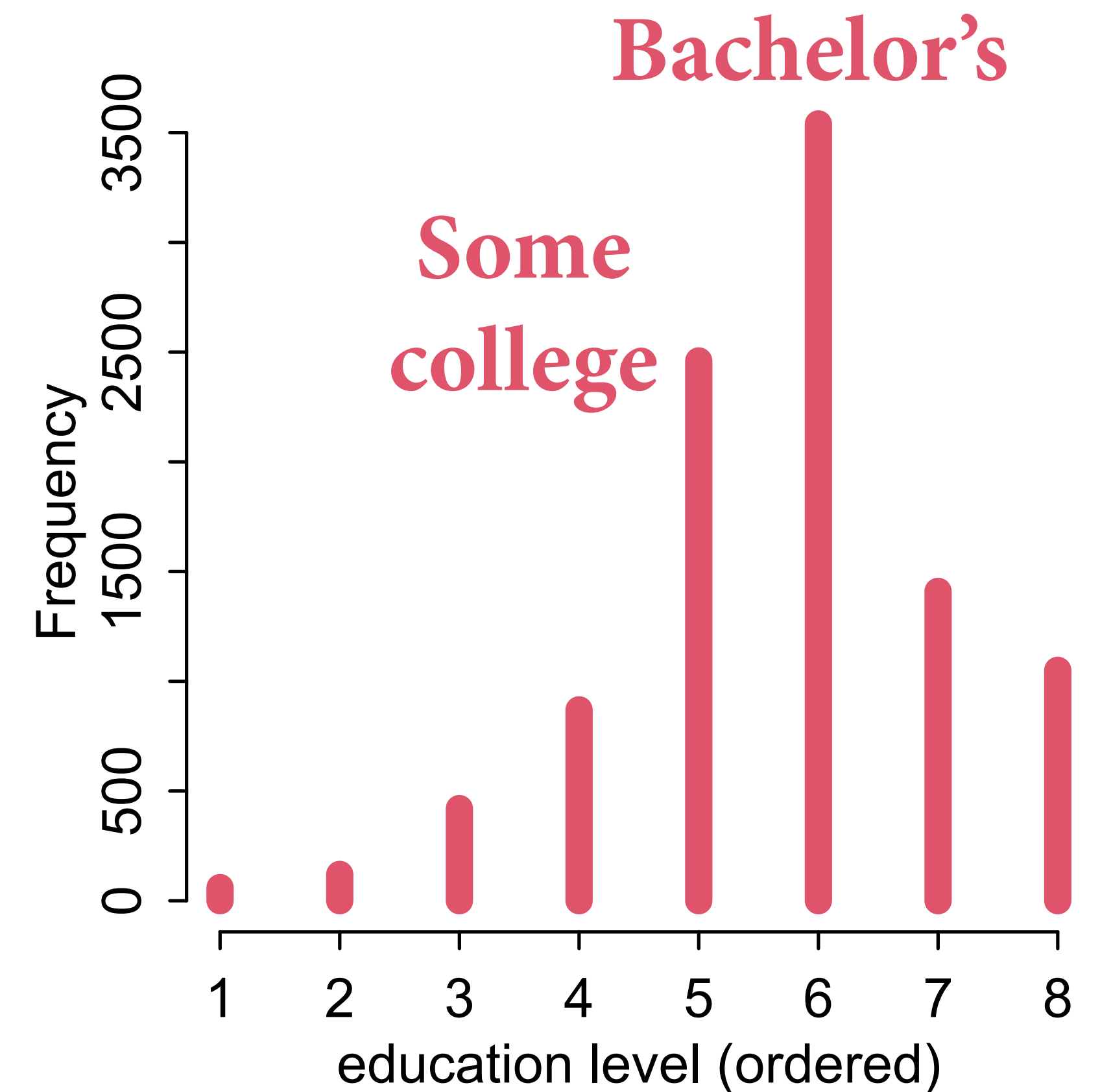
Ordered monotonic predictors

Education is an ordered category

Unlikely that each level has same effect

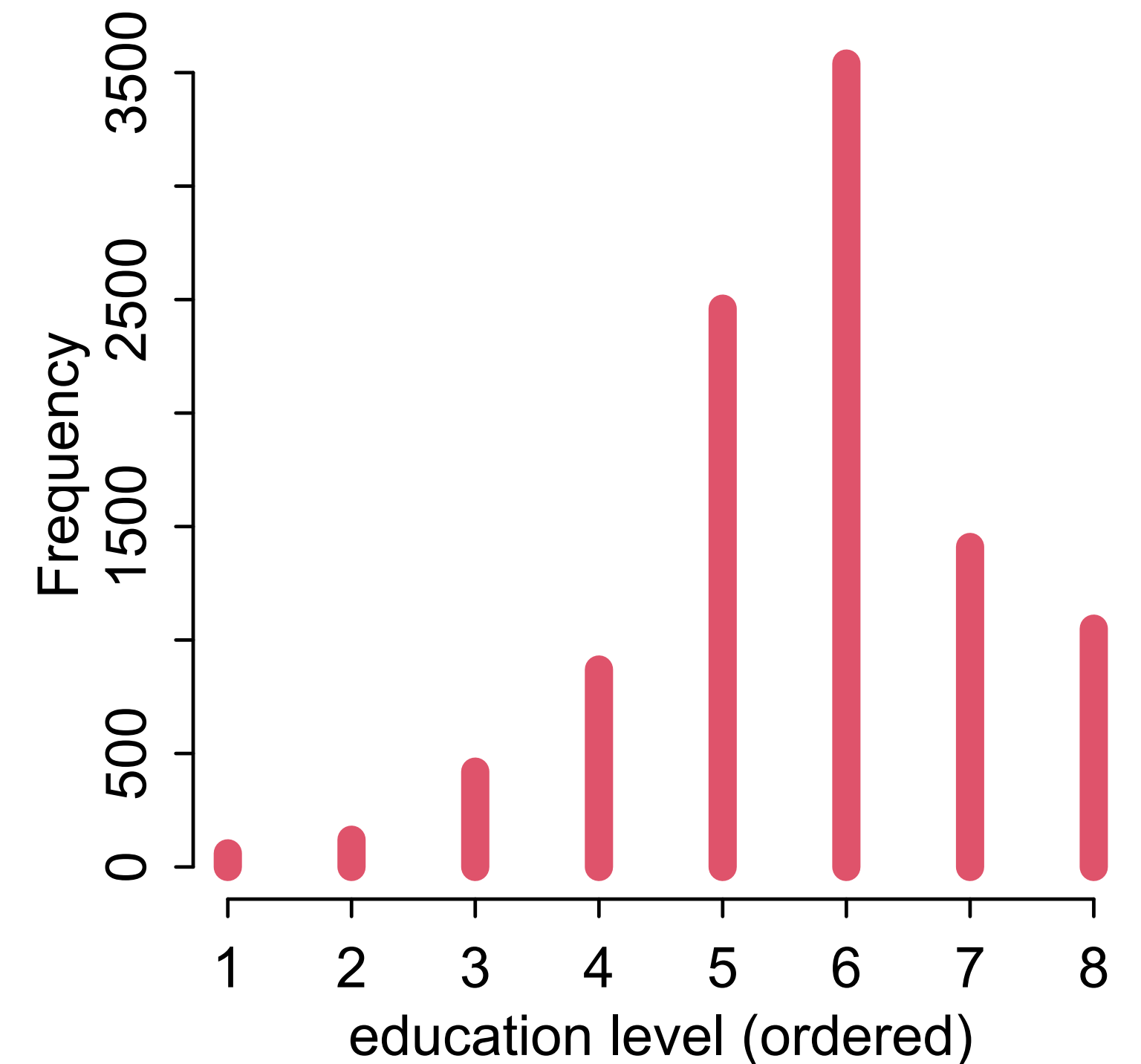
Want a parameter for each level

But how to **enforce ordering**, so that each level has larger (or smaller) effect than previous?



Ordered monotonic predictors

1 (elementary)	$\phi_i = 0$
2 (middle school)	$\phi_i = \delta_1$
3 (some high school)	$\phi_i = \delta_1 + \delta_2$
4 (high school)	$\phi_i = \delta_1 + \delta_2 + \delta_3$
5 (some college)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4$
6 (college)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5$
7 (master's)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6$
8 (doctorate)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6 + \delta_7$



Ordered monotonic predictors

1 (elementary)	$\phi_i = 0$	
2 (middle school)	$\phi_i = \delta_1$	
3 (some high school)	$\phi_i = \delta_1 + \delta_2$	
4 (high school)	$\phi_i = \delta_1 + \delta_2 + \delta_3$	
5 (some college)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4$	
6 (college)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5$	
7 (master's)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6$	<i>maximum effect of education</i>
8 (doctorate)	$\phi_i = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6 + \delta_7 = \beta_E$	

Ordered monotonic predictors

1 (elementary)

2 (middle school)

3 (some high school)

4 (high school)

5 (some college)

6 (college)

7 (master's)

8 (doctorate)

$$\delta_0 = 0$$

$$\sum_{j=0}^7 \delta_j = 1$$

Ordered monotonic predictors

1 (elementary)

2 (middle school)

3 (some high school)

4 (high school)

5 (some college)

6 (college)

7 (master's)

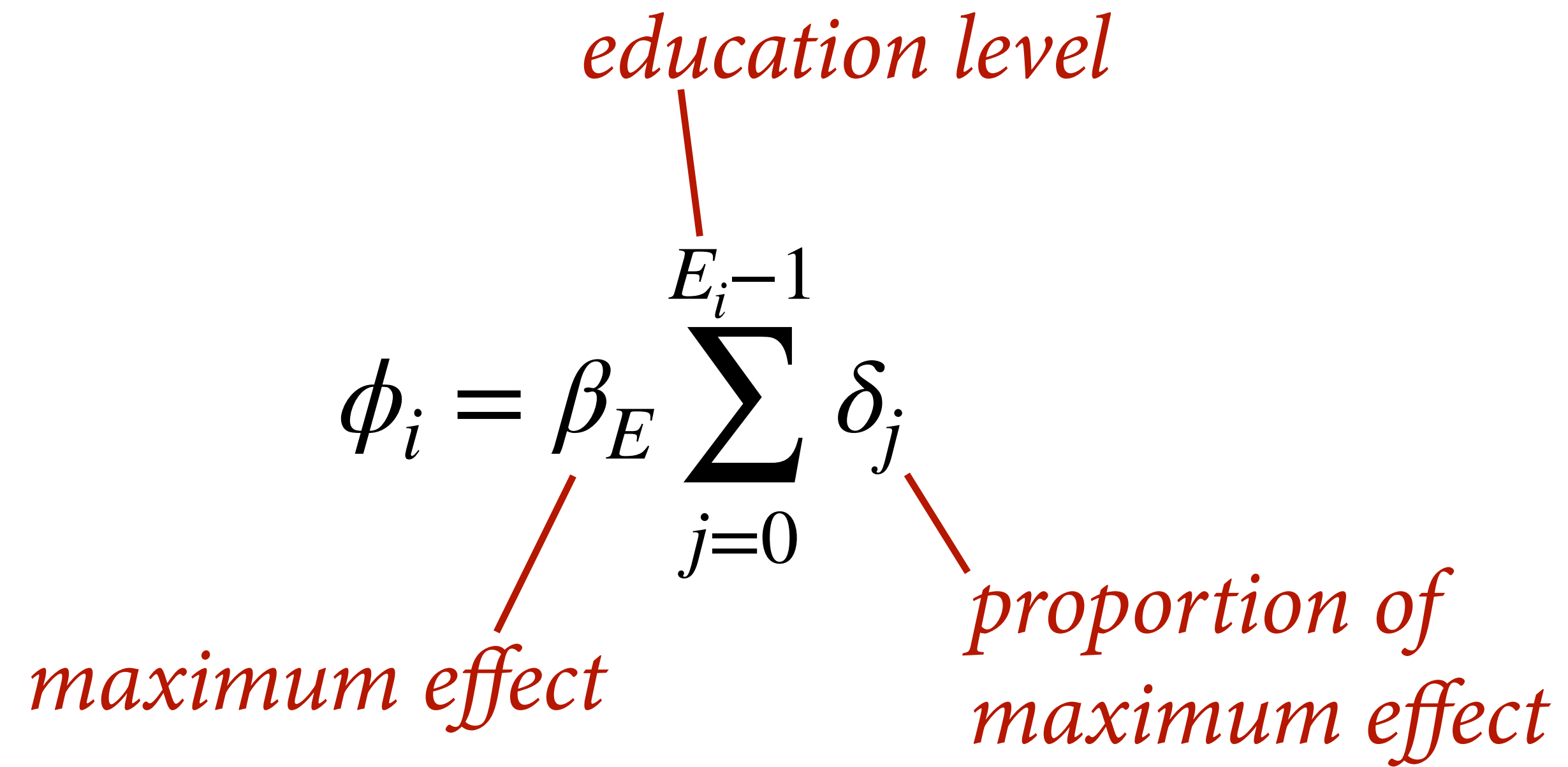
8 (doctorate)

$$\phi_i = \beta_E \sum_{j=0}^{E_i-1} \delta_j$$

education level

maximum effect

proportion of maximum effect



Ordered monotonic *priors*

How do we set priors for the delta parameters?

delta parameters form a **simplex**

Simplex: vector that sums to 1

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_E \sum_{j=0}^{E_i-1} \delta_j + \dots$$

$$\alpha_j \sim \text{Normal}(0,1)$$

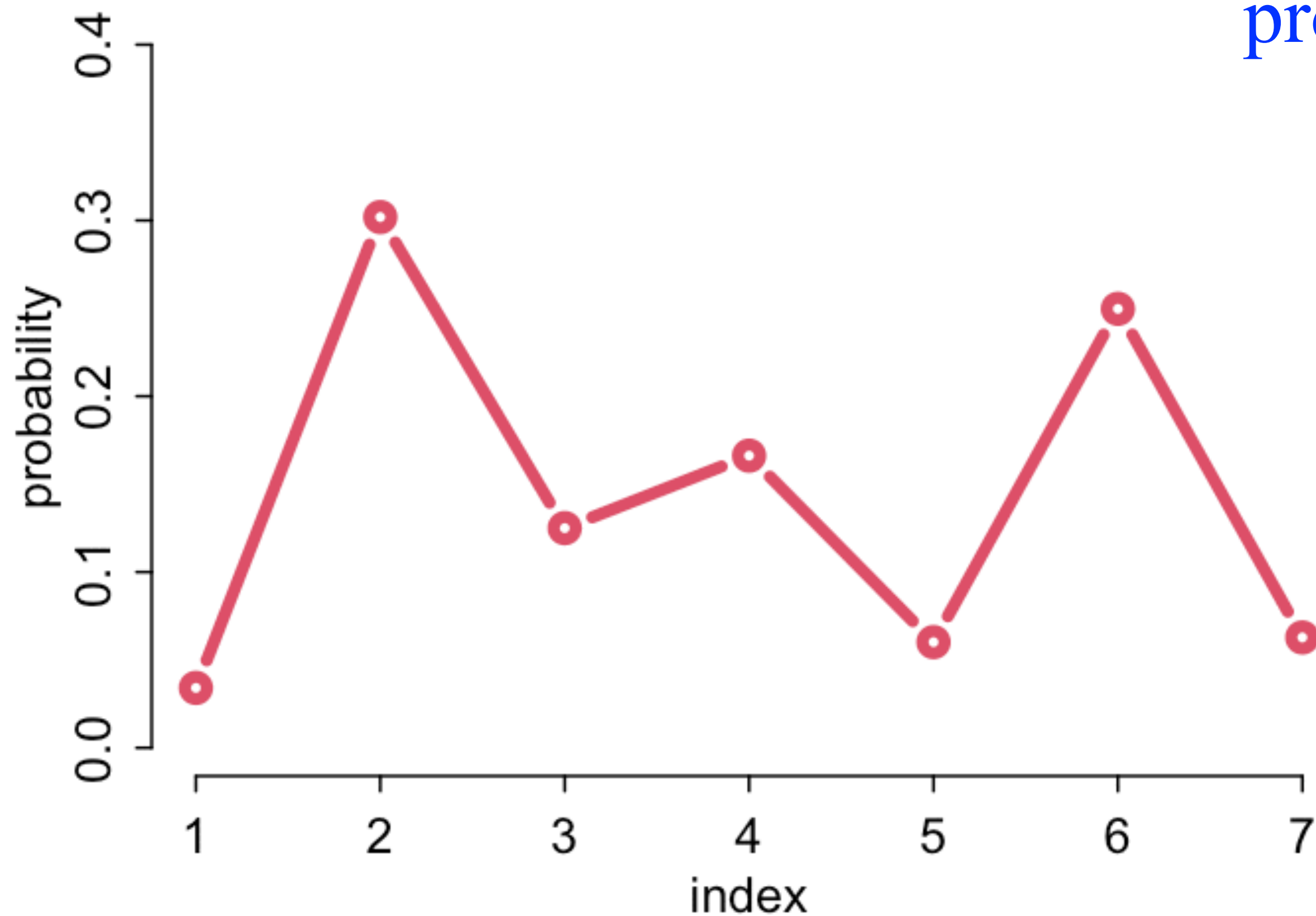
$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\delta_j \sim ?$$

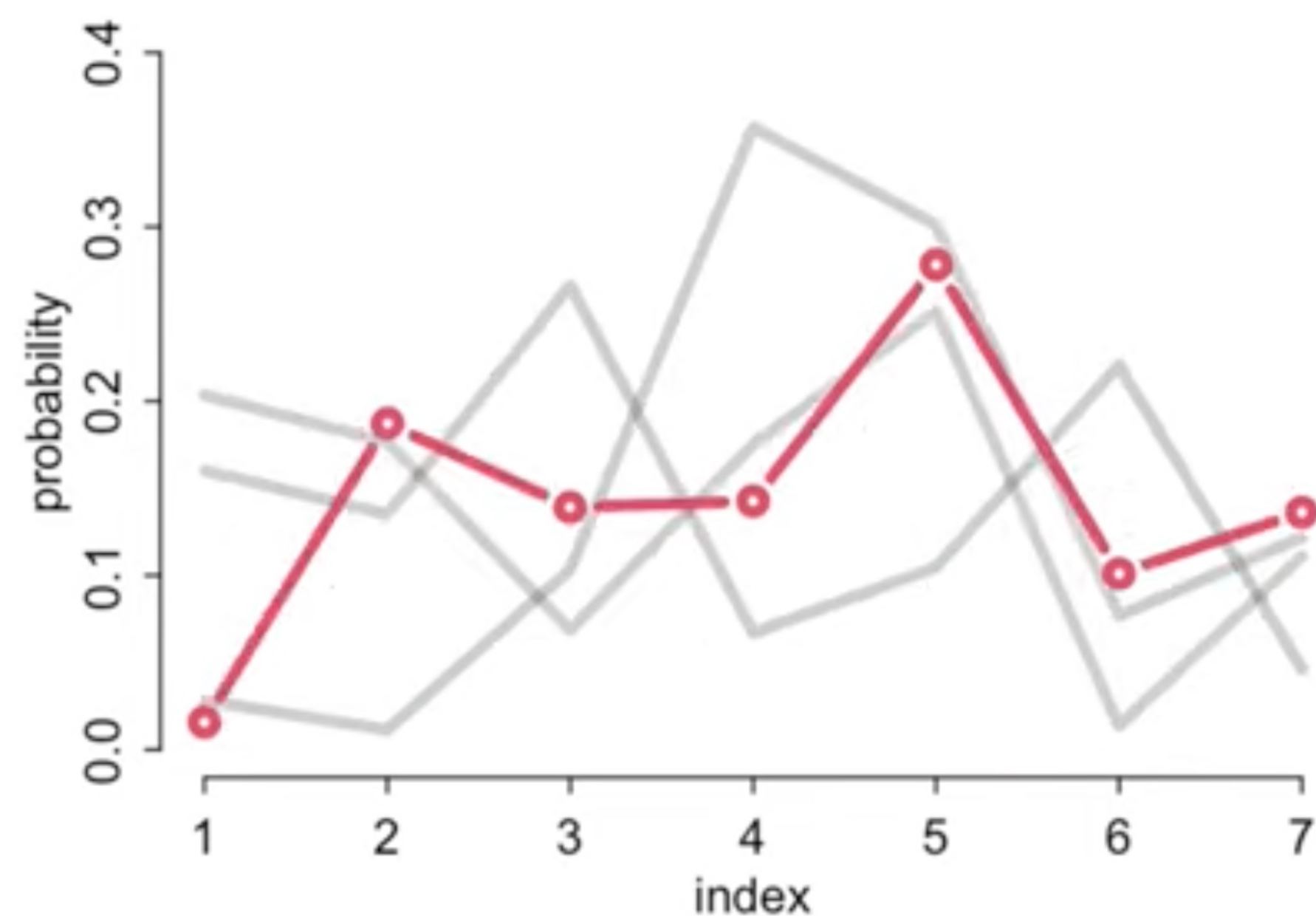
$$\delta \sim \text{Dirichlet}(a)$$

$$a = [2, 2, 2, 2, 2, 2, 2]$$

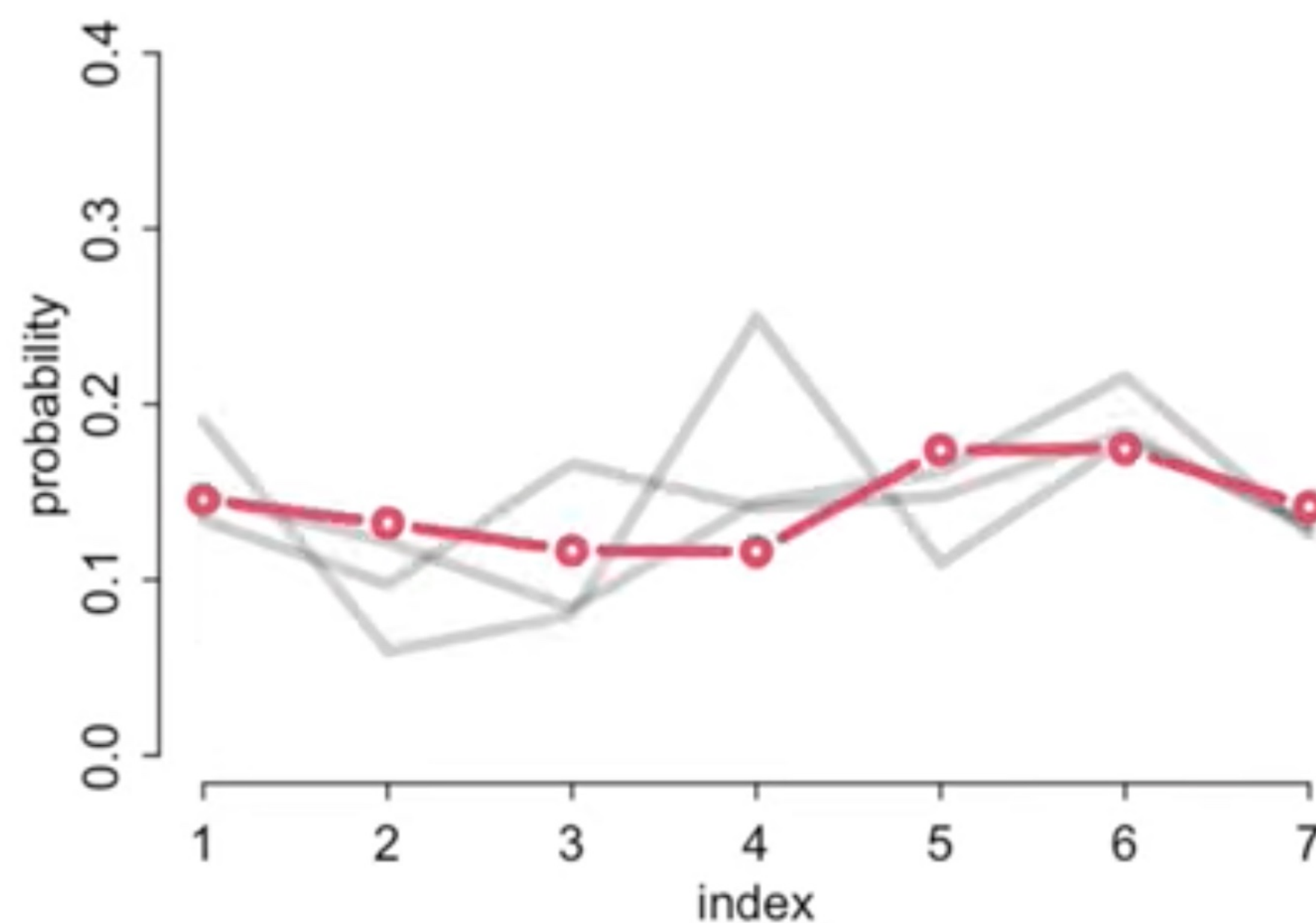
If all values in the vector “a” are equal,
then larger values mean that the
probabilities for each index are more similar



$\delta \sim \text{Dirichlet}(a)$
 $a = [2, 2, 2, 2, 2, 2, 2]$

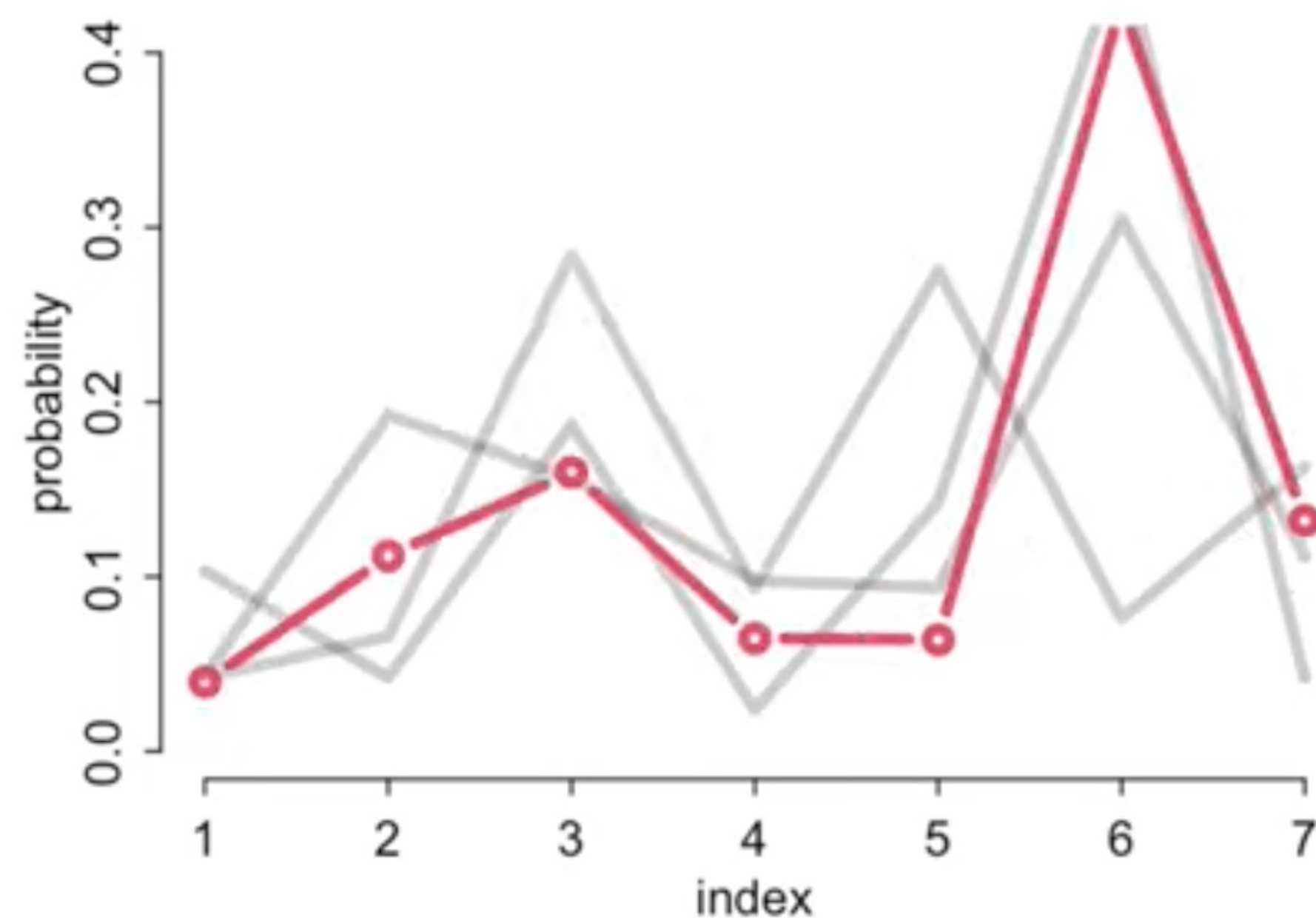


$\delta \sim \text{Dirichlet}(a)$
 $a = [10, 10, 10, 10, 10, 10, 10]$

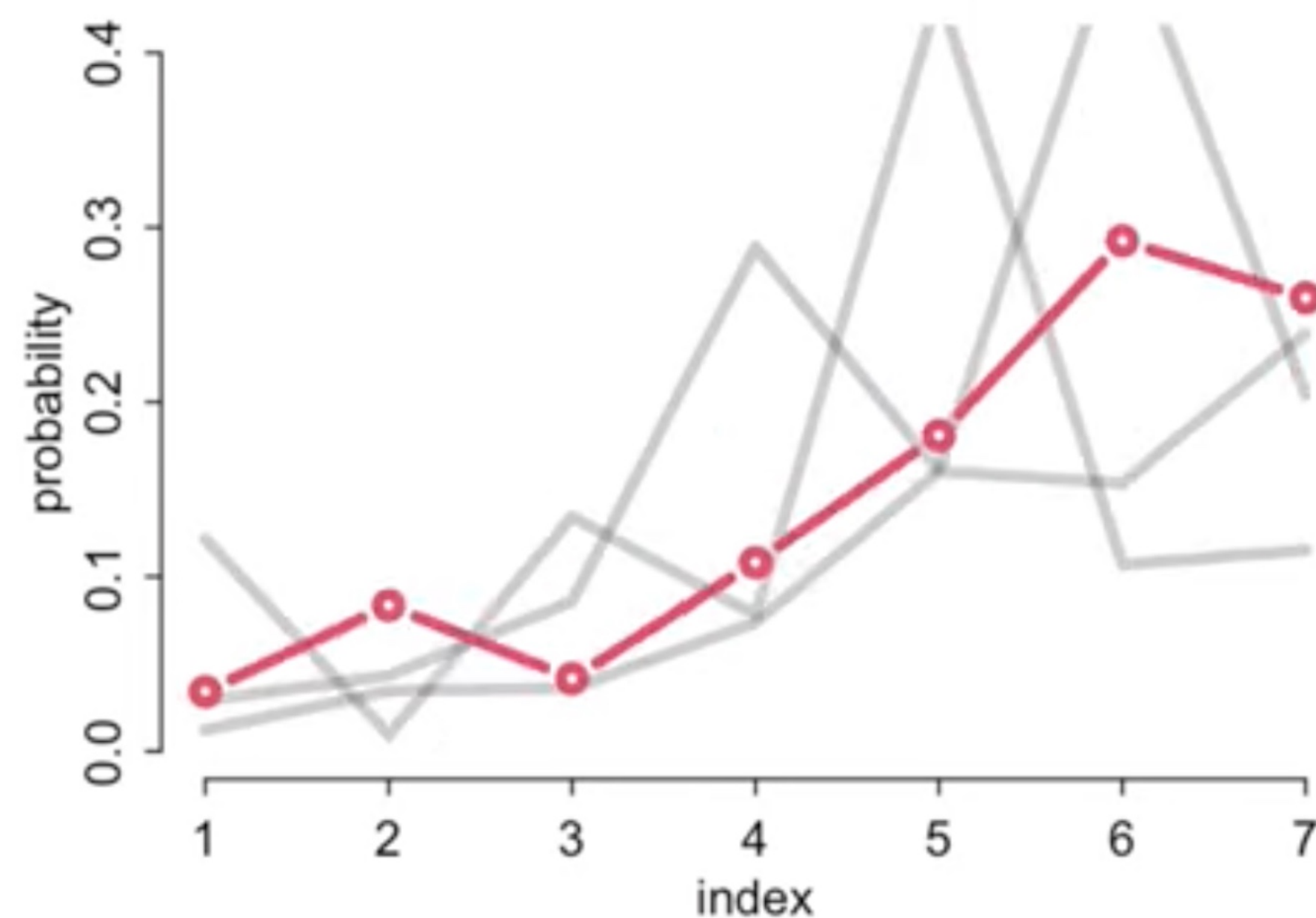


One can build prior information about delta sizes into the prior

$\delta \sim \text{Dirichlet}(a)$
 $a = [2, 2, 2, 2, 2, 2, 2]$



$\delta \sim \text{Dirichlet}(a)$
 $a = [1, 2, 3, 4, 5, 6, 7]$



```

edu_levels <- c( 6 , 1 , 8 , 4 , 7 , 2 , 5 , 3 )
edu_new <- edu_levels[ d$edu ]

dat$E <- edu_new
dat$a <- rep(2,7) # dirichlet prior

mRXE <- ulam(
  alist(
    R ~ ordered_logistic( phi , alpha ),
    phi <- bE*sum( delta_j[1:E] ) +
      bA*A + bI*I + bC*C,
    alpha ~ normal( 0 , 1 ),
    c(bA,bI,bC,bE) ~ normal( 0 , 0.5 ),
    vector[8]: delta_j <- append_row( 0 , delta ),
    simplex[7]: delta ~ dirichlet( a )

  ), data=dat , chains=4 , cores=4 )

```

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_E \sum_{j=0}^{E_i-1} \delta_j + \dots$$

$$\alpha_j \sim \text{Normal}(0,1)$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\delta \sim \text{Dirichlet}(a)$$

```

edu_levels <- c( 6 , 1 , 8 , 4 , 7 , 2 , 5 , 3 )
edu_new <- edu_levels[ d$edu ]

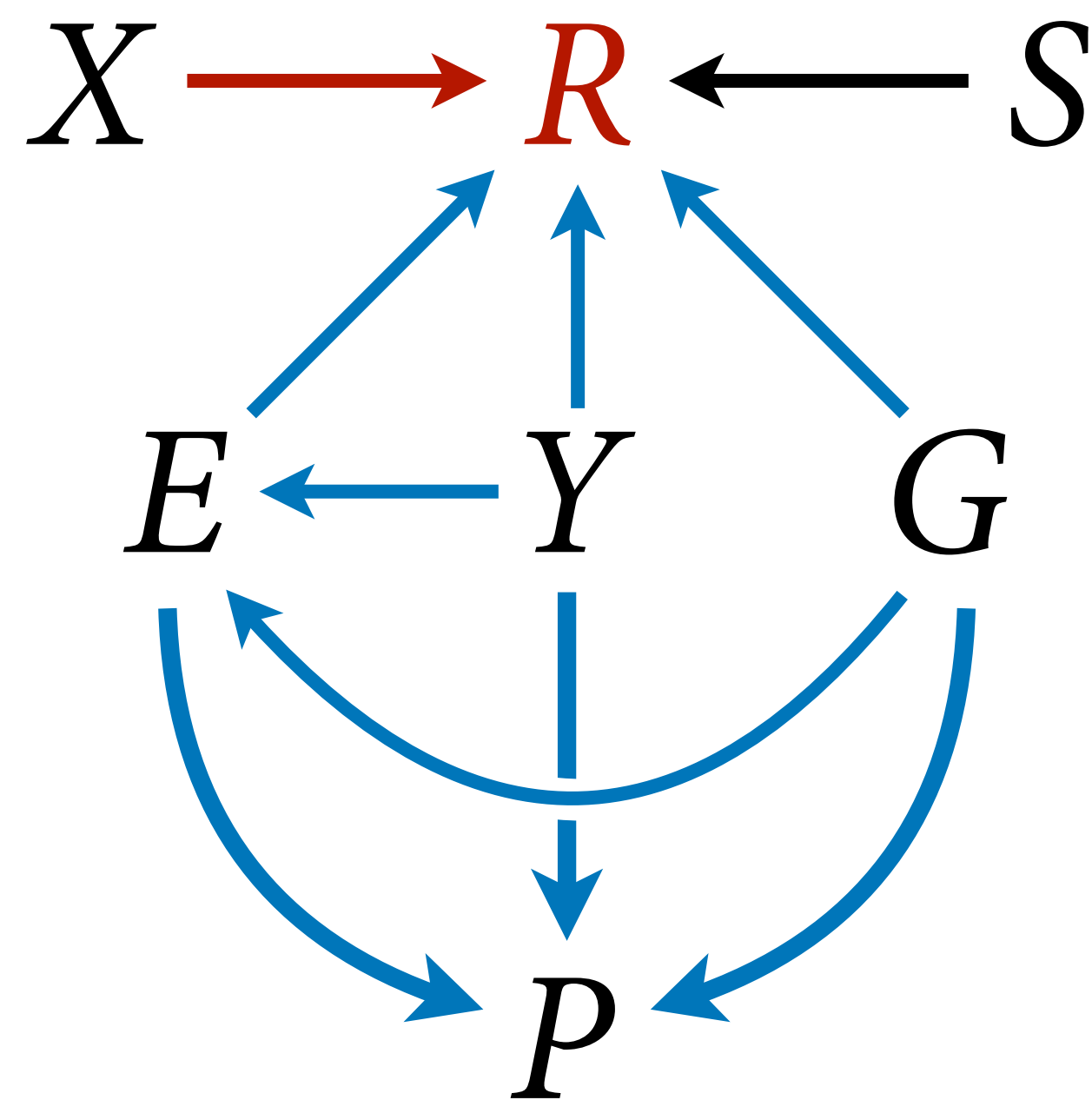
dat$E <- edu_new
dat$a <- rep(2,7) # dirichlet prior

mRXE <- ulam(
  alist(
    R ~ ordered_logistic( phi , alpha ),
    phi <- bE*sum( delta_j[1:E] ) +
      bA*A + bI*I + bC*C,
    alpha ~ normal( 0 , 1 ),
    c(bA,bI,bC,bE) ~ normal( 0 , 0.5 ),
    vector[8]: delta_j <- append_row( 0 ,
      simplex[7]: delta ~ dirichlet( a )
    ), data=dat , chains=4 , cores=4 )

```

```
> precis(mRXE,2)
```

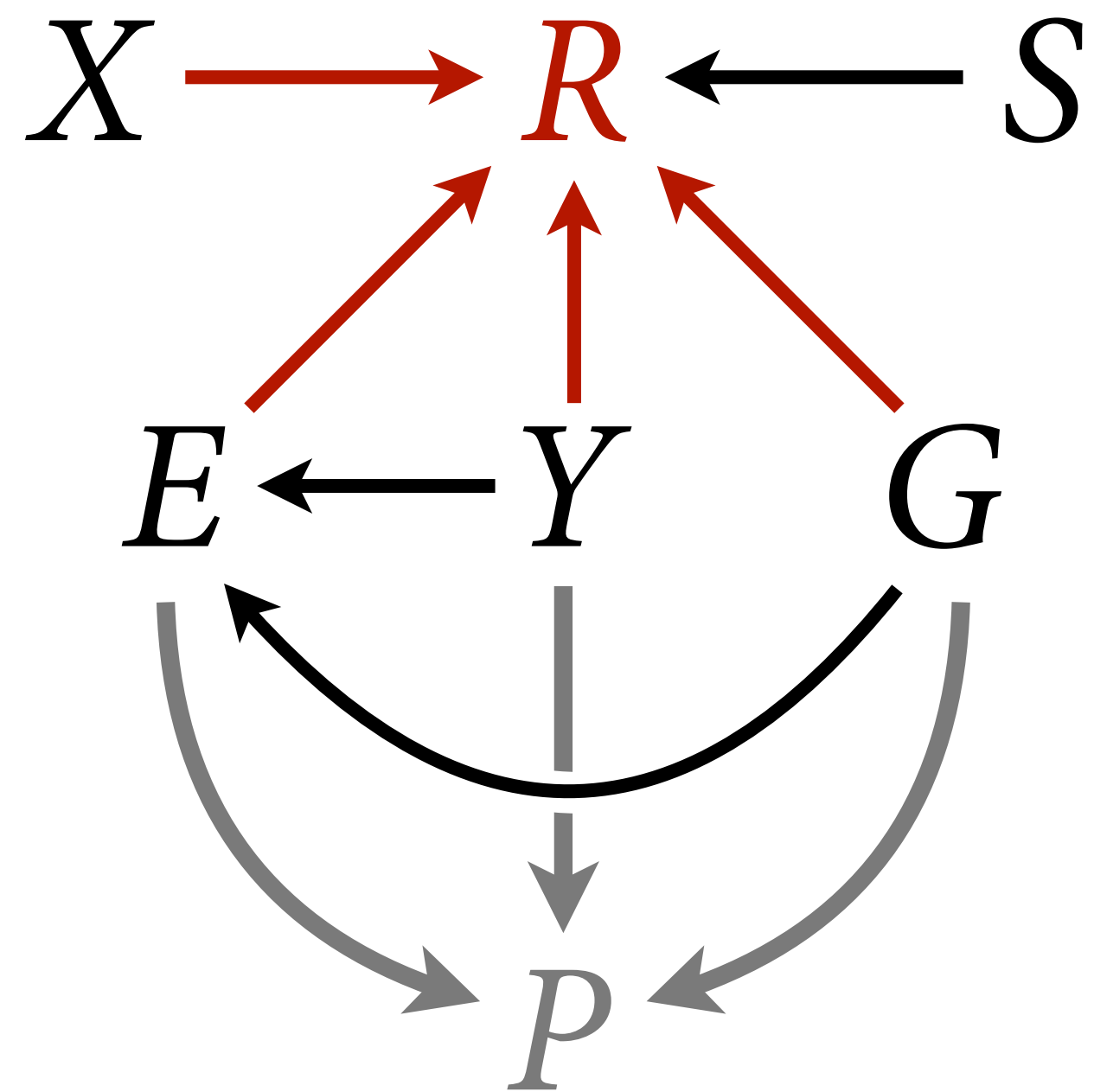
	mean	sd	5.5%	94.5%	n_eff	Rhat4
alpha[1]	-3.07	0.14	-3.32	-2.86	793	1
alpha[2]	-2.39	0.14	-2.63	-2.17	804	1
alpha[3]	-1.81	0.14	-2.05	-1.60	811	1
alpha[4]	-0.79	0.14	-1.03	-0.57	799	1
alpha[5]	-0.12	0.14	-0.36	0.10	804	1
alpha[6]	0.79	0.14	0.54	1.00	831	1
bE	-0.31	0.16	-0.57	-0.06	838	1
bC	-0.96	0.05	-1.04	-0.88	1757	1
bI	-0.72	0.04	-0.77	-0.66	1982	1
bA	-0.70	0.04	-0.77	-0.64	1779	1
delta[1]	0.22	0.13	0.05	0.47	1227	1
delta[2]	0.14	0.09	0.03	0.31	2258	1
delta[3]	0.20	0.11	0.05	0.38	2256	1
delta[4]	0.17	0.09	0.04	0.34	1926	1
delta[5]	0.04	0.05	0.01	0.12	945	1
delta[6]	0.10	0.07	0.02	0.23	1870	1
delta[7]	0.13	0.08	0.03	0.27	2335	1



bE not interpretable
 -> need to stratify by Gender

```
> precis(mRXE,2)
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
alpha[1]	-3.07	0.14	-3.32	-2.86	793	1
alpha[2]	-2.39	0.14	-2.63	-2.17	804	1
alpha[3]	-1.81	0.14	-2.05	-1.60	811	1
alpha[4]	-0.79	0.14	-1.03	-0.57	799	1
alpha[5]	-0.12	0.14	-0.36	0.10	804	1
alpha[6]	0.79	0.14	0.54	1.00	831	1
bE	-0.31	0.16	-0.57	-0.06	838	1
bC	-0.96	0.05	-1.04	-0.88	1757	1
bI	-0.72	0.04	-0.77	-0.66	1982	1
bA	-0.70	0.04	-0.77	-0.64	1779	1
delta[1]	0.22	0.13	0.05	0.47	1227	1
delta[2]	0.14	0.09	0.03	0.31	2258	1
delta[3]	0.20	0.11	0.05	0.38	2256	1
delta[4]	0.17	0.09	0.04	0.34	1926	1
delta[5]	0.04	0.05	0.01	0.12	945	1
delta[6]	0.10	0.07	0.02	0.23	1870	1
delta[7]	0.13	0.08	0.03	0.27	2335	1



$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{E,G[i]} \sum_{j=0}^{E_i-1} \delta_j +$$

$$\beta_{A,G[i]} A_i + \beta_{I,G[i]} I_i + \beta_{C,G[i]} C_i +$$

$$\beta_{Y,G[i]} Y_i$$

$$\alpha_j \sim \text{Normal}(0,1)$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\delta \sim \text{Dirichlet}(a)$$

```

dat$Y <- standardize(d$age)

mRXEYGt <- ulam(
  alist(
    R ~ ordered_logistic( phi , alpha ),
    phi <- bE[G]*sum( delta_j[1:E] ) +
      bA[G]*A + bI[G]*I + bC[G]*C +
      bY[G]*Y,
    alpha ~ normal( 0 , 1 ),
    bA[G] ~ normal( 0 , 0.5 ),
    bI[G] ~ normal( 0 , 0.5 ),
    bC[G] ~ normal( 0 , 0.5 ),
    bE[G] ~ normal( 0 , 0.5 ),
    bY[G] ~ normal( 0 , 0.5 ),
    vector[8]: delta_j <-<- append_row( 0 , delta ),
    simplex[7]: delta ~ dirichlet( a )
  ), data=dat , chains=4 , cores=4 , threads=2 )

```

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{E,G[i]} \sum_{j=0}^{E_i-1} \delta_j + \beta_{A,G[i],i} A_i + \beta_{I,G[i]} I_i + \beta_{C,G[i]} C_i + \beta_{Y,G[i]} Y_i$$

$$\alpha_j \sim \text{Normal}(0,1)$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\delta \sim \text{Dirichlet}(a)$$

```

dat$Y <- standardize(d$age)

mRXEYGt <- ulam(
  alist(
    R ~ ordered_logistic( phi , alpha ),
    phi <- bE[G]*sum( delta_j[1:E] ) +
      bA[G]*A + bI[G]*I + bC[G]*C +
      bY[G]*Y,
    alpha ~ normal( 0 , 1 ),
    bA[G] ~ normal( 0 , 0.5 ),
    bI[G] ~ normal( 0 , 0.5 ),
    bC[G] ~ normal( 0 , 0.5 ),
    bE[G] ~ normal( 0 , 0.5 ),
    bY[G] ~ normal( 0 , 0.5 ),
    vector[8]: delta_j <-<- append_row( 0 , delta ),
    simplex[7]: delta ~ dirichlet( a )
  ), data=dat , chains=4 , cores=4 , threads=2 )

```

4 chains times 2 threads each = 8 cores

$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{E,G[i]} \sum_{j=0}^{E_i-1} \delta_j + \beta_{A,G[i],i} A_i + \beta_{I,G[i]} I_i + \beta_{C,G[i]} C_i + \beta_{Y,G[i]} Y_i$$

$$\alpha_j \sim \text{Normal}(0,1)$$

$$\beta_- \sim \text{Normal}(0,0.5)$$

$$\delta \sim \text{Dirichlet}(a)$$


```

dat$Y <- standardize(d$age)

mRXEYGt <- ulam(
  alist(
    R ~ ordered_logistic( phi , alpha ),
    phi <- bE[G]*sum( delta_j[1:E] ) +
      bA[G]*A + bI[G]*I + bC[G]*C +
      bY[G]*Y,
    alpha ~ normal( 0 , 1 ),
    bA[G] ~ normal( 0 , 0.5 ),
    bI[G] ~ normal( 0 , 0.5 ),
    bC[G] ~ normal( 0 , 0.5 ),
    bE[G] ~ normal( 0 , 0.5 ),
    bY[G] ~ normal( 0 , 0.5 ),
    vector[8]: delta_j <-< append_row( 0 , delta ),
    simplex[7]: delta ~ dirichlet( a )
  ), data=dat , chains=4 , cores=4 , threads=2 )

```

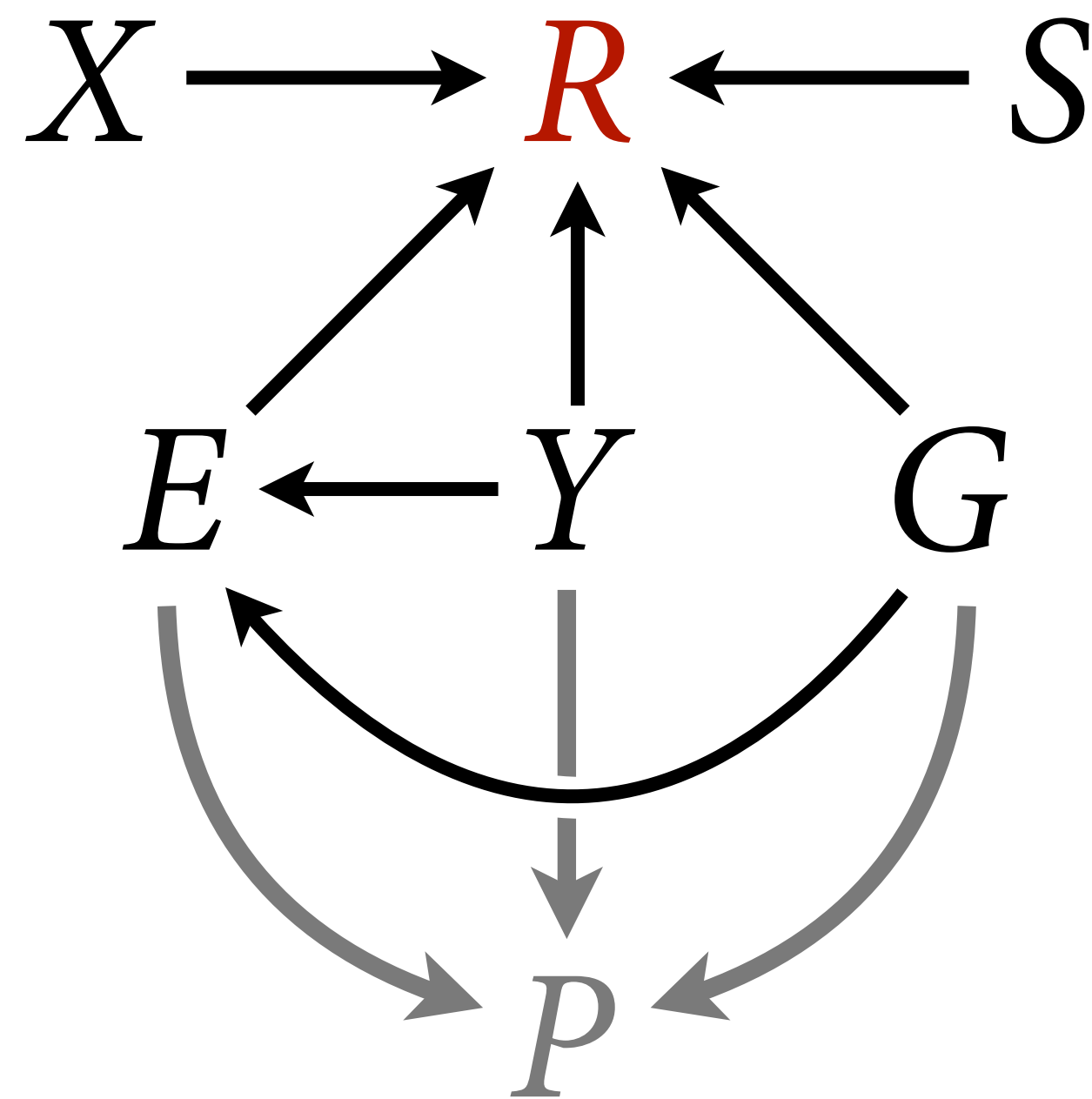
1 thread each

Sampling durations (minutes):			
	warmup	sample	total
chain:1	6.53	3.99	10.52
chain:2	7.33	2.66	9.99
chain:3	6.88	3.70	10.58
chain:4	6.40	2.63	9.03

2 threads each

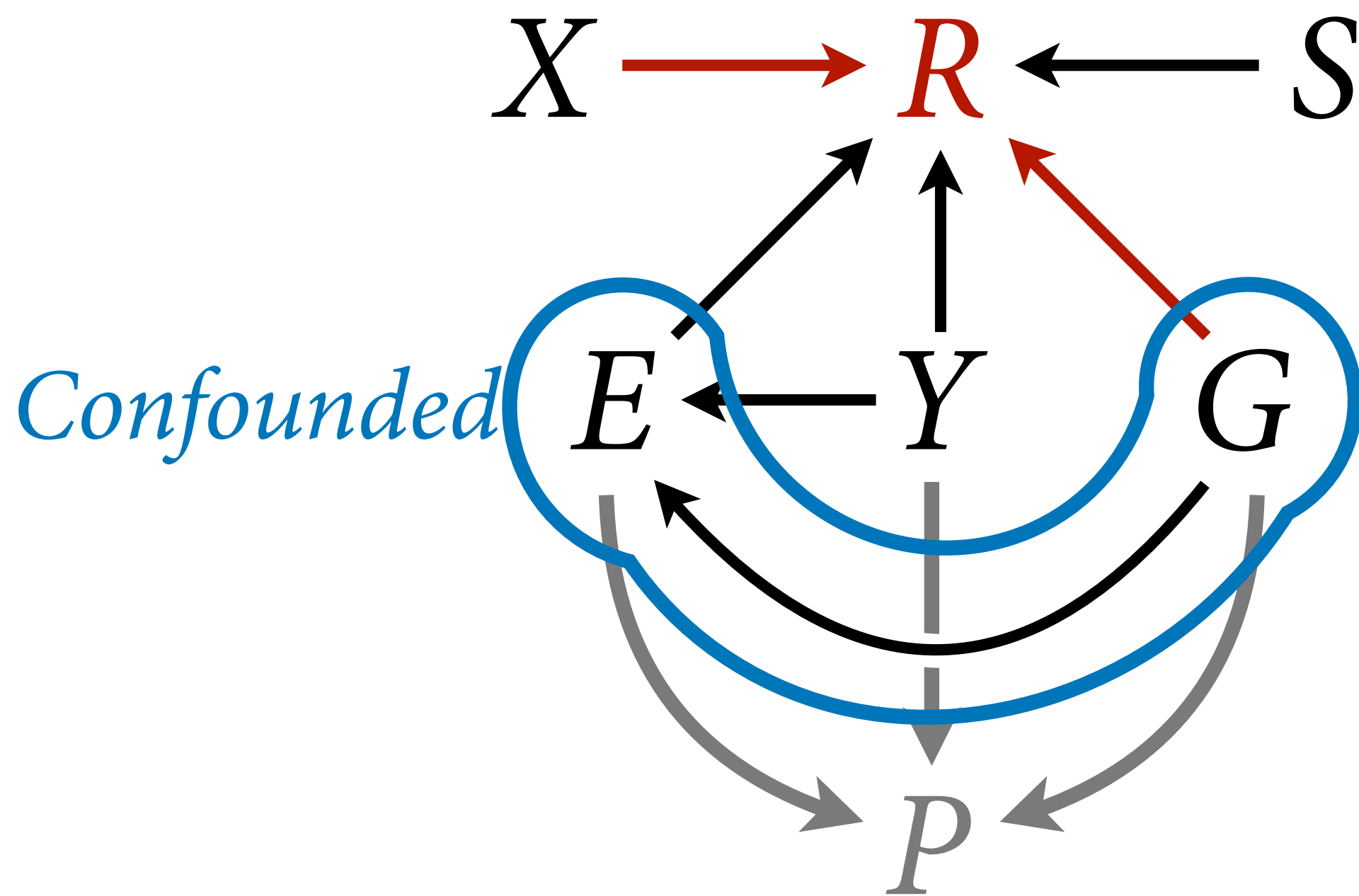
Sampling durations (minutes):			
	warmup	sample	total
chain:1	4.41	1.80	6.21
chain:2	4.69	1.87	6.56
chain:3	5.14	1.56	6.70
chain:4	4.21	1.84	6.05

4 chains times 2 threads each = 8 cores



```
> precis(mRXEYGt,2)
```

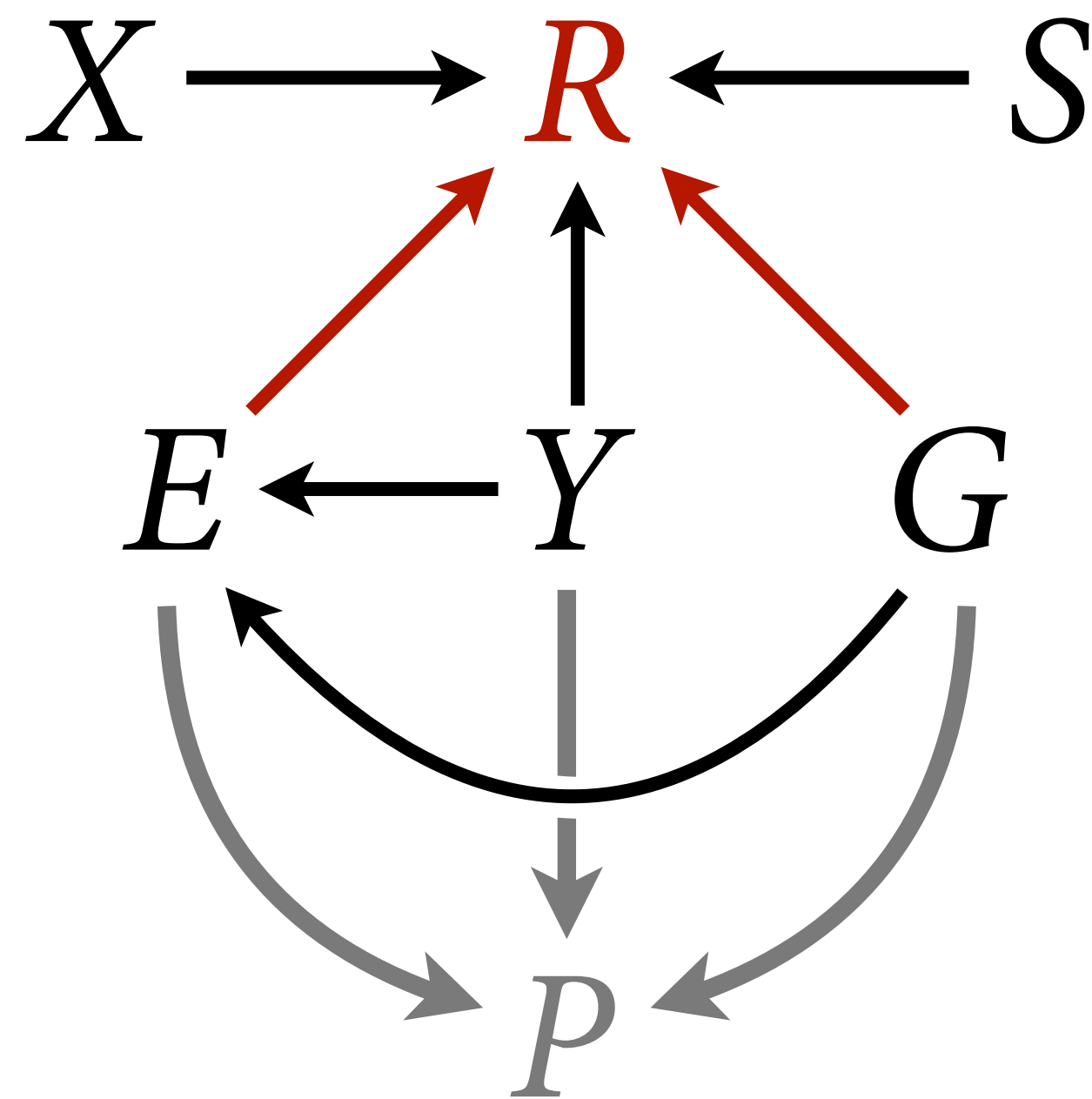
	mean	sd	5.5%	94.5%	n_eff	Rhat4
alpha[1]	-2.89	0.10	-3.06	-2.73	729	1
alpha[2]	-2.21	0.10	-2.37	-2.06	728	1
alpha[3]	-1.62	0.10	-1.78	-1.47	724	1
alpha[4]	-0.58	0.10	-0.74	-0.43	729	1
alpha[5]	0.11	0.10	-0.05	0.26	726	1
alpha[6]	1.03	0.10	0.87	1.18	746	1
bA[1]	-0.56	0.06	-0.65	-0.47	1932	1
bA[2]	-0.81	0.05	-0.90	-0.73	2013	1
bI[1]	-0.66	0.05	-0.74	-0.58	2539	1
bI[2]	-0.76	0.05	-0.84	-0.68	2283	1
bC[1]	-0.77	0.07	-0.88	-0.65	2029	1
bC[2]	-1.09	0.07	-1.20	-0.99	2012	1
bE[1]	-0.63	0.14	-0.85	-0.42	810	1
bE[2]	0.41	0.14	0.19	0.62	795	1
bY[1]	0.00	0.03	-0.05	0.05	2740	1
bY[2]	-0.13	0.03	-0.18	-0.09	1426	1
delta[1]	0.15	0.08	0.04	0.31	1759	1
delta[2]	0.15	0.09	0.04	0.30	2440	1
delta[3]	0.29	0.11	0.11	0.46	2001	1
delta[4]	0.08	0.05	0.02	0.17	2414	1
delta[5]	0.06	0.04	0.01	0.14	1087	1
delta[6]	0.24	0.07	0.13	0.34	2301	1
delta[7]	0.04	0.02	0.01	0.08	2755	1



*Only
direct
effect G*

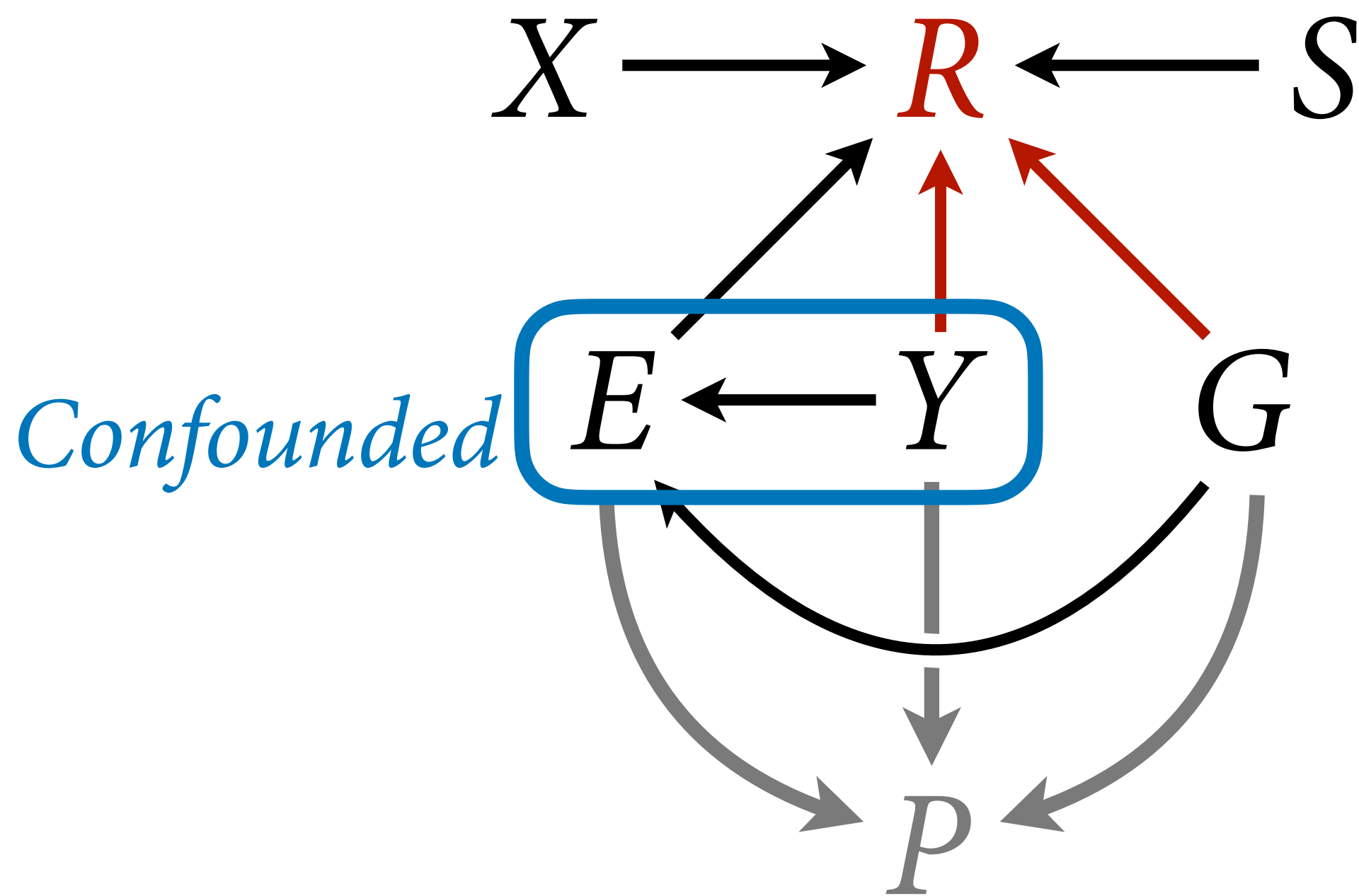
```
> precis(mRXEYGt,2)
```

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bC[1]	-0.77	0.07	-0.88	-0.65	2029	1
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delta[6]	0.24	0.07	0.13	0.34	2301	1
delta[7]	0.04	0.02	0.01	0.08	2755	1



```
> precis(mRXEYGt,2)
```

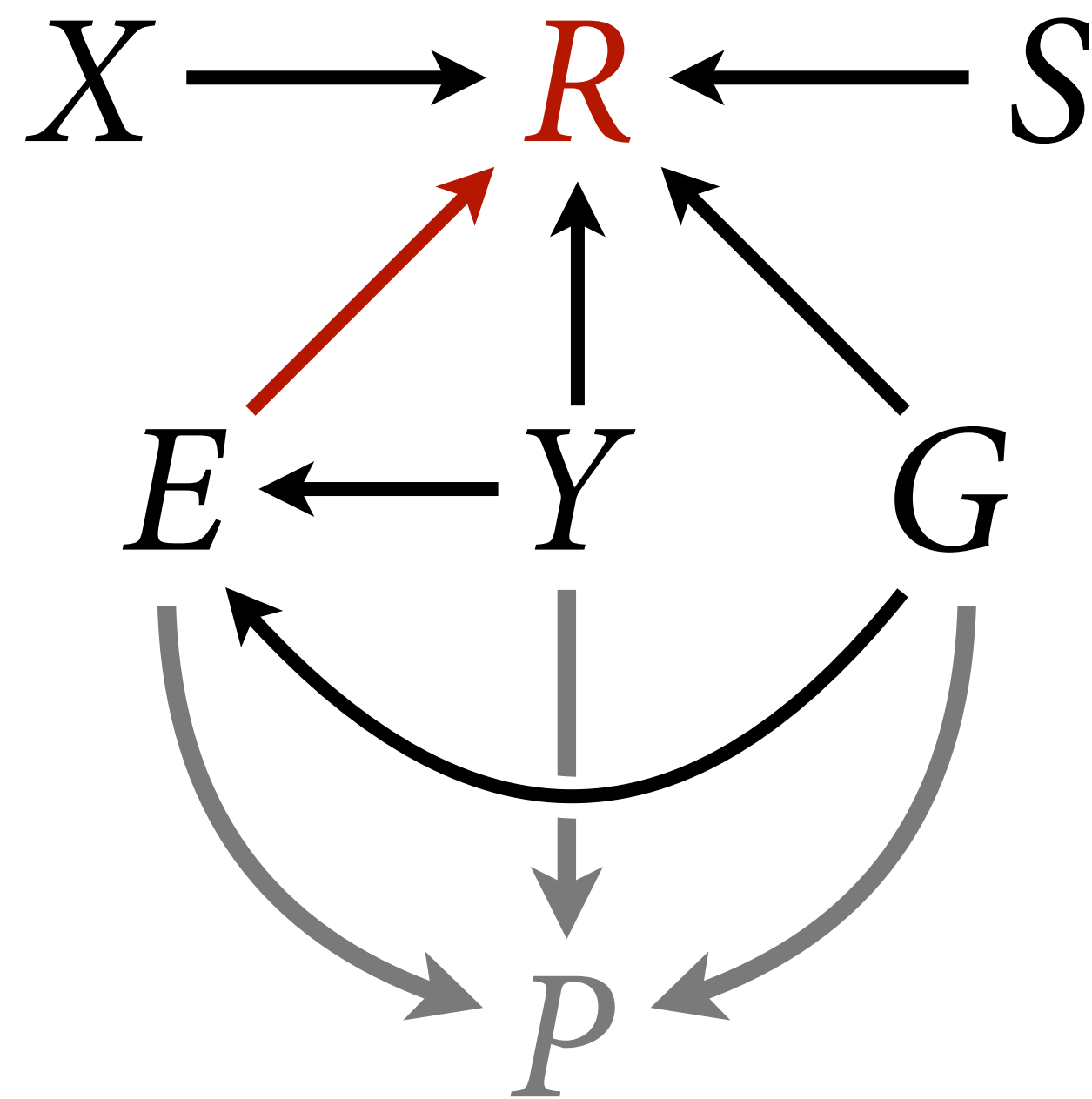
	mean	sd	5.5%	94.5%	n_eff	Rhat4
alpha[1]	-2.89	0.10	-3.06	-2.73	729	1
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bC[1]	-0.77	0.07	-0.88	-0.65	2029	1
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bE[2]	0.41	0.14	0.19	0.62	795	1
bY[1]	0.00	0.03	-0.03	0.03	2740	1
bY[2]	-0.13	0.03	-0.18	-0.09	1426	1
delta[1]	0.15	0.08	0.04	0.31	1759	1
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*Only
direct
effect*

```
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```

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bC[2]	-1.09	0.07	-1.20	-0.99	2012	1
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delta[7]	0.04	0.02	0.01	0.08	2755	1



```
> precis(mRXEYGt,2)
```

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Complex causal effects

Causal effects (predicted consequences of intervention) require **marginalization**

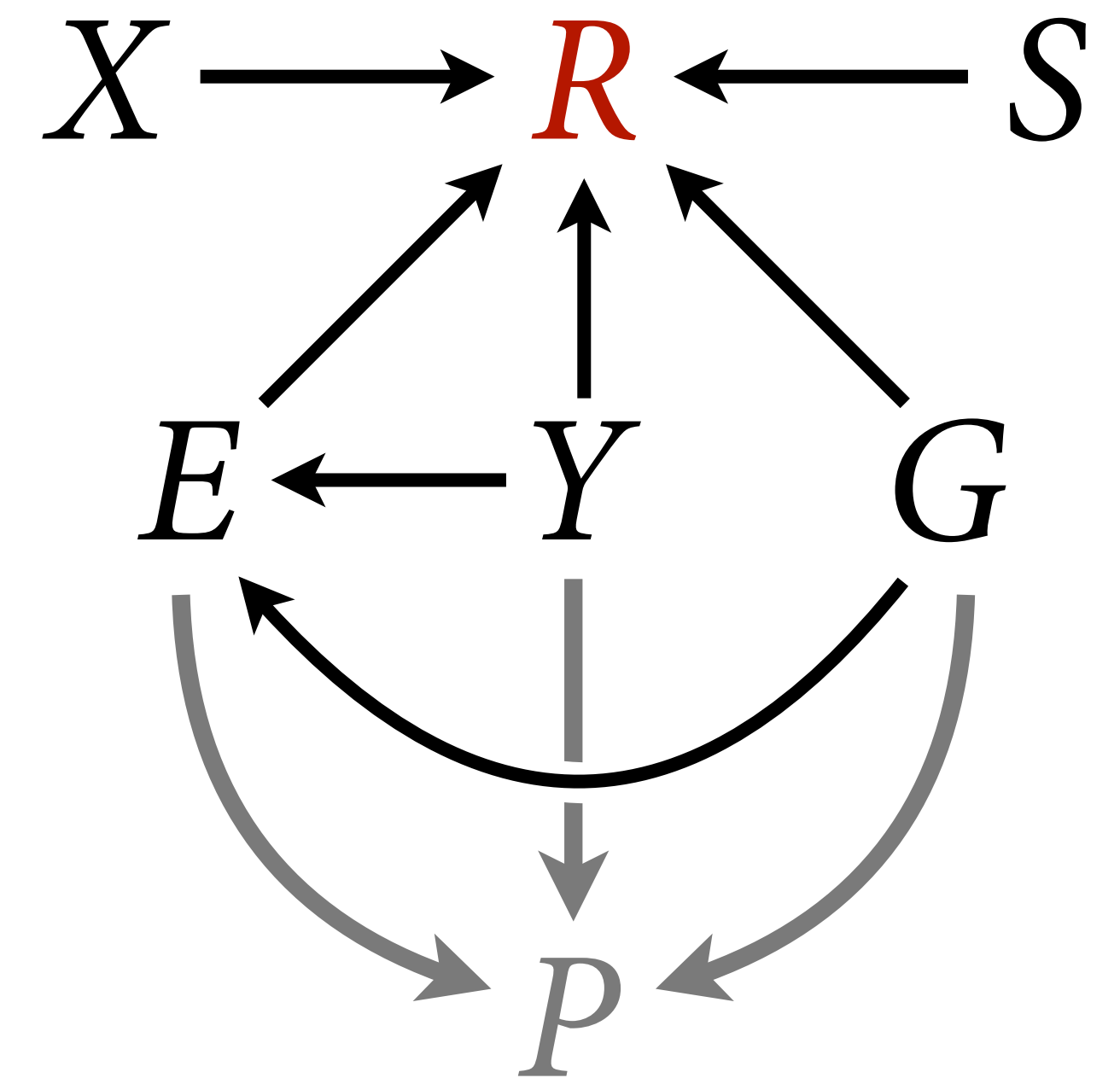
Example: Causal effect of E requires distribution of Y and G to average over

Problem 1: Should not marginalize over **this** sample—*cursed P !* Post-stratify to new target.

Problem 2: Should not set all Y to same E

-> A more representative Age, Gender, and Ed. distribution with appropriate covariance

Example: Causal effect of Y requires effect of Y on E , which we cannot estimate (P again!)



Complex causal effects

Causal effects (predicted consequences of
interventions)

Example
of Y and

Problem
sample

Problem

No matter how complex, still just a **generative simulation** using **posterior samples**

Need generative model to plan estimation

Need generative model to compute estimates

S

G

P

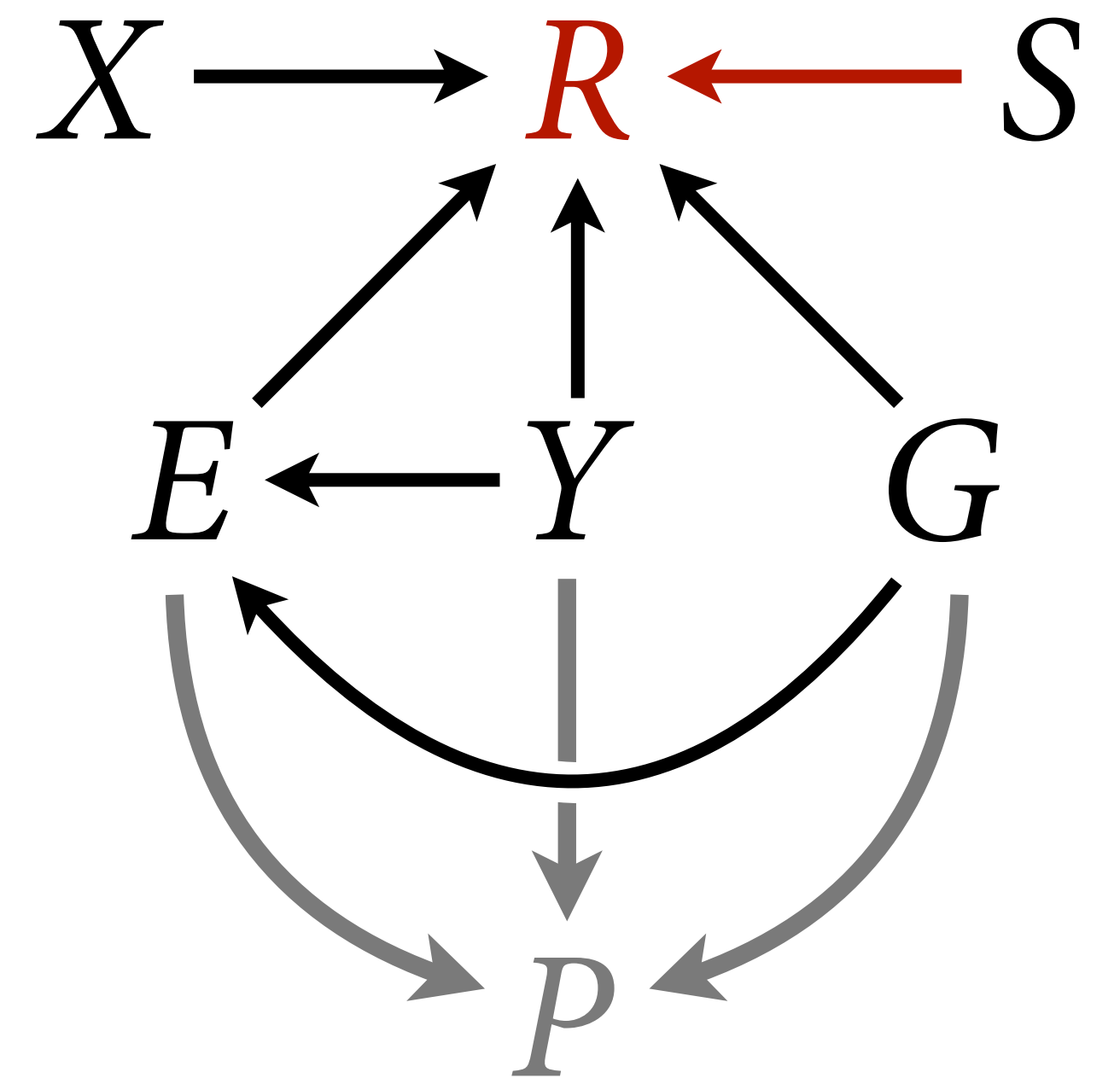
Example: Causal effect of Y requires effect of Y
on E , which we cannot estimate (P again!)

Repeat observations

30 stories (S)

```
> table(d$story)
```

aqu	boa	box	bur	car	che	pon	rub	sha	shi	spe	swi
662	662	1324	1324	662	662	662	662	662	662	993	993



Repeat observations

30 stories (S)

```
> table(d$story)
```

aqu	boa	box	bur	car	che	pon	rub	sha	shi	spe	swi
662	662	1324	1324	662	662	662	662	662	662	993	993

331 individuals (U)

```
> table(d$id)
```

96;434	96;445	96;451	96;456	96;458	96;466	96;467	96;474	96;480	96;481	96;497
30	30	30	30	30	30	30	30	30	30	30
96;498	96;502	96;505	96;511	96;512	96;518	96;519	96;531	96;533	96;538	96;547
30	30	30	30	30	30	30	30	30	30	30
96;550	96;553	96;555	96;558	96;560	96;562	96;566	96;570	96;581	96;586	96;591
30	30	30	30	30	30	30	30	30	30	30

